A Framework for Stochastic Dynamics, Coarse-Graining, and Learned Probability Flows

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Introduction

The starting point of both statistical physics and generative artificial intelligence is the relationship between the evolution of an underlying state and the induced evolution of observables and probability distributions. To describe this relationship, we distinguish three fundamental objects: the state x of a system, an observable f defined on its state space, and a probability measure μ describing the statistical distribution of its states. Once the underlying dynamics is specified, it induces two complementary descriptions: the evolution of observables and the transport of probability measures.

We begin by establishing the algebraic, measurable, and differentiable structures of the underlying state space and the corresponding observable space. We then specify deterministic and stochastic dynamics and investigate how they induce evolution on observables and probability measures. This provides the mathematical foundation for studying coarse-graining, where microscopic descriptions are mapped onto lower-dimensional collective variables, and conditional expectation provides a natural way to formulate reduced observables and effective dynamics.

The central question throughout this development is how these successive transformations are related and under what conditions they preserve the relevant dynamical and statistical structures. In particular, we examine the compatibility between flow composition, the pushforward of probability measures, conditional expectation, and the approximation of infinitesimal generators. These questions connect the mathematical description of microscopic dynamics with the construction of effective models and their numerical or data-driven approximations.

Original problem source and attribution. The point of departure is the supplied assessment portfolio [Assessment], primarily Project 9, together with the Hamiltonian application of Project 10 and selected questions from Projects 8 and 11–14. Core questions develop the theoretical framework; secondary tasks retain their statements and original solution source, with detailed answers temporarily deferred.

All numerical labels of the form "Project n, item m" refer to the *Student Assessment Exercise* problem portfolio supplied with this project [Assessment] (original PDF retained locally and excluded from the repository). A source marker appears at the relevant result, and the full question-to-section concordance is given in the Source Index (Table 1).

Source Index and Direct Links

Each blue project entry links to its retained statement, problem, or answer. *Core* identifies a question central to the theoretical framework; *Concise* identifies an auxiliary question with its necessary conclusions and a shortened argument; *Pending* identifies a task whose solution or proof has been commented out and whose entry leads to an explicit completion notice. A superscript $\star$ marks an entry with a solution, proof, or extension still to be completed, including every entry whose proof reads *Proof to be completed.* and the partially deferred extensions in Projects 9.11 and 13.1. The page number is also a clickable link to that location. Project and question numbers refer to the *original assessment PDF* [Assessment], not to this manuscript. An unnumbered project task is listed as "Main task." The separate entry for Project 9, item 4 (numerics) leads to its discrete-time analysis.

Shaded *Problem* and *Exercise* panels identify task statements. Orange *Later thought* panels contain optional pending ideas, not additional solved source problems. Theorem, proposition, and lemma statements use the same blue panels; their proofs follow outside the panels. Corollaries are displayed without an enclosing panel.

Table 1: Selected assessment questions and their locations.

Original source	**Page**	**Placement in the unified theory**
Project 8 – Main task	**??**	Concise: Boundary diagnosis and corrected no-flux Markov scheme
Project 8 – 1*	**??**	Pending: Printed ghost-cell stencil: positivity and mass failure
Project 9 – 1	11	Core: Affine ODE and variation of constants
Project 9 – 2	20	Core: Composition, inverse, and time-reversal of flows
Project 9 – 3	21	Core: Variational matrix and noncommuting-matrix correction
Project 9 – 4	21	Core: Liouville–Jacobi determinant theorem
Project 9 – 4 (numerics)	26	Concise: Euler Jacobian and scalar convergence validation
Project 9 – 5	38	Concise: Volume transport and Reynolds-type identity
Project 9 – 6*	39	Pending: Minimum-energy path and optimal velocity
Project 9 – 7	44	Core: Pushforward dynamics and continuity equation
Project 9 – 8	44	Core: Density change-of-variables formula

Source Index and Direct Links (continued)

Original source	Page	Placement in the unified theory
Project 9 – 9	49	Concise: Material derivative and transport theorem
Project 9 – 10	55	Core: Differential-entropy evolution
Project 9 – 11⋆	57	Concise: Score evolution; Hessian and long-time extensions deferred
Project 9 – 12	59	Concise: Relative-entropy evolution for two flows
Project 9 – 13	??	Concise: First variation of transported KL divergence
Project 9 – 14⋆	??	Pending: Transport from a Gaussian to a target law
Project 9 – 15⋆	??	Pending: Measure-valued kinetic optimal-control problem
Project 10 – 1	51	Concise: Hamilton equations from the mechanical Hamiltonian
Project 10 – 2	52	Concise: Energy conservation along trajectories
Project 10 – 3⋆	53	Pending: Time reversal and momentum reflection
Project 10 – 4	52	Concise: Symplecticity and preservation of phase volume
Project 11 – 1	58	Core: Nonnegativity of relative entropy
Project 11 – 2⋆	61	Pending: KL asymmetry and unequal-variance Gaussians
Project 11 – 3⋆	61	Pending: Truncated Gaussian: KL curve and quadratic constrained minimum
Project 11 – 4	61	Concise: Joint and separate convexity of relative entropy
Project 11 – 5	63	Core: Gibbs variational identity and exponential tilt
Project 11 – 6⋆	64	Core: Donsker–Varadhan dual representation
Project 11 – 7⋆	60	Core: Chain rule for relative entropy
Project 11 – 8⋆	57	Pending: Stein identity and its operator
Project 11 – 9⋆	66	Pending: f-divergence and NWJ form under attainment hypotheses
Project 11 – 10⋆	67	Pending: Exponential Fenchel dual and finite-state saddle proof
Project 11 – 11⋆	67	Pending: Shift optimization recovering Donsker–Varadhan
Project 11 – 12⋆	68	Pending: Logistic classification and density-ratio optimality
Project 11 – 13⋆	68	Pending: Reward-guided measure optimization and mean-field tilt
Project 12 – 1	??	Concise: Gaussian conditional distribution
Project 12 – 2⋆	??	Pending: Noisy conditioning and level-set approximation
Project 12 – 3	??	Core: Coarea formula and effective free energy
Project 12 – 4	??	Core: Conditional mean-force formula

Source Index and Direct Links (continued)

Original source	Page	Placement in the unified theory
Project 13 – 1*	??	Concise: Total covariance and Gaussian reconstruction
Project 13 – 2	??	Core: Itô equation for a collective variable
Project 13 – 3	??	Core: Markovian matching of time marginals
Project 14 – 1	??	Concise: Gaussian stochastic-interpolant statistics
Project 14 – 2	??	Core: Conditional interpolant velocity field
Project 14 – 3	??	Core: Conditional-score identity

Chapter 1

From Algebraic Structures to Probability Dynamics and Learned Flows

1.1 State Spaces, Observables, and Probability Evolution

1.1.1 States, measurable structure, and observables

Consider N classical particles in three spatial dimensions. A microscopic configuration, including positions and momenta, is represented by

$$x = (q_1, \dots, q_N, p_1, \dots, p_N) \in V = \mathbb{R}^{6N}.$$

Here N is the particle count and $q_i, p_i \in \mathbb{R}^3$ are particle-indexed vectors, not time-indexed states. The equations of motion determine the trajectory of a given configuration. Physical predictions, however, usually concern functions of that configuration. For example, a Hamiltonian $H(x)$ assigns an energy to a state, while

$$f_\varphi(x) = \sum_{i=1}^{N} \varphi(q_i)$$

measures a spatially weighted particle number for a prescribed probe φ. Such functions are *observables*. If the preparation of the system does not specify a unique initial configuration, a probability law instead describes how likely different configurations are. The three objects we must therefore distinguish are a *state* x, an *observable* f, and a *probability law* μ.

The distinction is more than terminology. A dynamical rule moves states; the same rule changes the values of observables and transports probability laws. We will construct these induced actions rather than postulate three unrelated evolution equations. This is useful both in statistical physics, where microscopic trajectories give rise to ensemble predictions, and in machine learning, where learned flows act on samples but are evaluated by statistics of their resulting distributions.

States and linear structure. A *state space* is the set of admissible physical states, equipped with the structures required by the question under study. Throughout the finite-dimensional foundation we use a single symbol $V = \mathbb{R}^d$: it is both the physical state space and the real vector space on which linear operations are defined. In the particle example, $d = 6N$. Other systems need not have a vector-space state space; a fixed-length spin, for instance, lies on the sphere $S^2 = \{x \in \mathbb{R}^3 : \|x\| = 1\}$. When a general measurable space or a nonlinear state space is needed later, we introduce it explicitly as E rather than use E as another name for V.

Definition 1.1 (Linear endomorphisms)

Let $V = \mathbb{R}^d$ be the real vector space of states, with its usual addition and scalar multiplication. A map $A : V \to V$ is a *linear endomorphism* if

$$A(ax + by) = aA(x) + bA(y) \quad (x, y \in V,\ a, b \in \mathbb{R}).$$

The notation

$$\mathrm{End}(V) = \mathrm{Hom}_{\mathbb{R}}(V, V)$$

denotes the space of all real-linear maps from V to itself. With

- pointwise addition $(A + B)(x) = A(x) + B(x)$;
- scalar multiplication $(cA)(x) = cA(x)$;
- multiplication given by composition $AB = A \circ B$,

this space is a **unital associative algebra** over $\mathbb{R}$. Its unit is the identity map $I_V(x) = x$. Choosing a basis identifies this algebra with $M_d(\mathbb{R})$, the algebra of real $d \times d$ matrices: addition and scalar multiplication correspond to the respective matrix operations, and composition corresponds to matrix multiplication.

Linear endomorphisms will reappear as linear ODE velocity fields and as derivatives of nonlinear flow maps. Their introduction does not constrain the dynamics to be linear. A general state transformation $F : V \to V$ need not belong to $\mathrm{End}(V)$. For example, $F(x) = x^3$ is a well-defined transformation of $\mathbb{R}$ but is not linear. To describe the effect of arbitrary state transformations, we next turn to functions on V.

Measurable structure and observable algebra. A physical observable assigns a numerical value to each state of a system. However, when the initial state is not known exactly, we must also be able to describe events involving possible states. For example, if $H : V \to \mathbb{R}$ denotes the energy of a state, the condition $H(x) \le h$ determines the subset

$$\{x \in V : H(x) \le h\}.$$

To assign probabilities to such events, we must first specify a collection of subsets of the state space on which probabilities can be consistently defined. This motivates the introduction of a measurable structure.

Definition 1.2 (σ-algebra and measurable space)

On the chosen state set V, a family of subsets $\mathcal{S} \subseteq 2^V$ is called a *σ-algebra* on V if it satisfies:

1. $V \in \mathcal{S}$;
2. If $A \in \mathcal{S}$, then $V \setminus A \in \mathcal{S}$;
3. If $(A_n)_{n\geq 1}$ is a sequence of sets in $\mathcal{S}$, then

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{S}.$$

The elements of $\mathcal{S}$ are called *measurable sets*. The pair $(V, \mathcal{S})$ is called a *measurable space*.

These conditions ensure that measurable sets remain measurable under complements and countable unions. They also imply that $\varnothing \in \mathcal{S}$ and that $\mathcal{S}$ is closed under countable intersections, since

$$\bigcap_{n=1}^{\infty} A_n = V \setminus \left(\bigcup_{n=1}^{\infty} (V \setminus A_n) \right).$$

In probability theory, measurable subsets of the state space represent events to which probabilities may be assigned. The measurable structure specifies which events are admissible; it does not itself assign probabilities.

For the continuous dynamics considered here, we also need a notion of neighborhoods: when do states approach one another, and what does it mean for an observable or a state transformation to be continuous? A *topology* supplies the neighborhood structure underlying these notions of convergence and continuity. This structure is used to describe continuous trajectories and flow maps; later, bounded continuous observables also serve as tests for weak convergence of probability laws.

On $V = \mathbb{R}^d$, we use the topology induced by the Euclidean distance. Generating a σ-algebra from the open sets then connects this topological structure to measurable events, so that probability laws can be studied together with continuous observables and dynamics.

Definition 1.3 (Euclidean open sets and Borel σ-algebra)

Let $V = \mathbb{R}^d$ with its Euclidean distance. For $x \in V$ and $r > 0$, the *open ball* centered

at x with radius r is

$$B_r(x) := \{y \in V : \|y - x\| < r\}.$$

The collection of *open subsets* of V is

$$\mathcal{O}(V) := \{O \subseteq V : \forall x \in O,\ \exists r > 0 :\ B_r(x) \subseteq O\}\,.$$

This collection is the *Euclidean topology* on V. The *Borel σ-algebra* on V is

$$\mathcal{B}(V) := \sigma(\mathcal{O}(V)),$$

where $\sigma(\mathcal{O}(V))$ denotes the smallest σ-algebra containing every open subset of V. The pair $(V, \mathcal{B}(V))$ is called the *Borel measurable space*.

The Borel σ-algebra contains all open and closed sets, as well as the sets obtained from them through countable unions, countable intersections, and complements. It provides a natural measurable structure compatible with the topology already introduced on the state space.

Having specified which subsets of V are measurable, we must next determine which functions preserve this structure.

Definition 1.4 (Measurable maps and observables)

Let $(V, \mathcal{S})$ and $(Y, \mathcal{G})$ be measurable spaces. A map $F : V \to Y$ is called *measurable* if $\forall B \in \mathcal{G}$, $F^{-1}(B) \in \mathcal{S}$, where

$$F^{-1}(B) := \{x \in V : F(x) \in B\}$$

denotes the inverse image of B under F. A real-valued observable on V is a Borel measurable function

$$f : V \to \mathbb{R},$$

where both V and $\mathbb{R}$ are equipped with their Borel σ-algebras.

Measurability ensures that conditions imposed on observable values determine measurable events in the state space. For example, $\forall a \in \mathbb{R}$,

$$\{x \in V : f(x) \leq a\} = f^{-1}((-\infty, a]) \in \mathcal{B}(V).$$

It also guarantees compatibility with state transformations. If $F : V \to V$ and $f : V \to \mathbb{R}$ are Borel measurable, then their composition is Borel measurable, since

$$(f \circ F)^{-1}(B) = F^{-1}(f^{-1}(B)) \in \mathcal{B}(V)$$

for every $B \in \mathcal{B}(\mathbb{R})$. Thus, composing an observable with a measurable state transformation produces another measurable observable. In particular, every continuous state transforma-

tion is Borel measurable.

We now introduce a convenient algebraic space of observables. Although physical observables need not be bounded, bounded observables provide a natural class that is closed under the basic pointwise operations.

Definition 1.5 (Algebra of observables)

Let $V = \mathbb{R}^d$ with its Borel σ-algebra. Define

$$\begin{aligned}\mathcal{A} &:= B_b(V) \\ &= \left\{ f : V \to \mathbb{R} : \big(\forall B \in \mathcal{B}(\mathbb{R}), f^{-1}(B) \in \mathcal{B}(V)\big) \wedge \big(\sup_{x\in V} |f(x)| < \infty\big) \right\}.\end{aligned}$$

For $f, g \in \mathcal{A}$ and $a \in \mathbb{R}$, define the pointwise operations

$$\begin{aligned}(f+g)(x) &= f(x) + g(x), \\ (af)(x) &= af(x), \\ (fg)(x) &= f(x)g(x).\end{aligned}$$

These operations preserve boundedness and Borel measurability. Consequently, $\mathcal{A}$ is a *commutative unital algebra* over $\mathbb{R}$, with multiplicative identity

$$\mathbf{1}(x) = 1, \quad \forall x \in V.$$

The multiplication in $\mathcal{A}$ is pointwise multiplication of functions, whereas multiplication in $\mathrm{End}(V)$ is composition of linear maps. These are distinct algebraic operations on different mathematical objects.

The boundedness restriction in $\mathcal{A}$ is a mathematical convenience, not a physical requirement. Important observables, such as a Hamiltonian $H(x)$ or $\|x\|^2$, may be unbounded and therefore need not belong to $\mathcal{A}$. Such observables remain measurable functions and can be used in statistical expressions whenever the required integrability conditions are satisfied.

The space $\mathcal{A}$ also admits the supremum norm

$$\|f\|_\infty := \sup_{x\in V} |f(x)|.$$

The completeness of this normed algebra will be used when discussing operator domains in the next section.

1.1.2 Probability laws and integration

A microscopic system is characterized by a state belonging to a physical state space V. However, the actual microscopic state need not be completely determined by the information available to us. In some situations, the system occupies a definite but unknown state, and a probabilistic description represents our uncertainty about its value. In other situations, a

preparation protocol may produce different microscopic states across repeated realizations, even when the same macroscopic conditions are imposed. The resulting variability can also be described probabilistically. These two situations have different physical interpretations: the former concerns incomplete knowledge of a state, whereas the latter concerns the statistical variability of a preparation procedure. Nevertheless, both can be represented mathematically by probability measures on the physical state space.

To formulate this description precisely, we distinguish the sample space Ω, whose elements represent possible outcomes of an experiment or an uncertainty model, from the physical state space V, whose elements represent the possible microscopic configurations of the system. A random state associates each outcome in the sample space with a corresponding physical state. By assigning probabilities to events in the sample space, we can consequently determine the probabilities of measurable events in V. Thus, the Borel measurable structure $(V, \mathcal{B}(V))$ introduced earlier provides the natural setting for describing statistical properties of microscopic states, while the sample space provides a general representation of the underlying uncertainty.

Definition 1.6 (Probability space, random state, and law)

A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$, where $\mathcal{F}$ is a σ-algebra on Ω and $\mathbb{P}$ is a countably additive map $\mathbb{P} : \mathcal{F} \to [0, 1]$ with $\mathbb{P}(\Omega) = 1$. A V-valued random state is a measurable map

$$X : (\Omega, \mathcal{F}) \longrightarrow (V, \mathcal{B}(V)).$$

Its law $\mu = \mathrm{Law}(X)$ is the Borel probability measure on V defined by $\mu(B) = \mathbb{P}(X \in B)$. Write $\mathcal{P}(V)$ for the set of all Borel probability measures on V.

For every $f \in \mathcal{A}$, the expectation of $f(X)$ depends only on its law:

$$\mathbb{E}[f(X)] = \int_V f(x)\, \mu(\mathrm{d}x) =: \langle f, \mu \rangle.$$

The bracket is an *integration pairing*. It is linear in the observable and extends to a bilinear pairing if probabilities are replaced by finite signed measures. One should not identify $\mathcal{P}(V)$ itself with a vector space: it is a convex subset of the space of signed measures. Similarly, a probability law is not necessarily a density. Only when μ is absolutely continuous with respect to Lebesgue measure do we write $\mu(\mathrm{d}x) = \rho(x)\ \mathrm{d}x$.

1.1.3 Transformations, duality, and evolution

A Borel measurable state transformation $F : V \to V$ induces two actions. An observable f is evaluated after the transformation as $f \circ F$; a law μ assigns the transformed event B the probability $\mu(F^{-1}(B))$. The integration pairing relates these actions. Neither linearity nor invertibility of F is needed.

Definition 1.7 (Composition operator and pushforward measure)

For a Borel measurable $F : V \to V$, its composition (Koopman) operator is the linear map $\mathcal{K}_F : \mathcal{A} \to \mathcal{A}$, defined by

$$(\mathcal{K}_F f)(x) := f(F(x)), \quad f \in \mathcal{A},$$

and its pushforward map $\mathsf{T}_F : \mathcal{P}(V) \to \mathcal{P}(V)$ by

$$\mathsf{T}_F \mu := F_{\#}\mu, \quad (F_{\#}\mu)(B) := \mu(F^{-1}(B)), \quad B \in \mathcal{B}(V).$$

For a measurable map $\xi : V \to \mathcal{Z}$ into another measurable state space $\mathcal{Z}$, the same definition gives $\xi_{\#}\mu \in \mathcal{P}(\mathcal{Z})$; here $\mathcal{P}(\mathcal{Z})$ uses its specified sigma-algebra.

Theorem 1.8 (Observable–measure duality)**.** *On $(V, \mathcal{B}(V))$, let $F : V \to V$ be Borel measurable. Then $\mathcal{K}_F$ is a positive, unital algebra homomorphism on $B_b(V)$ and $\|\mathcal{K}_F f\|_\infty \le \|f\|_\infty$. For every $f \in B_b(V)$ and $\mu \in \mathcal{P}(V)$,*

$$\boxed{\langle \mathcal{K}_F f, \mu \rangle = \langle f, \mathsf{T}_F \mu \rangle.} \tag{1.1}$$

Moreover, for measurable $F, G : V \to V$,

$$\mathcal{K}_F \mathcal{K}_G = \mathcal{K}_{G \circ F}, \quad \mathsf{T}_G \mathsf{T}_F = \mathsf{T}_{G \circ F}. \tag{1.2}$$

Proof. Linearity and preservation of products follow by evaluating at x:

$$\mathcal{K}_F(af + bg)(x) = af(F(x)) + bg(F(x)),$$
$$\mathcal{K}_F(fg)(x) = f(F(x))g(F(x)).$$

Positivity, preservation of the constant function $\mathbf{1}$, and the supremum-norm bound follow from the same definition. For an indicator $f = \mathbf{1}_B$, the defining identity of the pushforward reads

$$\int_V \mathbf{1}_B(y)\,(F_{\#}\mu)(\,\mathrm{d}y) = \mu(F^{-1}(B)) = \int_V \mathbf{1}_B(F(x))\,\mu(\,\mathrm{d}x).$$

Linearity proves this for simple functions. Monotone approximation and the monotone convergence theorem extend it to nonnegative measurable functions, and taking positive and negative parts proves (1.1) for bounded real observables. Finally, $\mathcal{K}_F \mathcal{K}_G f = f \circ G \circ F$ and $(G_{\#}F_{\#}\mu)(B) = \mu(F^{-1}(G^{-1}(B)))$, proving (1.2). □

Thus a nonlinear state map still induces a linear operator on observables, dual to its action on laws. No probability density is required.

1.1.3.1 Families of state transformations

An evolution is a compatible family of state transformations, not merely one map. Suppose maps $\Phi_{s,t} : V \to V$ are defined for $s \le t$ and satisfy

$$\Phi_{s,s} = \mathrm{id}_V, \quad \Phi_{r,t} \circ \Phi_{s,r} = \Phi_{s,t} \quad (s \le r \le t).$$

Define $\mathcal{K}_{s,t} := \mathcal{K}_{\Phi_{s,t}} : \mathcal{A} \to \mathcal{A}$ and $\mathsf{T}_{s,t} := \mathsf{T}_{\Phi_{s,t}} : \mathcal{P}(V) \to \mathcal{P}(V)$. The preceding theorem immediately yields

$$\begin{gathered} \mathcal{K}_{s,r}\mathcal{K}_{r,t} = \mathcal{K}_{s,t}, \quad \mathsf{T}_{r,t}\mathsf{T}_{s,r} = \mathsf{T}_{s,t}, \\ \langle \mathcal{K}_{s,t} f, \mu_s \rangle = \langle f, \mathsf{T}_{s,t}\mu_s \rangle. \end{gathered} \tag{1.3}$$

The order of the composition operators is reversed because an observable at time t is pulled back to the initial state, whereas a probability law is pushed forward in chronological order. The next section will *construct* such $\Phi_{s,t}$ by solving an ODE, rather than assume their existence. In particular, the existence interval and invertibility of a flow require hypotheses on the underlying vector field.

A deterministic flow does not require a probability space: $\Phi_{s,t}(x)$ is defined for a specified initial state. If one subsequently introduces a random initial condition X_s with law μ_s, the random state $X_t = \Phi_{s,t}(X_s)$ has law $\mathsf{T}_{s,t}\mu_s$. Thus probability enters to describe an *ensemble of initial conditions*, not to define the underlying deterministic equation.

The remainder follows the construction from trajectories to transported laws (Project 9), its Hamiltonian application (Project 10), entropy and duality (Project 11), conditional reduction (Projects 12–13), and learned or numerical evolution (Projects 14 and 8). The Source Index records the individual questions; each result states its own assumptions.

1.2 Dynamical Flows and Probability Transport

This section develops a common framework for describing **state-space dynamics** and **the evolution of probability laws**. Taking flows generated by ordinary differential equations (ODEs) as the starting point, we study how the motion of individual states induces the transport of observables, measures, and densities. We then illustrate the geometric structure through Hamiltonian dynamics and turn to entropy identities and variational principles. The discussion is organized around the following topics and questions:

- *State flows and volume deformation.* Construct the flow $\Phi_{s,t}$ from an ODE with velocity field v_t and study its dependence on initial conditions.
  - How does the flow derivative $M_{s,t}(x)$ describe local deformation?
  - How does its determinant $J_{s,t}(x)$ determine changes in volume and transported densities?
- *Evolution of observables and probability laws.* Study the observable evolution operator $\mathcal{K}_{s,t}$, the law transport operator $\mathsf{T}_{s,t}$, and the generator $\mathcal{L}_t$.

  - How does the generator $\mathcal{L}_t$ relate to the velocity field v_t and the flow $\Phi_{s,t}$?
  - How does the continuity equation $\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0$ describe the evolution of densities?

- *Hamiltonian dynamics.* Illustrate the framework with the Hamiltonian $H(q, p)$ and its flow ϕ_t.
  - How does Hamiltonian structure lead to energy conservation and symplecticity?
  - What does volume preservation imply for probability transport?

- *Entropy and variational principles.* Develop transport identities for differential entropy $S(\rho)$ and relative entropy $D_{\mathrm{KL}}(\mu\|\nu)$, then connect convex duality and the log-partition functional $\Lambda_Q(g)$ to variational principles and free energy $F(\rho)$.
  - How do entropy functionals evolve under transport?
  - How do the variational representations of relative entropy and the free-energy gap characterize equilibrium and deviations from it?

1.2.1 State flows and geometric properties

Consider a moving fluid whose motion is described by a time-dependent velocity field $v_t(x)$. At each time t and position x, the vector $v_t(x)$ specifies the instantaneous velocity of the fluid at that location. A fluid element moving with the surrounding fluid follows the local velocity field. If x_t denotes its position at time t, its motion is therefore governed by

$$\dot{x}_t = v_t(x_t).$$

Assume that v is sufficiently smooth in time and space. Given an initial time s and an initial position x, we seek a trajectory x_t that describes the subsequent motion of this fluid element.

However, understanding fluid motion involves more than determining individual trajectories. We are also interested in how the motion depends on the initial position. In particular, two fluid elements starting from nearby positions may move apart or approach each other. Their relative motion determines the local deformation of the fluid, including stretching, rotation, and expansion or contraction. These effects are closely related to the evolution of infinitesimal volume elements and, through mass conservation, the redistribution of fluid density.

This motivates a systematic study of the initial-value problem: how does a given initial state determine a unique trajectory, and how does that trajectory depend on its initial condition? We begin by formulating this problem precisely.

1.2.1.1 ODE initial-value problems

Let $V = \mathbb{R}^d$, where $d \geq 1$ is an integer, be the Euclidean state space, equipped with its Euclidean norm. The time-dependent velocity field specifies the fluid velocity $v(t, x)$ at

each time t and position x. We write it as the map

$$v : \mathbb{R} \times \mathbb{R}^d \to \mathbb{R}^d, \quad (t, x) \mapsto v_t(x),$$

and assume that it is smooth and satisfies $v_t \in C_0(\mathbb{R}^d; \mathbb{R}^d)$, for all $t \in \mathbb{R}$. Given initial data $(s, x) \in \mathbb{R} \times \mathbb{R}^d$, the corresponding initial-value problem is to find a trajectory $x_t \in C^1(I, V)$ satisfying

$$\begin{cases} \dot{x}_t = v_t(x_t), \quad t \in [s, +\infty), \\ x_s = x. \end{cases} \tag{1.4}$$

We next state an existence and uniqueness theorem for the initial-value problem.

Theorem 1.9 (Picard–Lindelöf Theorem)**.** *Consider the initial-value problem*

$$\begin{cases} \dot{x}(t) = f(t, x(t)), \\ x(t_0) = x_0, \end{cases}$$

where $f : \mathbb{R} \times \mathbb{R}^d \to \mathbb{R}^d$. Suppose that:

1. *f is continuous in (t, x).*
2. *f is locally Lipschitz continuous with respect to x, locally uniformly in t. That is, for every compact interval $J \subset \mathbb{R}$ and every $R > 0$, there exists $L_{J,R} < \infty$ such that*

$$\|f(t, y) - f(t, z)\| \le L_{J,R}\|y - z\|, \quad t \in J, \quad y, z \in \overline{B}_R(0).$$

Then, for every initial condition $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^d$, there exists an open interval containing t_0 on which the initial-value problem admits a unique C^1 solution. Moreover, any two solutions with the same initial data agree on the intersection of their intervals of definition.

The proof is given in Appendix A.1. The theorem provides the existence and uniqueness of the trajectory, which allows us to define the corresponding flow map in the following discussion.

1.2.1.2 A linear inhomogeneous initial-value problem

Example 1.10 (Solution of a time-dependent linear inhomogeneous ODE)**.**

Source exercise: Project 9, item 1. *Student Assessment Exercise* [Assessment] (adapted).

Problem

Let $A : I \to \mathrm{End}(V)$ and $B : I \to V$ be continuous functions. Consider the time-dependent affine vector field

$$v_t(y) = A_t y + B_t, \quad y \in V.$$

For a prescribed initial time $s \in I$ and initial state $x \in V$, consider the initial-value problem

$$\begin{cases} \dot{x}_t = A_t x_t + B_t, \\ x_s = x. \end{cases}$$

Prove that this initial-value problem admits a unique solution x_t on I, and derive an explicit representation of the solution in terms of the given functions A_t, B_t, and the initial state x.[a]

[a] In particular, construct the evolution operator associated with the homogeneous system and use it to obtain the solution of the inhomogeneous equation.

Solution.

We now establish the conditions under which this ODE admits a globally defined solution on I and prove its uniqueness.

1. **Existence.**

   On every compact subinterval $J \subset I$ containing s, continuity bounds A and B. Hence, the affine vector field $v_t(x) = A_t x + B_t$ is globally Lipschitz in the state variable and has at most linear growth, uniformly on J. By Theorem 1.9, a unique solution exists on each such interval.

2. **Global extension and uniqueness.**

   Since any two such intervals containing s overlap, uniqueness guarantees that their solutions agree on the overlap. Consequently, these solutions can be consistently joined into a single trajectory defined on all of I. The same uniqueness argument ensures that this trajectory is unique.

Since the unknown x_t describes the evolution of a state vector over time, the object to be determined is a trajectory rather than a single element of V. We therefore introduce the trajectory space

$$\mathcal{X} = C^1(I, V), \quad \mathcal{Y} = C(I, V).$$

Here, $\mathcal{X}$ consists of continuously differentiable V-valued trajectories, while $\mathcal{Y}$ consists of continuous V-valued functions. Both inherit their vector-space structures from V, with addition and scalar multiplication defined pointwise.

To express the differential equation in this functional framework, we introduce the

- differentiation operator:

$$\partial_t : \mathcal{X} \to \mathcal{Y}, \quad (\partial_t \gamma)(t) = \dot{\gamma}(t),$$

- operator induced by the time-dependent family A_t:

$$\mathsf{M}_A : \mathcal{X} \to \mathcal{Y}, \quad (\mathsf{M}_A \gamma)(t) = A_t \gamma(t).$$

Both are linear maps, and the given continuous forcing term satisfies $B \in \mathcal{Y}$. Defining $\mathsf{D}_A := \partial_t - \mathsf{M}_A$, the differential equation can be written as an equation for the entire trajectory:

$$\begin{cases} \mathsf{D}_A \gamma = B, \\ \gamma(s) = x. \end{cases}$$

Here, $\mathsf{D}_A \gamma = B$ means that

$$\dot{\gamma}(t) - A_t \gamma(t) = B_t$$

for every $t \in I$. Thus, the operator equation is simply the trajectory-space formulation of the original ODE, supplemented by its initial-value constraint.

1. The homogeneous equation. To solve the inhomogeneous equation, we first consider the associated homogeneous equation[1]:

$$\begin{cases} \mathsf{D}_A \eta = 0, \\ \eta(s) = v. \end{cases}$$

Allowing the initial vector v to vary, all solutions of the homogeneous differential equation form the kernel of D_A:

$$\mathcal{H} := \ker \mathsf{D}_A = \{\eta \in \mathcal{X} : \mathsf{D}_A \eta = 0\}.$$

Since D_A is linear, $\mathcal{H}$ is a linear subspace of the trajectory space $\mathcal{X}$. To understand the structure of $\mathcal{H}$, we associate each homogeneous trajectory with its state at a fixed time. By uniqueness of the initial-value problem, this state uniquely determines the entire trajectory. Thus, we can identify the trajectory space $\mathcal{H}$ with the finite-dimensional state space V, without directly constructing a basis for $\mathcal{H}$.

1. **Evaluation at an arbitrary time t.** For each $t \in I$, define

$$E_t : \mathcal{H} \to V, \quad E_t(\eta) = \eta(t).$$

By pointwise linearity, E_t is linear. Moreover, existence and uniqueness of the homogeneous initial-value problem at time t ensure that E_t is bijective. Hence, E_t is a

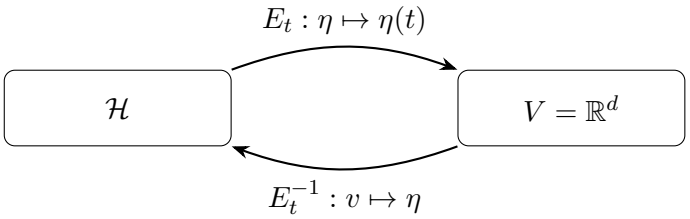
[1] Equivalently, $\dot{\eta}(t) - A_t \eta(t) = 0,\ \eta(s) = v$, where $v \in V$ is arbitrary.

linear isomorphism for every $t \in I$.

2. **The inverse solution map and the dimension of the solution space.** For each $t \in I$, define the solution map as the inverse of evaluation:

$$S_t := E_t^{-1} : V \to \mathcal{H}.$$

The output $S_t(v)$ is the unique complete homogeneous trajectory whose value at time t is v.



The inverse identities are $E_t \circ S_t = I_V$ and $S_t \circ E_t = I_{\mathcal{H}}$. Since E_t is a linear isomorphism, its inverse S_t is linear as well:

$$S_t(av + bw) = aS_t(v) + bS_t(w), \quad v, w \in V.$$

This also expresses the superposition principle for homogeneous solutions. In particular,

$$\mathcal{H} \cong V, \quad \dim \mathcal{H} = d.$$

Although the ambient trajectory space $\mathcal{X}$ is generally infinite-dimensional, its homogeneous solution subspace has exactly d degrees of freedom, parameterized by the initial vector.

For a fixed initial time $s \in I$, first use S_s to associate the initial state with its unique homogeneous trajectory, and then use E_t to evaluate that trajectory at time t:

$$V \xrightarrow{S_s} \mathcal{H} \xrightarrow{E_t} V.$$

This composition defines the homogeneous evolution operator

$$U(t, s) := E_t \circ S_s \in \mathrm{End}(V).$$

Thus every homogeneous trajectory $\eta \in \mathcal{H}$ satisfies

$$\begin{cases} \eta(t) = U(t, s)\eta(s), \\ \eta(s) = v. \end{cases}$$

In particular, $U(s, s) = I_V$.

For fixed s and any $v \in V$, the trajectory $\eta(t) : t \mapsto U(t, s)v$ is precisely the homogeneous solution whose value at time s is v. It therefore satisfies

$$\partial_1 U(t, s) = A_t U(t, s), \quad U(s, s) = I_V. \tag{1.5}$$

Here ∂_1 denotes differentiation with respect to the first time argument. The equation is an identity in $\mathrm{End}(V)$: it follows because the corresponding vector identity holds for every $v \in V$. Thus $U(t,s)$ describes the finite-time evolution generated by the instantaneous linear dynamics A_t.

We now establish the basic algebraic properties of the evolution operator $U(t,s)$.

- **Composition law.**

  For any $v \in V$, the trajectories $\tau \mapsto U(\tau,r)U(r,s)v$ and $\tau \mapsto U(\tau,s)v$ solve the same homogeneous equation and have the same value at $\tau = r$. By uniqueness,

$$U(t,r)U(r,s) = U(t,s), \quad r,s,t \in I.$$

- **Invertibility.**

  Taking the intermediate time to be either endpoint in the composition law gives

$$U(t,s)U(s,t) = U(t,t) = I_V,$$
$$U(s,t)U(t,s) = U(s,s) = I_V.$$

  Hence,

$$U(t,s)^{-1} = U(s,t), \quad U(t,s) \in GL(V).$$

Thus, the homogeneous evolution is an invertible linear transformation. Further details on the construction and algebraic properties of U are given in Appendices A.3 and A.4. Its time-ordered representation for noncommuting A_t is developed in Appendix A.5.

2. The inhomogeneous equation. Now consider the inhomogeneous equation

$$\mathsf{D}_A\gamma = B.$$

We already know that the homogeneous solution space is $\mathcal{H} = \ker \mathsf{D}_A$. Suppose that we have found one particular solution γ_{p} satisfying $\mathsf{D}_A\gamma_{\mathrm{p}} = B$. If γ is any other solution, then

$$\mathsf{D}_A(\gamma - \gamma_{\mathrm{p}}) = \mathsf{D}_A\gamma - \mathsf{D}_A\gamma_{\mathrm{p}} = B - B = 0,$$

so $\gamma - \gamma_{\mathrm{p}} \in \mathcal{H}$. Conversely, for every $\eta \in \mathcal{H}$, linearity gives

$$\mathsf{D}_A(\gamma_{\mathrm{p}} + \eta) = \mathsf{D}_A\gamma_{\mathrm{p}} + \mathsf{D}_A\eta = B.$$

Therefore all solutions of the inhomogeneous equation form

$$\mathcal{S}_B = \gamma_{\mathrm{p}} + \mathcal{H}.$$

This is an affine subspace of $\mathcal{X}$, obtained by translating the homogeneous solution space $\mathcal{H}$ by γ_{p}. Having determined this algebraic structure, it remains to construct one particular solution.

Constructing the particular solution.

To simplify the construction, choose the particular solution with zero initial value:

$$\begin{cases} \dot{\gamma}_{\mathrm{p}}(t) = A_t \gamma_{\mathrm{p}}(t) + B_t, \\ \gamma_{\mathrm{p}}(s) = 0. \end{cases}$$

Its existence and uniqueness follow from Theorem 1.9. The linear term $A_t\gamma_{\mathrm{p}}(t)$ is governed by the same operator A_t as in the homogeneous system, whose evolution is described by $U(t,s)$; B_t is the additional forcing.

To construct this particular solution, we use the homogeneous evolution operator established above. For each fixed pair of times, $U(t,s)$ is a linear map from the entire state space V to itself. Its inverse $U(s,t)$ is therefore also defined on all of V. Since $\gamma_{\mathrm{p}}(t) \in V$, we may apply $U(s,t)$ to $\gamma_{\mathrm{p}}(t)$, even though γ_{p} itself solves the inhomogeneous equation.

We introduce the auxiliary trajectory $c(t)$ by

$$c(t) = U(s,t)\gamma_{\mathrm{p}}(t). \tag{1.6}$$

The auxiliary trajectory $c \in \mathcal{X}$ represents the evolution of the particular solution in a reference frame that removes the homogeneous dynamics generated by A_t. Its time dependence is therefore governed solely by the external driving B_t, transformed by the inverse homogeneous evolution operator $U(s,t)$. Equivalently, by the inverse identity,

$$\gamma_{\mathrm{p}}(t) = U(t,s)c(t).$$

This is a time-dependent change of representation relative to the homogeneous evolution; it does not assume that γ_{p} itself follows the homogeneous dynamics.

To determine the evolution of $c(t)$, we first note that for fixed s, the map $t \mapsto U(t,s)$ is continuously differentiable and takes values in $GL(V)$. Since inversion is a smooth operation on $GL(V)$, its inverse $t \mapsto U(s,t)$ is also continuously differentiable. We can therefore differentiate the inverse identity $U(s,t)U(t,s) = I_V$ with respect to t. Using (1.5), we obtain

$$0 = \frac{\partial}{\partial t}\big(U(s,t)U(t,s)\big) = \partial_2 U(s,t)\, U(t,s) + U(s,t)\partial_1 U(t,s).$$

Right-multiplying by $U(s,t)$ and using $U(t,s)U(s,t) = I_V$ gives

$$\partial_2 U(s,t) = -U(s,t)A_t.$$

The product rule applied to (1.6), together with $\dot{\gamma}_{\mathrm{p}}(t) = A_t\gamma_{\mathrm{p}}(t) + B_t$, now yields

$$\begin{aligned} \dot{c}(t) &= -U(s,t)A_t\gamma_{\mathrm{p}}(t) + U(s,t)\dot{\gamma}_{\mathrm{p}}(t) \\ &= -U(s,t)A_t\gamma_{\mathrm{p}}(t) + U(s,t)\big(A_t\gamma_{\mathrm{p}}(t) + B_t\big) \\ &= U(s,t)B_t. \end{aligned}$$

The homogeneous terms cancel exactly, leaving only the forcing term in the transformed

coordinates. Since $\gamma_{\mathrm{p}}(s) = 0$ and $U(s,s) = I_V$, we have $c(s) = 0$. Integrating from s to t gives

$$c(t) = \int_s^t U(s,\tau)B_\tau \, \mathrm{d}\tau.$$

Using $U(t,s)U(s,\tau) = U(t,\tau)$ to return to the original coordinates gives

$$\begin{aligned}\gamma_{\mathrm{p}}(t) &= U(t,s)\int_s^t U(s,\tau)B_\tau \, \mathrm{d}\tau \\ &= \int_s^t U(t,s)U(s,\tau)B_\tau \, \mathrm{d}\tau \\ &= \int_s^t U(t,\tau)B_\tau \, \mathrm{d}\tau.\end{aligned}$$

Hence the zero-initial-value particular solution is

$$\gamma_{\mathrm{p}}(t) = \int_s^t U(t,\tau)B_\tau \, \mathrm{d}\tau.$$

The homogeneous trajectory with initial state x is $\eta_x(t) := U(t,s)x$. Adding it to the particular solution gives

$$x_t = \eta_x(t) + \gamma_{\mathrm{p}}(t).$$

Therefore,

$$\boxed{x_t = U(t,s)x + \int_s^t U(t,\tau)B_\tau \, \mathrm{d}\tau.} \tag{1.7}$$

This is the variation-of-constants formula.

The solution has a natural algebraic interpretation. For fixed $s, t \in I$, its dependence on the initial state is given by

$$x \longmapsto U(t,s)x + \gamma_{\mathrm{p}}(t).$$

The chosen γ_{p} depends on the reference time s through $\gamma_{\mathrm{p}}(s) = 0$, but is independent of x. Since $U(t,s)$ is invertible, this is an invertible affine transformation of the state space.

The homogeneous equation generates linear transformations of V, while the forcing introduces a time-dependent translation. Correspondingly, the homogeneous solutions form a vector space, whereas the solutions of the inhomogeneous equation with fixed forcing form an affine space modeled on that homogeneous solution space. We can make this geometric structure more explicit within the trajectory space. Extend evaluation from $\mathcal{H}$ to the ambient trajectory space by defining

$$\widetilde{E}_s : \mathcal{X} \to V, \quad \widetilde{E}_s(\gamma) = \gamma(s),$$

whose restriction to $\mathcal{H}$ is E_s. Define its kernel by

$$\mathcal{K}_s := \ker \widetilde{E}_s = \{\gamma \in \mathcal{X} : \gamma(s) = 0\}.$$

Thus, $\mathcal{K}_s$ consists of all trajectories that vanish at the initial time. For any $\gamma \in \mathcal{X}$, the trajectory

$$\eta(t) = U(t,s)\gamma(s).$$

is homogeneous and has the same initial value as γ. Therefore, $\gamma - \eta \in \mathcal{K}_s$. Moreover, a homogeneous trajectory that vanishes at s must vanish identically. Hence

$$\mathcal{H} \cap \mathcal{K}_s = \{0\}.$$

We consequently obtain the direct-sum decomposition $\mathcal{X} = \mathcal{H} \oplus \mathcal{K}_s$.

Recall the chosen particular solution γ_{p}, with $\gamma_{\mathrm{p}}(s) = 0$, and the homogeneous trajectory $\eta_x(t) = U(t,s)x$ associated with the prescribed initial state x.The inhomogeneous solution set is

$$\mathcal{S}_B = \gamma_{\mathrm{p}} + \mathcal{H},$$

while the set of all trajectories satisfying the initial-value constraint is

$$\mathcal{X}_{s,x} := \{\gamma \in \mathcal{X} : \gamma(s) = x\} = \eta_x + \mathcal{K}_s.$$

The trajectory $\gamma_{\mathrm{p}} + \eta_x$ belongs to both sets: it solves the inhomogeneous equation and has initial value x. To see that it is the only element of the intersection, suppose

$$\gamma = \gamma_{\mathrm{p}} + \eta = \eta_x + k,$$

where $\eta \in \mathcal{H}$ and $k \in \mathcal{K}_s$. Then

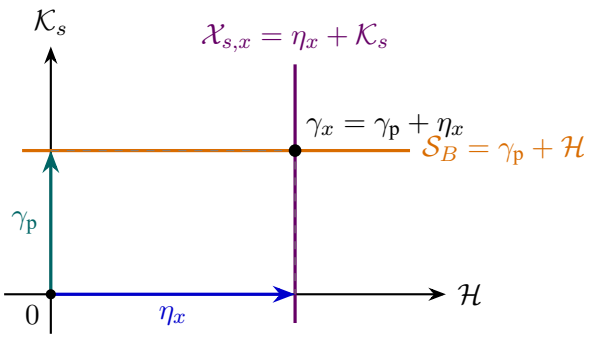
$$\eta - \eta_x = k - \gamma_{\mathrm{p}}.$$

The left-hand side belongs to $\mathcal{H}$, while the right-hand side belongs to $\mathcal{K}_s$, because $\gamma_{\mathrm{p}}(s) = 0$. Since $\mathcal{H} \cap \mathcal{K}_s = \{0\}$, both sides must vanish. Thus $\eta = \eta_x$ and $\gamma = \gamma_{\mathrm{p}} + \eta_x$. Therefore the intersection is exactly

$$\mathcal{S}_B \cap \mathcal{X}_{s,x} = \{\gamma_{\mathrm{p}} + \eta_x\}.$$

Denote this selected trajectory by $\gamma_x = \gamma_{\mathrm{p}} + \eta_x$, so that $\gamma_x(t) = x_t$.

The following diagram illustrates this geometry in a two-dimensional schematic slice of the trajectory space. The horizontal and vertical directions represent the complementary subspaces $\mathcal{H}$ and $\mathcal{K}_s$, respectively. They are not assumed to be orthogonal.



The homogeneous solution space determines the directions of the inhomogeneous solution set, while the particular solution determines its affine displacement. The initial-value constraint selects exactly one trajectory from this affine family. ♦

Example 1.11 (A concrete special case: Newton's law of cooling)**.** We now illustrate the preceding construction with a simple physical example. Consider a body whose temperature T_t relaxes toward a constant ambient temperature T_{env}. Under the lumped-temperature approximation and a constant heat-transfer coefficient, Newton's law of cooling takes the form

$$\dot{T}_t = -\lambda\big(T_t - T_{\text{env}}\big), \quad T_0 = T_0,$$

where $\lambda > 0$ is the relaxation rate and $t \geq 0$.

Rearranging the equation,

$$\dot{T}_t = -\lambda T_t + \lambda T_{\text{env}}.$$

This is a special case of the original problem with

$$V = \mathbb{R}, \quad A_t = -\lambda, \quad B_t = \lambda T_{\text{env}},$$

where we identify $\text{End}(\mathbb{R})$ with $\mathbb{R}$ through scalar multiplication.

Since A_t is constant, the evolution operator is simply

$$U(t, s) = e^{-\lambda(t-s)}.$$

The homogeneous contribution associated with theinitial temperature is

$$\eta_{T_0}(t) = U(t, 0)T_0 = T_0 e^{-\lambda t}.$$

For this illustration the reference time is $s = 0$. The zero-initial-value particular solution is

$$\begin{aligned}\gamma_{\text{p}}(t) &= \int_0^t U(t, \tau)B \, \mathrm{d}\tau \\ &= \lambda T_{\text{env}} \int_0^t e^{-\lambda(t-\tau)} \, \mathrm{d}\tau \\ &= T_{\text{env}} \left(1 - e^{-\lambda t}\right).\end{aligned}$$

Adding the homogeneous and particular contributions, we obtain

$$T_t = T_0 e^{-\lambda t} + T_{\text{env}} \left(1 - e^{-\lambda t}\right).$$

Equivalently,

$$T_t = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-\lambda t}.$$

For a concrete illustration, choose

$$T_0 = 80\,^\circ\text{C}, \quad T_{\text{env}} = 20\,^\circ\text{C}, \quad \lambda = 0.5\,\text{min}^{-1}.$$

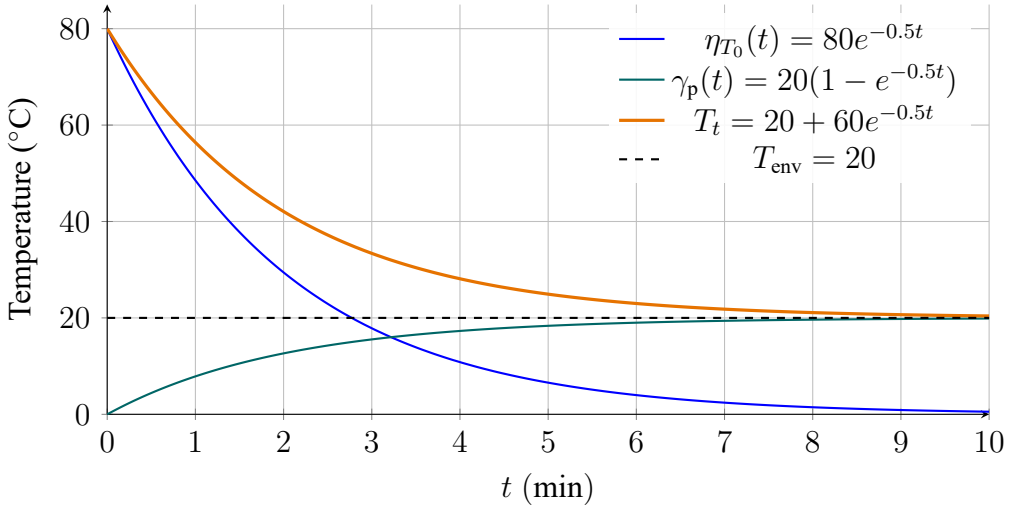
Then

$$\eta_{T_0}(t) = 80e^{-0.5t}, \quad \gamma_{\mathrm{p}}(t) = 20(1 - e^{-0.5t}),$$

and the complete trajectory is

$$T_t = 20 + 60e^{-0.5t}.$$

The following plot displays the homogeneous contribution, the zero-initial-value particular solution, and their sum. The horizontal dashed line indicates the ambient equilibrium temperature.



The homogeneous contribution describes the decay of the initial-state contribution, whereas the particular solution describes the accumulated effect of the ambient-temperature forcing. Their sum gives the actual temperature trajectory, which approaches T_{env} as $t \to +\infty$.

This example demonstrates that the affine-shift construction is directly applicable to physical relaxation processes. The decomposition is not merely a formal manipulation: it separates the contribution of the initial state from the accumulated response to external driving.

1.2.1.3 Two-time flows and variational matrices

In the homogeneous linear case, the evolution operator induces the flow map $\Phi_{s,t}(x) = U(t,s)x$. We now extend this construction to a general time-dependent velocity field $v_t(x)$, for which the flow is not necessarily linear in the initial state. Under the standing assumptions of Section 1.2.1.1, the initial-value problem admits a unique global solution for every initial time s and initial state $x \in V$. This follows from the Picard–Lindelöf theorem together with the growth control ensuring global existence. Denoting this solution by $x_t^{s,x}$, we define the associated two-time flow by

$$\Phi_{s,t} : V \to V, \quad x \mapsto \Phi_{s,t}(x) := x_t^{s,x}.$$

Figure 1.1 illustrates how the flow transports both an individual state and a material region.

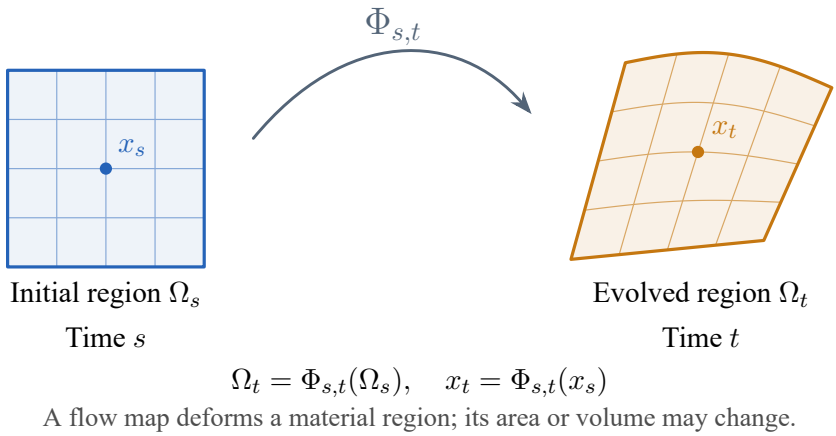


Figure 1.1: Transport of a material region by the two-time flow $\Phi_{s,t}$. The initial region Ω_s (left) and the marked state x_s are mapped to Ω_t (right) and x_t. The grid shows the deformation under the same smooth map; the area or volume need not be preserved.

The following theorem establishes the existence and basic structural properties of this two-time flow.

Theorem 1.12 (Existence, composition, and reversibility of the flow)**.**

Source exercise: Project 9, item 2. *Student Assessment Exercise* [Assessment] (adapted).

Let $V = \mathbb{R}^d$. Assume that the velocity field $v : \mathbb{R} \times V \to V$ is jointly continuous, continuously differentiable in space with jointly continuous derivative, and globally Lipschitz in space uniformly on compact time intervals. Denote by $x_t^{s,x}$ the unique global solution of (1.4) with $x_s^{s,x} = x$, and define the two-time flow

$$\Phi_{s,t} : V \to V, \quad x \mapsto \Phi_{s,t}(x) := x_t^{s,x}.$$

Then the following statements hold:

1. ***Well-defined flow.*** *By Theorem 1.9 and the linear growth bound implied by the assumptions, the two-time flow $\Phi_{s,t} : V \to V$ is well-defined for every pair of times, including $t < s$.*

2. ***Composition and invertibility.*** *Without any ordering restriction on r, s, t, the flow satisfies*

$$\Phi_{s,s} = \mathrm{id}_V, \quad \Phi_{r,t} \circ \Phi_{s,r} = \Phi_{s,t}, \quad \Phi_{s,t}^{-1} = \Phi_{t,s}.$$

   Moreover, each $\Phi_{s,t} : V \to V$ is a C^1 diffeomorphism.

3. ***Autonomous flows.*** *If, in addition, $v_t = v$ is autonomous[a] and its solutions extend*

to all real times, then $\phi_t := \Phi_{0,t}$ defines a one-parameter group satisfying

$$\phi_{t+r} = \phi_t \circ \phi_r, \quad \phi_{-t} = \phi_t^{-1}.$$

4. ***Time reversal.*** *For any solution x_t on $[0, T]$, the time-reversed trajectory*

$$y_t = x_{T-t}, \quad 0 \le t \le T,$$

satisfies $\dot{y}_t = -v_{T-t}(y_t)$, $y_0 = x_T$.

[a] An autonomous dynamical system is one whose vector field has no explicit time dependence, i.e., $v_t(x) = v(x)$. Consequently, its evolution depends only on the elapsed time $t - s$, rather than on the initial time s itself.

Proof. (Proof in Appendix A.2.) □

Having established the existence, uniqueness, and algebraic properties of the flow, we now examine how its evolution depends on the initial state. In particular, the derivative of the flow with respect to the initial condition describes the evolution of infinitesimal perturbations, while its determinant characterizes the corresponding local volume change. These quantities satisfy the following evolution equations.

Proposition 1.13 (Variational equation and Jacobian evolution)**.**

Source exercise: Project 9, item 3–4. *Student Assessment Exercise* [Assessment] (adapted).

For the flow $\Phi_{s,t}$ under the standing assumptions, prove the following properties:

- ***Jacobian matrix.*** *The Jacobian matrix of the flow,*

$$M_{s,t}(x) := D_x \Phi_{s,t}(x) \in GL(V),$$

satisfies

$$\begin{cases} \partial_t M_{s,t}(x) = Dv_t(\Phi_{s,t}(x)) M_{s,t}(x), \\ M_{s,s}(x) = I_d. \end{cases} \tag{1.8}$$

- ***Jacobian determinant.*** *The determinant of the Jacobian matrix satisfies*

$$J_{s,t}(x) := \det M_{s,t}(x) = \exp\left(\int_s^t \nabla \cdot v_r(\Phi_{s,r}(x)) \, \mathrm{d}r \right). \tag{1.9}$$

Proof. Fix an initial time s and state x. By the definition of the flow and (1.4), we have

$$\partial_t \Phi_{s,t}(x) = v_t(\Phi_{s,t}(x)), \quad \Phi_{s,s}(x) = x. \tag{1.10}$$

By the continuous differentiability of the vector field in the state variable, the flow is differ-

entiable with respect to its initial state. Differentiating the flow equation (1.10) with respect to x and applying the chain rule gives

$$\partial_t D_x \Phi_{s,t}(x) = Dv_t(\Phi_{s,t}(x)) D_x \Phi_{s,t}(x).$$

By the definition $M_{s,t}(x) := D_x \Phi_{s,t}(x)$, we obtain

$$\partial_t M_{s,t}(x) = Dv_t(\Phi_{s,t}(x)) M_{s,t}(x).$$

Since $\Phi_{s,s} = \mathrm{id}_V$, its derivative is the identity operator:

$$M_{s,s}(x) = D_x \Phi_{s,s}(x) = D_x \,\mathrm{id}_V(x) = I_d.$$

Thus, we obtain the matrix initial-value problem (1.8)

$$\begin{cases} \partial_t M_{s,t}(x) = Dv_t(\Phi_{s,t}(x)) M_{s,t}(x), \\ M_{s,s}(x) = I_d. \end{cases}$$

Algebraic evolution in $\mathrm{End}(V)$**.** For a fixed initial state x, define

$$A_t := Dv_t(\Phi_{s,t}(x)) \in \mathrm{End}(V), \quad M_t := D_x \Phi_{s,t}(x) \in \mathrm{GL}(V).$$

The variational equation can then be written as an evolution equation in the associative algebra $\mathrm{End}(V)$:

$$\dot{M}_t = A_t \circ M_t, \quad M_s = I_V,$$

where multiplication is given by composition of linear operators. Its equivalent integral equation is

$$M_t = I_V + \int_s^t A_r \circ M_r \, \mathrm{d}r.$$

Successive substitution yields the Peano–Baker series in $\mathrm{End}(V)$:

$$M_t = I_V + \int_s^t A_{r_1} \, \mathrm{d}r_1 + \int_s^t \int_s^{r_1} A_{r_1} \circ A_{r_2} \, \mathrm{d}r_2 \, \mathrm{d}r_1 + \cdots .$$

More generally, its n-th order term is

$$\int_{s \le r_n \le \cdots \le r_1 \le t} A_{r_1} \circ \cdots \circ A_{r_n} \, \mathrm{d}r_n \cdots \mathrm{d}r_1.$$

Under the continuity of A_t on compact time intervals, this series converges absolutely in $\mathrm{End}(V)$.

The operator products retain their chronological order. Since $\mathrm{End}(V)$ is generally non-commutative, operators at different times need not commute:

$$A_{r_1} \circ A_{r_2} \neq A_{r_2} \circ A_{r_1}.$$

Consequently, the evolution generally cannot be represented by an ordinary exponential. Instead, it is expressed through the time-ordered exponential in $\mathrm{End}(V)$:

$$M_{s,t}(x) = \mathcal{T}\exp_{\mathrm{End}(V)}\left(\int_s^t A_r \, \mathrm{d}r\right).$$

If the operators commute pairwise at different times, namely

$$A_t \circ A_r = A_r \circ A_t, \quad s \le r, t,$$

the time ordering is unnecessary, and the evolution reduces to the ordinary exponential in the associative algebra:

$$M_{s,t}(x) = \exp_{\mathrm{End}(V)}\left(\int_s^t A_r \, \mathrm{d}r\right).$$

Furthermore, differentiating the flow composition identity

$$\Phi_{s,t} = \Phi_{r,t} \circ \Phi_{s,r}$$

with respect to the initial state gives

$$M_{s,t}(x) = M_{r,t}(\Phi_{s,r}(x)) \circ M_{s,r}(x).$$

Thus, the induced linear evolution inherits the composition structure of the original flow, known as the cocycle property.

Jacobian determinant and volume evolution. The flow composition and inverse identities imply

$$\Phi_{t,s} \circ \Phi_{s,t} = \mathrm{id}_V .$$

Differentiating with respect to the initial state x and applying the chain rule gives

$$D\Phi_{t,s}(\Phi_{s,t}(x)) \circ M_{s,t}(x) = I_V.$$

Since $M_{s,t}(x) \in \mathrm{End}(V)$ and V is finite-dimensional, the existence of a left inverse implies that $M_{s,t}(x)$ is invertible. Thus, $M_{s,t}(x) \in \mathrm{GL}(V)$. Define the scalar Jacobian

$$J_{s,t}(x) := \det M_{s,t}(x).$$

For a differentiable curve of invertible linear operators, Jacobi's formula gives

$$\partial_t J_{s,t}(x) = J_{s,t}(x)\,\mathrm{tr}\left(M_{s,t}(x)^{-1} \circ \partial_t M_{s,t}(x)\right).$$

Using the variational equation established above,

$$\partial_t M_{s,t}(x) = A_t \circ M_{s,t}(x),$$

we obtain

$$\begin{aligned}\partial_t J_{s,t}(x) &= J_{s,t}(x)\,\mathrm{tr}\left(M_{s,t}(x)^{-1} \circ A_t \circ M_{s,t}(x)\right)\\ &= J_{s,t}(x)\,\mathrm{tr}(A_t).\end{aligned}$$

The second equality follows from the cyclic invariance of the trace. Since

$$A_t = Dv_t(\Phi_{s,t}(x)), \quad \mathrm{tr}(Dv_t(y)) = \nabla \cdot v_t(y),$$

the Jacobian satisfies the scalar initial-value problem

$$\begin{cases}\partial_t J_{s,t}(x) = \nabla \cdot v_t(\Phi_{s,t}(x)) J_{s,t}(x),\\ J_{s,s}(x) = 1.\end{cases}$$

Solving this scalar linear equation yields

$$J_{s,t}(x) = \exp\left(\int_s^t \nabla \cdot v_r(\Phi_{s,r}(x))\,\mathrm{d}r\right) > 0.$$

This establishes the Liouville–Jacobi formula. In particular, the determinant evolution is governed by an ordinary scalar exponential, even when the operators A_t at different times do not commute. □

Figure 1.2 summarizes the local volume interpretation of the determinant formula (1.9).

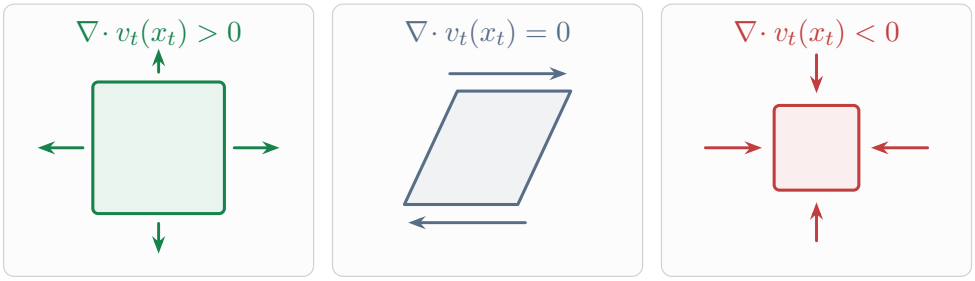


Figure 1.2: Divergence and instantaneous local volume change along $x_t = \Phi_{s,t}(x_s)$. From left to right, outward arrows illustrate expansion, shear illustrates zero instantaneous volume change, and inward arrows illustrate contraction. The Jacobian determinant gives the local volume ratio (area ratio in two dimensions); its value over a finite time interval depends on the integral of divergence along the trajectory.

Example 1.14. Consider the autonomous system

$$\begin{cases} \dot{x}_t = 0.2x_t, \\ \dot{y}_t = 0.1y_t + 0.3x_t^2, \end{cases} \qquad (x_{t=0}, y_{t=0}) = (x_0, y_0).$$

The second equation couples the y-motion nonlinearly to the x-coordinate. The velocity field is smooth but is not globally Lipschitz on $\mathbb{R}^2$ because of its quadratic term. The explicit solution below nevertheless exists for every real time and defines a smooth complete flow. Thus this example verifies the identity of Proposition 1.13 for this explicitly constructed flow, without asserting the global Lipschitz hypothesis used to guarantee existence in the preceding results.

The first equation gives $x_t = e^{0.2t}x_0$. Substitution into the second equation and multiplication by the integrating factor $e^{-0.1t}$ yield

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(e^{-0.1t}y_t\right) = 0.3x_0^2 e^{0.3t}, \quad e^{-0.1t}y_t - y_0 = x_0^2(e^{0.3t} - 1).$$

Consequently $y_t = e^{0.1t}y_0 + x_0^2(e^{0.4t} - e^{0.1t})$, and the flow map is

$$\Phi_{0,t}(x_0, y_0) = \begin{pmatrix} e^{0.2t}x_0 \\ e^{0.1t}y_0 + x_0^2(e^{0.4t} - e^{0.1t}) \end{pmatrix}.$$

Using the notation of Proposition 1.13 with $s = 0$, differentiation with respect to the initial state (x_0, y_0) gives

$$M_{0,t}(x_0, y_0) = D_{(x_0,y_0)}\Phi_{0,t}(x_0, y_0) = \begin{pmatrix} e^{0.2t} & 0 \\ 2x_0(e^{0.4t} - e^{0.1t}) & e^{0.1t} \end{pmatrix}.$$

Its determinant is therefore

$$J_{0,t}(x_0, y_0) = \det M_{0,t}(x_0, y_0) = e^{0.3t} > 0.$$

For the velocity field and its divergence, we have

$$v(x, y) = \begin{pmatrix} 0.2x \\ 0.1y + 0.3x^2 \end{pmatrix}, \quad \nabla \cdot v = \frac{\partial(0.2x)}{\partial x} + \frac{\partial(0.1y + 0.3x^2)}{\partial y} = 0.3.$$

The divergence integral therefore gives

$$J_{0,t}(x_0, y_0) = \exp\left(\int_0^t \nabla \cdot v(\Phi_{0,r}(x_0, y_0))\,\mathrm{d}r\right) = e^{0.3t},$$

in exact agreement with the direct determinant calculation.

Take the initial rectangle $R = [-1, 1] \times [-0.7, 0.7]$, whose area is $|R| = 2.8$. Since the flow is invertible and its Jacobian determinant is positive, the change-of-variables formula gives

$$|\Phi_{0,t}(R)| = \int_R J_{0,t}(x_0, y_0)\, \mathrm{d}x_0\, \mathrm{d}y_0 = 2.8e^{0.3t}.$$

In particular, at $t = 1.5$,

$$J_{0,1.5}(x_0, y_0) = e^{0.45} \approx 1.568312, \quad |\Phi_{0,1.5}(R)| \approx 4.391274.$$

The area has therefore increased by approximately 56.83%. Figure 1.3 visualizes this evolution from $t = 0$ to $t = 1.5$.

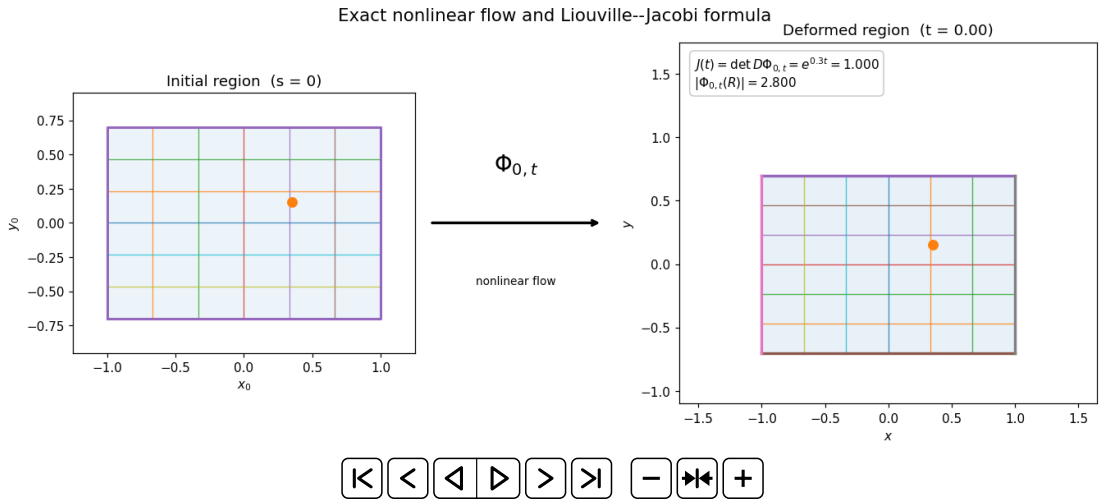


Figure 1.3: Nonlinear deformation of the initial rectangle under $\Phi_{0,t}$. The Jacobian determinant is $J_{0,t}(x_0, y_0) = e^{0.3t}$, in agreement with the Liouville–Jacobi formula. The first frame is the static preview.

The initially rectangular grid stretches and bends under the flow: the nonlinear term $0.3x^2$ changes its shape, while the divergence determines the instantaneous relative rate of local area change. Because $\nabla \cdot v = 0.3$ is independent of position, every initial point has the same local area multiplier $e^{0.3t}$, even though the deformation depends nonlinearly on position. Shape change and area change are distinct features of the flow.

Numerical flow Jacobians (Euler and higher-order solvers)

Source exercise: Project 9, item 4. *Student Assessment Exercise* [Assessment] (adapted).

Problem — Discrete Jacobian and convergence

Consider the differential equation[a]

$$\dot{x}_t = v_t(x_t).$$

Define

$$\widehat{x}_{k+1} = V_k(\widehat{x}_k), \quad \widehat{x}_0 = x_0, \quad V_k(x) = x + hv_{t_k}(x).$$

(a) **Single-step Jacobian.** Prove that

$$DV_k(x) = I_d + hDv_{t_k}(x)$$
$$\det DV_k(x) = 1 + h\nabla \cdot v_{t_k}(x) + O(h^2).$$

On this unit interval $h = 1/K$ and $t_k = kh$.

(b) **Discrete flow Jacobian.** Define $\widehat{\Phi}_{h,K} = V_{K-1} \circ \cdots \circ V_0$ and $\widehat{J}_{h,K}(x_0) = \det D\widehat{\Phi}_{h,K}(x_0)$. Show that

$$\widehat{J}_{h,K}(x_0) = \prod_{k=0}^{K-1} \det\big(I_d + hDv_{t_k}(\widehat{x}_k)\big).$$

(c) **Global convergence order.** With $J_1(x_0) := J_{0,1}(x_0) = \det D\Phi_{0,1}(x_0)$, use Proposition 1.13 to prove

$$E(h) = |\widehat{J}_{h,K}(x_0) - J_1(x_0)| = O(h).$$

Account separately for accumulated single-step errors, trajectory errors, and quadrature errors. Write $e_k := \widehat{x}_k - x_{t_k}$ for the state error.

(d) **Higher-order solvers.** What if you use a better ODE solver? Validate this convergence order numerically using an illustrative example.

[a] on time interval $[0, 1]$, with initial state $x_0 \in V = \mathbb{R}^d$. Assume v is C^2 on a neighbourhood of $[0, 1] \times B$, where B is a compact convex set containing the exact trajectory and all sufficiently fine Euler trajectories. All derivative bounds below are uniform on this set; part (d) requires the additional regularity appropriate to the solver.

Proof.

(a) Single-step Jacobian. Recall that the forward Euler map is

$$V_k(x) = x + hv_{t_k}(x).$$

To compute its derivative with respect to the state variable x, we first write the velocity field in component form:

$$v_{t_k}(x) = \begin{pmatrix} v^1_{t_k}(x) \\ v^2_{t_k}(x) \\ \vdots \\ v^d_{t_k}(x) \end{pmatrix}.$$

Consequently, the Euler map can be expressed componentwise as

$$V_k(x) = \begin{pmatrix} x^1 + hv^1_{t_k}(x) \\ x^2 + hv^2_{t_k}(x) \\ \vdots \\ x^d + hv^d_{t_k}(x) \end{pmatrix}.$$

The Jacobian matrix is obtained by differentiating each component of V_k with respect to each coordinate of x. Its (i, j)-th entry is

$$\frac{\partial V^i_k(x)}{\partial x^j} = \delta_{ij} + h\frac{\partial v^i_{t_k}(x)}{\partial x^j},$$

where δ_{ij} is the Kronecker delta, arising from $\partial x^i/\partial x^j = \delta_{ij}$. Therefore, the Jacobian matrix takes the form

$$DV_k(x) = \begin{pmatrix} 1 + h\frac{\partial v^1_{t_k}}{\partial x^1} & h\frac{\partial v^1_{t_k}}{\partial x^2} & \cdots & h\frac{\partial v^1_{t_k}}{\partial x^d} \\ h\frac{\partial v^2_{t_k}}{\partial x^1} & 1 + h\frac{\partial v^2_{t_k}}{\partial x^2} & \cdots & h\frac{\partial v^2_{t_k}}{\partial x^d} \\ \vdots & \vdots & \ddots & \vdots \\ h\frac{\partial v^d_{t_k}}{\partial x^1} & h\frac{\partial v^d_{t_k}}{\partial x^2} & \cdots & 1 + h\frac{\partial v^d_{t_k}}{\partial x^d} \end{pmatrix}.$$

Recognizing the identity matrix and the Jacobian of the velocity field, we obtain

$$DV_k(x) = I_d + hDv_{t_k}(x).$$

To derive the determinant expansion, let $A = Dv_{t_k}(x) \in \mathrm{End}(V)$ have columns $a_1, \ldots, a_d$. By the multilinearity of the determinant,

$$\begin{aligned} \det(I_d + hA) &= \det(e_1 + ha_1, \ldots, e_d + ha_d) \\ &= 1 + h\sum_{j=1}^{d} \det(e_1, \ldots, a_j, \ldots, e_d) + R^{\det}_A(h) \\ &= 1 + h\,\mathrm{tr}(A) + R^{\det}_A(h), \end{aligned}$$

where $R^{\det}_A(h)$ collects all terms containing at least two columns from hA. Substituting $A = Dv_{t_k}(x)$ and using $\mathrm{tr}(Dv_{t_k}(x)) = \nabla \cdot v_{t_k}(x)$, we obtain

$$\det DV_k(x) = 1 + h\nabla \cdot v_{t_k}(x) + R^{\det}_k(h, x),$$

where $R^{\det}_k(h, x) := R^{\det}_{Dv_{t_k}(x)}(h)$. For the logarithmic expansion, define

$$u_k(h, x) := h\nabla \cdot v_{t_k}(x) + R^{\det}_k(h, x).$$

Then

$$\det DV_k(x) = 1 + u_k(h, x).$$

For sufficiently small h, the determinant $\det DV_k(x)$ is positive uniformly for $x \in B$ and $k = 0, \dots, K-1$, as established in Appendix A.6. Recall the determinant expansion

$$\det DV_k(x) = 1 + h\nabla \cdot v_{t_k}(x) + R_k^{\det}(h, x),$$

where $R_k^{\det}(h, x) = O(h^2)$ uniformly. Consequently,

$$u_k(h, x) := \det DV_k(x) - 1 = O(h)$$

uniformly. We may therefore apply the scalar Taylor expansion $\log(1+u) = u + O(u^2)$ for sufficiently small h. More precisely, define

$$r_{\log}(u) := \log(1+u) - u.$$

The combined logarithmic remainder is then

$$r_k^{\log}(h, x) := R_k^{\det}(h, x) + r_{\log}(u_k(h, x)).$$

It follows that

$$\begin{aligned} \log\det DV_k(x) &= \log(1 + u_k(h, x)) \\ &= h\nabla \cdot v_{t_k}(x) + r_k^{\log}(h, x). \end{aligned}$$

Since $u_k(h, x) = O(h)$ and $r_{\log}(u) = O(u^2)$, both contributions to the combined remainder are of order h^2. In particular, there exists a constant $C > 0$, independent of h, k, and x, such that

$$\sup_{x \in B} |r_k^{\log}(h, x)| \le Ch^2.$$

(b) Discrete flow Jacobian. Recall that the forward Euler scheme is defined by

$$\widehat{x}_{k+1} = V_k(\widehat{x}_k), \quad V_k(x) = x + hv_{t_k}(x).$$

Starting from the initial state x_0, successive applications of the single-step maps give

$$\widehat{x}_K = V_{K-1} \circ \cdots \circ V_1 \circ V_0(x_0).$$

$$x_0 = \widehat{x}_0 \xrightarrow{V_0} \widehat{x}_1 \xrightarrow{V_1} \cdots \xrightarrow{V_{K-2}} \widehat{x}_{K-1} \xrightarrow{V_{K-1}} \widehat{x}_K.$$

Thus, the discrete flow map is

$$\widehat{\Phi}_{h,K} = V_{K-1} \circ \cdots \circ V_1 \circ V_0.$$

To determine how the final state depends on the initial state, we differentiate this composition with respect to x_0.

By the chain rule, for the first two steps,

$$D(V_1 \circ V_0)(x_0) = DV_1(\widehat{x}_1) DV_0(x_0),$$

where $\widehat{x}_1 = V_0(x_0)$. Applying the chain rule successively yields

$$D\widehat{\Phi}_{h,K}(x_0) = DV_{K-1}(\widehat{x}_{K-1}) \cdots DV_1(\widehat{x}_1) DV_0(x_0).$$

The order of matrix multiplication is essential, since Jacobian matrices at different steps need not commute.

By definition, the Jacobian determinant of the discrete flow is

$$\widehat{J}_{h,K}(x_0) := \det D\widehat{\Phi}_{h,K}(x_0).$$

Using the multiplicative property of determinants, $\det(AB) = \det(A)\det(B)$, we obtain

$$\widehat{J}_{h,K}(x_0) = \prod_{k=0}^{K-1} \det DV_k(\widehat{x}_k).$$

From part (a), the Jacobian matrix of each single-step map is

$$DV_k(\widehat{x}_k) = I_d + hDv_{t_k}(\widehat{x}_k).$$

Substituting this expression gives

$$\widehat{J}_{h,K}(x_0) = \prod_{k=0}^{K-1} \det DV_k(\widehat{x}_k) = \prod_{k=0}^{K-1} \det\big(I_d + hDv_{t_k}(\widehat{x}_k)\big). \tag{1.11}$$

Hence, the Jacobian determinant of the entire discrete flow equals the product of the Jacobian determinants of its individual Euler steps.

(c) Global convergence order. We estimate the global error between the discrete and exact Jacobian determinants. Since both determinants are positive for sufficiently small h, define the logarithmic error

$$\delta(h) := \log \widehat{J}_{h,K}(x_0) - \log J_1(x_0).$$

Our goal is to prove that $|\delta(h)| = O(h)$ and then transfer this estimate to the Jacobian determinants themselves.

By the single-step expansion in part (a) and the product formula (1.11) for the discrete Jacobian,

$$\log \widehat{J}_{h,K}(x_0) = h \sum_{k=0}^{K-1} \nabla \cdot v_{t_k}(\widehat{x}_k) + R_{\log}(h), \quad |R_{\log}(h)| \le Ch,$$

whereas the continuous-time formula gives

$$\log J_1(x_0) = \int_0^1 \nabla \cdot v_t(x_t)\, \mathrm{d}t.$$

To compare these expressions, decompose the logarithmic error as

$$\delta(h) = R_{\log}(h) + E_{\text{traj}}(h) + E_{\text{quad}}(h),$$

where

$$E_{\text{traj}}(h) := h \sum_{k=0}^{K-1} \left[\nabla \cdot v_{t_k}(\widehat{x}_k) - \nabla \cdot v_{t_k}(x_{t_k})\right],$$
$$E_{\text{quad}}(h) := h \sum_{k=0}^{K-1} \nabla \cdot v_{t_k}(x_{t_k}) - \int_0^1 \nabla \cdot v_t(x_t)\, \mathrm{d}t.$$

Thus, it suffices to estimate these three contributions.

1. **Trajectory error.** By the uniform spatial Lipschitz bound L_x for $\nabla \cdot v_t$ and the standard Euler state estimate

   $$\max_{0 \le k \le K} \|\widehat{x}_k - x_{t_k}\| \le C_E h,$$

   we obtain

   $$|E_{\text{traj}}(h)| \le hKL_xC_Eh = L_xC_Eh.$$

2. **Quadrature error.** Let $g(t) := \nabla \cdot v_t(x_t)$. If L_t bounds $|\partial_t \nabla \cdot v_t(x)|$ and V_* bounds $\|v_t(x)\|$, then

   $$|g(t) - g(s)| \le L_t|t-s| + L_x\|x_t - x_s\| \le (L_t + L_xV_*)|t-s|.$$

   Writing $L_g := L_t + L_xV_*$, the Lipschitz estimate gives

   $$|E_{\text{quad}}(h)| \le \sum_{k=0}^{K-1} \int_{t_k}^{t_{k+1}} |g(t_k) - g(t)|\, \mathrm{d}t \le \frac{L_g}{2} Kh^2 = \frac{L_g}{2} h.$$

   Together with $|R_{\log}(h)| \le Ch$, these estimates yield

   $$|\delta(h)| \le \left(C + L_xC_E + \frac{L_g}{2}\right) h = C_*h.$$

By the continuous-time Jacobian formula and the uniform bound $|\nabla \cdot v_t(x)| \le B_{\text{div}}$,

$$0 < J_1(x_0) \le e^{B_{\text{div}}}.$$

Since $\widehat{J}_{h,K}(x_0) = J_1(x_0)e^{\delta(h)}$, the mean value estimate $|e^z - 1| \le e^{|z|}|z|$ gives

$$|\widehat{J}_{h,K}(x_0) - J_1(x_0)| \le J_1(x_0)e^{|\delta(h)|}|\delta(h)|$$

Therefore,

$$|\widehat{J}_{h,K}(x_0) - J_1(x_0)| = O(h).$$

Thus, the discrete Jacobian determinant converges to the exact continuous-time Jacobian determinant with first-order global accuracy.

(d) Higher-order solvers. Consider the augmented initial-value problem

$$\begin{cases} \dot{x}_t = v_t(x_t), & x_{t=0} = x_0, \\ \dot{M}_t = Dv_t(x_t)M_t, & M_0 = I_d. \end{cases}$$

Assume sufficient smoothness and stability for the chosen method to have order p on this augmented system, with bounded exact and numerical solutions. Its matrix approximation then satisfies

$$\|\widehat{M}_{h,K} - M_1\| = O(h^p), \quad M_1 := M_{0,1}(x_0) = D\Phi_{0,1}(x_0).$$

To identify $\widehat{M}_{h,K}$ with $D\widehat{\Phi}_{h,K}(x_0)$, the variational update must be obtained by differentiating the same numerical state update. For Runge–Kutta methods, differentiating every stage produces exactly the corresponding Runge–Kutta method on the augmented system. The determinant is a polynomial in the matrix entries and is Lipschitz on bounded matrix sets, so

$$|\det \widehat{M}_{h,K} - \det M_1| \le C\|\widehat{M}_{h,K} - M_1\| = O(h^p).$$

Thus classical RK4 gives fourth-order Jacobian and determinant convergence when its stage derivatives are computed consistently and the required regularity holds. State accuracy alone does not establish derivative accuracy; combining an RK4 state step with the Euler factor $I_d + hDv_{t_k}(\widehat{x}_k)$ does not generally give a fourth-order Jacobian. □

Problem — Numerical validation

Consider the scalar initial-value problem

$$\dot{x}_t = x_t, \quad x_0 = 1, \quad t \in [0, 1],$$

whose exact flow is $\Phi_{0,t}(x) = e^t x$ and whose Jacobian determinant at $t = 1$ is $J_1 = e$. Numerically validate the predicted convergence orders of the Jacobian determinants of the numerical flow maps by performing the following experiments:

1. Construct the numerical flow maps using forward Euler and classical RK4, and compute their Jacobian determinants $\widehat{J}_{h,K} = \det D\widehat{\Phi}_{h,K}(x_0)$.
2. Choose $K = 4, 8, 16, 32, 64, 128$ and set $h = 1/K$. Compute the absolute errors
$$E(h) = |\widehat{J}_{h,K} - e|$$
for both methods.
3. Plot the errors against h on a log–log scale.

4. Estimate the observed convergence orders using consecutive mesh refinements and compare them with the theoretical predictions.

Solution.

For the scalar field $v(x) = x$, differentiation of the Euler and RK4 updates gives

$$\widehat{J}^{\text{Euler}}_{h,K} = (1+h)^K, \quad \widehat{J}^{\text{RK4}}_{h,K} = \left(1 + h + \frac{h^2}{2} + \frac{h^3}{6} + \frac{h^4}{24}\right)^K.$$

Here $Kh = 1$ and the exact Jacobian is e. The leading errors are $(e/2)h$ and $(e/120)h^4$, respectively. The existing high-precision validation is retained below.

1. Construction of the numerical flow maps. We first construct the numerical flow maps and differentiate them with respect to the initial state. For the given equation $\dot{x}_t = x_t$, the exact flow is

$$\Phi_{0,t}(x) = e^t x,$$

and hence

$$J_1 = \det D\Phi_{0,1}(x_0) = e,$$

Let $t_k = kh$, where $h = 1/K$, and denote the numerical state at time t_k by X_k, starting from an arbitrary initial state $X_0 = x$.

- For the forward Euler method, the numerical update is

  $$X_{k+1} = X_k + hX_k = (1+h)X_k.$$

  Iterating over K time steps gives

  $$\widehat{\Phi}^{\text{Euler}}_{h,K}(x) = (1+h)^K x.$$

  Differentiating with respect to the initial state yields

  $$\widehat{J}^{\text{Euler}}_{h,K} = \det D\widehat{\Phi}^{\text{Euler}}_{h,K}(x_0) = (1+h)^K.$$

- For the classical fourth-order Runge–Kutta method (RK4), the four intermediate stages are

  $$k_1 = X_k, \quad k_2 = X_k + \frac{h}{2}k_1, \quad k_3 = X_k + \frac{h}{2}k_2, \quad k_4 = X_k + hk_3.$$

  Substituting the preceding stages successively gives

  $$k_1 = X_k, \quad k_2 = \left(1 + \frac{h}{2}\right) X_k,$$
  $$k_3 = \left(1 + \frac{h}{2} + \frac{h^2}{4}\right) X_k, \quad k_4 = \left(1 + h + \frac{h^2}{2} + \frac{h^3}{4}\right) X_k.$$

The RK4 update is

$$X_{k+1} = X_k + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

Substituting the four stages and collecting powers of h, we obtain

$$X_{k+1} = \left(1 + h + \frac{h^2}{2} + \frac{h^3}{6} + \frac{h^4}{24}\right) X_k.$$

Define the one-step amplification factor

$$R_4(h) = 1 + h + \frac{h^2}{2} + \frac{h^3}{6} + \frac{h^4}{24}.$$

After K steps, the numerical flow map is

$$\widehat{\Phi}^{\mathrm{RK4}}_{h,K}(x) = R_4(h)^K x.$$

Differentiating with respect to the initial state gives

$$\widehat{J}^{\mathrm{RK4}}_{h,K} = \det D\widehat{\Phi}^{\mathrm{RK4}}_{h,K}(x_0) = R_4(h)^K.$$

Thus, both discrete Jacobian determinants are obtained directly from the derivatives of their respective numerical flow maps.

2. Asymptotic error analysis. We now determine the leading-order errors of the two numerical Jacobians. Since $Kh = 1$, we have $K = h^{-1}$. For $h > 0$, the inequalities

$$1 + h < e^h, \quad R_4(h) < e^h$$

follow from the exponential series. Raising both inequalities to the power K gives

$$0 < \widehat{J}^{\mathrm{Euler}}_{h,K} < e, \quad 0 < \widehat{J}^{\mathrm{RK4}}_{h,K} < e.$$

Therefore, the absolute errors can be written as

$$E(h) = e - \widehat{J}_{h,K}$$

for both methods.

1. *Euler method.* Taking the logarithm of the discrete Jacobian determinant gives

$$\log \widehat{J}^{\mathrm{Euler}}_{h,K} = \log(1 + h)^K = \frac{1}{h} \log(1 + h).$$

Using the Taylor expansion

$$\log(1 + h) = h - \frac{h^2}{2} + \frac{h^3}{3} + O(h^4),$$

we obtain

$$\log \widehat{J}_{h,K}^{\text{Euler}} = 1 - \frac{h}{2} + \frac{h^2}{3} + O(h^3).$$

Exponentiating both sides and using $\exp(z) = 1 + z + O(z^2)$ as $z \to 0$, we find

$$\widehat{J}_{h,K}^{\text{Euler}} = e \exp\left(-\frac{h}{2} + \frac{h^2}{3} + O(h^3)\right) = e\left(1 - \frac{h}{2} + O(h^2)\right).$$

Subtracting from the exact Jacobian determinant yields

$$E_{\text{Euler}}(h) = \frac{e}{2}h + O(h^2).$$

Thus, the Euler approximation exhibits first-order convergence for the Jacobian determinant.

2. *Classical RK4 method.* Similarly, taking the logarithm gives

$$\log \widehat{J}_{h,K}^{\text{RK4}} = \frac{1}{h} \log R_4(h).$$

To determine its leading-order error, we compare the RK4 amplification factor with the Taylor expansion of the exact exponential:

$$\begin{aligned} e^h &= 1 + h + \frac{h^2}{2} + \frac{h^3}{6} + \frac{h^4}{24} + \frac{h^5}{120} + O(h^6) \\ &= R_4(h) + \frac{h^5}{120} + O(h^6). \end{aligned}$$

Consequently,

$$R_4(h) = e^h - \frac{h^5}{120} + O(h^6).$$

Factoring out e^h gives

$$R_4(h) = e^h\left(1 - \frac{h^5}{120} + O(h^6)\right).$$

Taking logarithms and using $\log(1 + z) = z + O(z^2)$ as $z \to 0$, we obtain

$$\log R_4(h) = h + \log\left(1 - \frac{h^5}{120} + O(h^6)\right) = h - \frac{h^5}{120} + O(h^6).$$

Dividing by h yields

$$\log \widehat{J}_{h,K}^{\text{RK4}} = 1 - \frac{h^4}{120} + O(h^5).$$

Exponentiating,

$$\widehat{J}_{h,K}^{\text{RK4}} = e \exp\left(-\frac{h^4}{120} + O(h^5)\right) = e\left(1 - \frac{h^4}{120} + O(h^5)\right).$$

Subtracting from the exact Jacobian determinant gives

$$E_{\mathrm{RK4}}(h) = \frac{e}{120}h^4 + O(h^5).$$

Thus, the RK4 approximation exhibits fourth-order convergence for the Jacobian determinant.

The two asymptotic expansions predict that the limiting slopes of the corresponding log–log error curves are one for Euler and four for RK4.

3. Numerical validation. We now verify these predictions numerically. Choose

$$K = 4, 8, 16, 32, 64, 128, \quad h = \frac{1}{K}.$$

For each mesh, evaluate the numerical flow maps and their Jacobians derived above, and compute the absolute errors

$$E_{\mathrm{Euler}}(h) = \left| \widehat{J}^{\mathrm{Euler}}_{h,K} - e \right|, \quad E_{\mathrm{RK4}}(h) = \left| \widehat{J}^{\mathrm{RK4}}_{h,K} - e \right|.$$

To estimate the convergence order from consecutive meshes, suppose that the error has the asymptotic form

$$E(h) = Ch^p + O(h^{p+1}),$$

where $C > 0$ and p is the convergence order. Replacing h by $h/2$ gives

$$E(h/2) = C\left(\frac{h}{2}\right)^p + O(h^{p+1}).$$

Taking the ratio of the two expressions, we obtain

$$\frac{E(h)}{E(h/2)} = 2^p\big(1 + O(h)\big).$$

Taking logarithms and dividing by $\log 2$ motivates the observed-order estimate

$$p_{\mathrm{obs}}(h) = \frac{\log(E(h)/E(h/2))}{\log 2}. \tag{1.12}$$

As $h \to 0$, this estimate approaches the theoretical order p.

We evaluate both numerical flow Jacobians and the reference value e using 80-digit decimal arithmetic. This precision is particularly important for RK4, whose errors become very small and would otherwise be affected by cancellation when subtracting nearly equal quantities. Table 1.1 reports the resulting errors and observed orders. Each reported order compares its row with the next finer mesh; no order is reported for the final row. Figure 1.4 plots both error curves against h on logarithmic axes. For comparison, we also include the leading-order error predictions derived above:

$$E^{\mathrm{lead}}_{\mathrm{Euler}}(h) = \frac{e}{2}h, \quad E^{\mathrm{lead}}_{\mathrm{RK4}}(h) = \frac{e}{120}h^4.$$

Table 1.1: Scalar flow-Jacobian errors and observed orders from (1.12). Errors are rounded from the 80-digit calculation.

K	h	E_{Euler}	E_{RK4}	p_{Euler}	p_{RK4}
4	0.25	2.768756×10^{-1}	7.188926×10^{-5}	0.8605	3.8504
8	0.125	1.524973×10^{-1}	4.984042×10^{-6}	0.9244	3.9250
16	0.0625	8.035333×10^{-2}	3.281185×10^{-7}	0.9605	3.9625
32	0.03125	4.129170×10^{-2}	2.104785×10^{-8}	0.9798	3.9812
64	0.015625	2.093688×10^{-2}	1.332722×10^{-9}	0.9898	3.9906
128	0.0078125	1.054281×10^{-2}	8.383902×10^{-11}	–	–

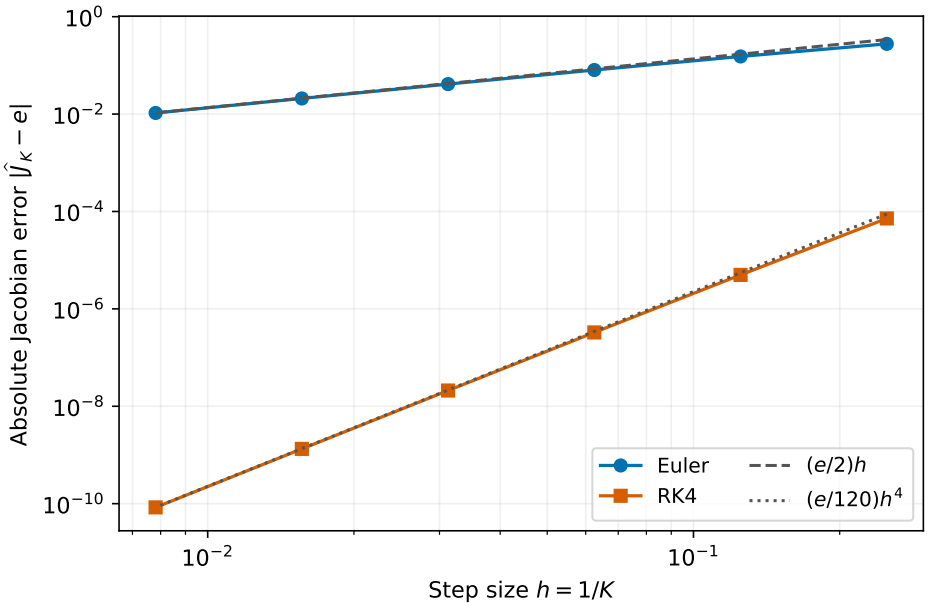


Figure 1.4: Scalar flow-Jacobian errors for Euler and RK4 on six successively refined meshes. The dashed and dotted lines show the leading terms $(e/2)h$ and $(e/120)h^4$, respectively.

The observed orders approach one for Euler and four for RK4, in agreement with the theoretical predictions. Moreover, the numerical error curves approach their respective leading-order reference curves as the time step decreases.

The reproducible calculation is `python experiments/scalar_jacobian.py`; the full-precision values are saved in `results/benchmarks/scalar_jacobian_convergence.csv`. Only the plotting coordinates are converted to ordinary floating-point numbers. ♦

1.2.1.4 Volume transport and minimum action

Let $\Omega_0 \subset \mathbb{R}^d$ be a bounded Lebesgue-measurable region, and let Φ_t denote the flow map generated by the ODE (1.4), with $\Phi_0 = \mathrm{id}$. The deformed domain at time t is defined by

$$\Omega_t := \Phi_t(\Omega_0) = \{\Phi_t(x_0) : x_0 \in \Omega_0\}\,.$$

We define its volume by

$$A_t := \mathrm{vol}(\Omega_t) := \lambda^d(\Omega_t) = \int_{\mathbb{R}^d} \mathbf{1}_{\Omega_t}(x)\,\mathrm{d}x,$$

where λ^d denotes the d-dimensional Lebesgue measure and $\mathbf{1}_{\Omega_t}$ is the indicator function of Ω_t. Assume that Φ_t is a smooth diffeomorphism. By the change-of-variables formula,

$$A_t = \int_{\Omega_0} |\det D\Phi_t(x_0)|\,\mathrm{d}x_0.$$

For a flow generated by an ODE with $\Phi_0 = \mathrm{id}$, the Jacobian determinant remains positive, so the absolute value can be omitted. Moreover, differentiating the volume with respect to time gives

$$\dot{A}_t = \int_{\Omega_t} \nabla \cdot v_t(x)\,\mathrm{d}x. \tag{1.13}$$

Corollary 1.15 (Volume transport and divergence)**.**

Source exercise: Project 9, item 5. *Student Assessment Exercise* [Assessment] (adapted).

Let $\mathcal{D}_0 \subset \mathbb{R}^d$ be a bounded Lebesgue-measurable region, and let $\Phi_{0,t}$ denote the flow map generated by (1.4), with $\Phi_{0,0} = \mathrm{id}$. Define the deformed domain by

$$\mathcal{D}_t := \Phi_{0,t}(\mathcal{D}_0) = \{\Phi_{0,t}(x_0) : x_0 \in \mathcal{D}_0\}.$$

Its volume is defined as the d-dimensional Lebesgue measure

$$|\mathcal{D}_t| := \lambda^d(\mathcal{D}_t) = \int_{\mathbb{R}^d} \mathbf{1}_{\mathcal{D}_t}(y)\,\mathrm{d}y,$$

where λ^d denotes the d-dimensional Lebesgue measure and $\mathbf{1}_{\mathcal{D}_t}$ is the indicator function of $\mathcal{D}_t$. Assume that $\Phi_{0,t}$ is a smooth diffeomorphism and that differentiation under the integral is legitimate. Then the volume evolves according to

$$\frac{\mathrm{d}}{\mathrm{d}t}|\mathcal{D}_t| = \int_{\mathcal{D}_t} \nabla \cdot v_t(y)\,\mathrm{d}y. \tag{1.14}$$

Proof. Let

$$M_{0,t}(x_0) := D\Phi_{0,t}(x_0), \quad J_{0,t}(x_0) := \det M_{0,t}(x_0).$$

By the change-of-variables formula,

$$|\mathcal{D}_t| = \int_{\mathcal{D}_0} J_{0,t}(x_0)\,\mathrm{d}x_0,$$

where $J_{0,t}(x_0) > 0$ because $\Phi_{0,t}$ is a flow starting from the identity map.

Differentiating under the integral gives

$$\frac{\mathrm{d}}{\mathrm{d}t}|\mathcal{D}_t| = \int_{\mathcal{D}_0} \frac{\partial}{\partial t} J_{0,t}(x_0)\,\mathrm{d}x_0.$$

The variational equation for the flow yields

$$\frac{\partial}{\partial t} M_{0,t}(x_0) = Dv_t(\Phi_{0,t}(x_0)) M_{0,t}(x_0).$$

Applying the Jacobi formula for the derivative of a determinant, we obtain

$$\begin{aligned}\frac{\partial}{\partial t} J_{0,t}(x_0) &= J_{0,t}(x_0)\,\mathrm{tr}\left(M_{0,t}(x_0)^{-1}\frac{\partial M_{0,t}(x_0)}{\partial t}\right)\\ &= J_{0,t}(x_0)\,\mathrm{tr}\left(Dv_t(\Phi_{0,t}(x_0))\right)\\ &= J_{0,t}(x_0)\nabla\cdot v_t(\Phi_{0,t}(x_0)).\end{aligned}$$

Consequently,

$$\frac{\mathrm{d}}{\mathrm{d}t}|\mathcal{D}_t| = \int_{\mathcal{D}_0} J_{0,t}(x_0)\nabla\cdot v_t(\Phi_{0,t}(x_0))\,\mathrm{d}x_0.$$

Finally, applying the change-of-variables formula $y = \Phi_{0,t}(x_0)$, we arrive at

$$\frac{\mathrm{d}}{\mathrm{d}t}|\mathcal{D}_t| = \int_{\mathcal{D}_t} \nabla\cdot v_t(y)\,\mathrm{d}y,$$

which completes the proof. □

Example 1.16 (Minimum weighted kinetic action between two states)**.**

Source exercise: Project 9, item 6. *Student Assessment Exercise* [Assessment] (adapted).

Problem

Let $w : [0,1] \to (0,\infty)$ be continuous, and fix two points $y, z \in \mathbb{R}^d$. Consider a time-dependent velocity field $v_t : \mathbb{R}^d \to \mathbb{R}^d$ generating a flow map $\Phi_{0,t}$ through

$$\frac{\partial}{\partial t}\Phi_{0,t}(x) = v_t(\Phi_{0,t}(x)), \quad \Phi_{0,0}(x) = x.$$

Find an optimal velocity field v_t^* that minimizes

$$I[v] = \int_0^1 w_t \|v_t(\Phi_{0,t}(y))\|_2^2\,\mathrm{d}t,$$

subject to the endpoint constraint $\Phi_{0,1}(y) = z$. Determine the corresponding optimal flow map $\Phi_{0,t}^*(x)$ for $t \in [0,1]$.

To be completed.

1.2.2 Observable evolution and probability transport

In statistical physics, the dynamics of a physical system governs not only the evolution of individual trajectories but also the evolution of probability distributions over the state space, describing the statistical ensemble of its possible states. To formulate such ensembles

mathematically, we first introduce a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let X_t be a measurable random variable taking values in the state space $(E, \mathcal{S})$, where $\mathcal{S}$ is the Borel σ-algebra. Thus,

$$X_t : \Omega \to E, \quad \omega \mapsto X_t(\omega).$$

For each sample point $\omega \in \Omega$, the value $X_t(\omega)$ specifies the state of the system at time t corresponding to that realization.

Trajectories space and Probability space In the classical dynamical setting, we may identify the sample space with a trajectory space, whose elements are entire state trajectories. More precisely, for a flow generated by $\dot{x}_t = v_t(x_t)$, define the trajectories space:

$$\mathcal{X} := \{\gamma \in C([0,T]; E) \mid \gamma(t) = \Phi_{0,t}(\gamma(0)), \quad t \in [0,T]\}.$$

Each $\gamma \in \mathcal{X}$ represents a possible trajectory of the system, and the random variable X_t is the corresponding evaluation map:

$$X_t : \mathcal{X} \to E, \quad \gamma \mapsto \gamma(t) = x_t.$$

We equip Ω with the σ-algebra generated by these evaluation maps:

$$\mathcal{F} := \sigma\left(\{X_t^{-1}(A) \;\middle|\; t \in [0,T],\ A \in \mathcal{S}\}\right).$$

Thus, the probability measure $\mathbb{P}$ assigns probabilities to different trajectories, while X_t extracts the state of a trajectory at a specified time,

$$\mathbb{P} : \mathcal{F} \to [0,1], \quad A \mapsto \mathbb{P}(A), \quad A \in \mathcal{F}.$$

Pushforward measures $\mathbb{P}$ is the trajectory-space probability measure. The law at time s is the pushforward $\mu_s = (X_s)_{\#}\mathbb{P}$, so that for every $A \in \mathcal{S}$,

$$\mu_s(A) = \mathbb{P}\big(X_s^{-1}(A)\big) = \mathbb{P}\big(\{\gamma \in \Omega : \gamma(s) \in A\}\big).$$

This operation merely transfers a measure from trajectory space to state space; it does not by itself describe temporal evolution. Two distinct operations are involved.

- **Evaluation map.** The map X_t extracts the state at time t and pushes the trajectory law forward to a law on the state space,

$$\mu_t = (X_t)_{\#}\mathbb{P} \in \mathcal{P}(E, \mathcal{S}).$$

- **Dynamical flow.** The deterministic flow $\Phi_{s,t} : E \to E$, which moves states along $x_t = \Phi_{s,t}(x_s)$, transports a measure on the state space from time s to time t,

$$(\Phi_{s,t})_{\#} : \mathcal{P}(E, \mathcal{S}) \to \mathcal{P}(E, \mathcal{S}), \quad \mu \mapsto (\Phi_{s,t})_{\#}\mu.$$

The two operations are compatible: because every trajectory follows the flow, the evaluation maps satisfy, for $0 \le s \le t \le T$,

$$X_t = \Phi_{s,t} \circ X_s,$$

i.e. the following diagram commutes:

$$\begin{array}{ccc} \mathcal{X} & \xrightarrow{X_s} & E \\ & \searrow^{X_t} & \downarrow{\scriptstyle \Phi_{s,t}} \\ & & E \end{array}$$

Consequently both routes give the same law:

$$\mu_t = (X_t)_{\#}\mathbb{P} = (\Phi_{s,t} \circ X_s)_{\#}\mathbb{P} = (\Phi_{s,t})_{\#}(X_s)_{\#}\mathbb{P} = (\Phi_{s,t})_{\#}\mu_s,$$

or, equivalently, for every $A \in \mathcal{S}$,

$$\mu_t(A) = \mu_s\big(\Phi_{s,t}^{-1}(A)\big).$$

The relationship between these objects and their evolution is summarized by

$$\underbrace{(\mathcal{X}, \mathcal{F}, \mathbb{P})}_{\text{trajectory space}} \xrightarrow{X_s} \underbrace{(E, \mathcal{S}, \mu_s)}_{\text{state space, law at time } s} \xrightarrow{\Phi_{s,t}} \underbrace{(E, \mathcal{S}, \mu_t)}_{\text{state space, law at time } t}.$$

This description corresponds to the classical Schrödinger picture, in which probability measures evolve under the dynamics while observables are kept fixed. Rather than following individual trajectories, we study how their statistical distribution is transported through the state space.

When the probability measures are absolutely continuous with respect to Lebesgue measure on $\mathbb{R}^d$, we may write

$$\mu_t(\,\mathrm{d}x) = \rho_t(x)\,\mathrm{d}x,$$

where ρ_t denotes the probability density at time t. Our objective is then to characterize the evolution of ρ_t in terms of the velocity field v_t. For a sufficiently regular invertible flow, the density can be expressed using the flow map and its Jacobian. Alternatively, by studying the action of μ_t on test functions, we can derive a differential equation for probability transport directly from v_t, without explicitly computing the flow map.

1.2.2.1 Generators and observable evolution

The observable–measure duality has a dynamical consequence that is central to statistical physics and Markov processes. Given a sufficiently smooth observable f, define the Koopman operator

$$(\mathcal{K}_{s,t} f)(x) = f(\Phi_{s,t}(x)).$$

It describes the evolution of observables induced by the state flow. Differentiating with respect to the terminal time and applying the chain rule gives

$$\partial_t(\mathcal{K}_{s,t}f) = \mathcal{K}_{s,t}\mathcal{L}_t f, \quad \mathcal{L}_t f = v_t \cdot \nabla f.$$

Here $\mathcal{L}_t$ is the infinitesimal generator induced by the velocity field. Differentiating instead with respect to the initial time, using the flow composition identity, yields the backward equation

$$\partial_s \mathcal{K}_{s,t} f = -\mathcal{L}_s(\mathcal{K}_{s,t}f).$$

The different operator orders reflect the distinction between terminal-time and initial-time differentiation.

The generator is a linear differential operator acting on a suitable domain of smooth observables. It is neither the velocity field itself nor the finite-time flow, but describes their induced infinitesimal action. More precisely, for sufficiently regular f,

$$\mathcal{L}_t f = \lim_{h \downarrow 0} \frac{\mathcal{K}_{t,t+h} f - f}{h}.$$

The existence of this limit depends on the chosen function space and topology; throughout, the differential expression is understood on the appropriate smooth test functions.

The same operator framework extends to stochastic dynamics. Let $(E, \mathcal{S})$ be a measurable state space, which need not have a vector space structure. A Markov transition kernel $\mathsf{K}_{s,t}(x, \mathrm{d}y)$ assigns a probability measure to each initial state and satisfies the Chapman–Kolmogorov composition rule. It induces an evolution operator on bounded measurable observables:

$$(\mathsf{P}_{s,t} f)(x) = \int_E f(y) \mathsf{K}_{s,t}(x, \mathrm{d}y).$$

The corresponding operators satisfy $\mathsf{P}_{s,r}\mathsf{P}_{r,t} = \mathsf{P}_{s,t},\ s \le r \le t$. Meanwhile, the probability law evolves according to

$$\mu_t(B) = \int_E \mathsf{K}_{s,t}(x, B) \mu_s(\mathrm{d}x).$$

Thus, observables and probability measures evolve in dual ways, related by the integral pairing

$$\langle f, \mu_t \rangle = \langle \mathsf{P}_{s,t} f, \mu_s \rangle.$$

A deterministic flow is recovered by taking

$$\mathsf{K}_{s,t}(x, \mathrm{d}y) = \delta_{\Phi_{s,t}(x)}(\mathrm{d}y),$$

for which $\mathsf{P}_{s,t} = \mathcal{K}_{s,t}$.

When the corresponding infinitesimal limit exists, the Markov evolution has a generator

$$\mathcal{L}_t f = \lim_{h \downarrow 0} \frac{\mathsf{P}_{t,t+h} f - f}{h}.$$

For suitable test functions and integrability conditions, the dual evolution then gives the weak forward equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle f, \mu_t\rangle = \langle \mathcal{L}_t f, \mu_t\rangle.$$

When a probability density exists, this weak equation can be expressed through the adjoint generator $\mathcal{L}_t^*$ as $\partial_t \rho_t = \mathcal{L}_t^* \rho_t$.

The form of the generator depends on the underlying dynamical mechanism. For example, consider a diffusion process

$$\mathrm{d}X_t = b_t(X_t)\mathrm{d}t + \sigma_t(X_t)\mathrm{d}W_t.$$

Its differential generator, obtained from Itô's formula, is

$$\mathcal{L}_t f = b_t \cdot \nabla f + \frac{1}{2}\,\mathrm{tr}\left(\sigma_t \sigma_t^\top \nabla^2 f\right).$$

For a finite-state jump process with transition rates $q_t(x, y)$, the generator instead takes the form

$$\mathcal{L}_t f(x) = \sum_{y \neq x} q_t(x, y)[f(y) - f(x)].$$

These examples illustrate how different microscopic mechanisms lead to different generator expressions while retaining the same underlying dual structure. The corresponding adjoint operators and probability evolution equations will be derived when the respective dynamics are introduced.

1.2.2.2 Probability transport and observable evolution

Density transport and the classical continuity equation. We first describe probability conservation through the geometry of the deterministic flow, taking $E = \mathbb{R}^d$. Recall the ODE $\dot{x}_t = v_t(x_t)$, that state dynamics

$$x_t = \Phi_{0,t}(x_0), \quad 0 \leq t \leq T.$$

Differentiability with respect to initial conditions requires additional spatial regularity; here we assume that v_t is C^1 in space with jointly continuous spatial derivatives. For the change of variables below, we further assume that each $\Phi_{0,t}$ is a C^1 diffeomorphism of $\mathbb{R}^d$ onto itself. This global invertibility requires the corresponding backward trajectories to exist and does not follow from forward existence and uniqueness alone.

Let $X_0 \sim \rho_0(x)\ \mathrm{d}x$,

$$X_t = \Phi_{0,t}(X_0), \quad \mu_t = (\Phi_{0,t})_{\#}\mu_0.$$

The pushforward relation established above conserves the probability assigned to a region transported by the flow. The following result expresses this conservation in terms of the local volume deformation.

Proposition 1.17 (Density along trajectories)**.**

Source exercise: Project 9, item 7–8. *Student Assessment Exercise* [Assessment] (adapted).

Under the preceding diffeomorphism assumptions, the transported law has a Lebesgue density ρ_t. Set

$$M_{0,t}(x_0) := D\Phi_{0,t}(x_0), \quad J_{0,t}(x_0) := \det M_{0,t}(x_0).$$

Then $J_{0,t}(x_0) > 0$, and a representative of ρ_t can be chosen so that, for almost every x_0,

$$\rho_t(\Phi_{0,t}(x_0)) = \rho_0(x_0)\big|J_{0,t}(x_0)\big|^{-1} = \rho_0(x_0)\exp\left(-\int_0^t \nabla\cdot v_s(\Phi_{0,s}(x_0))\,\mathrm{d}s\right).$$

If, in addition, this representative $\rho(t,x) = \rho_t(x)$ is C^1 in (t,x), then it satisfies the continuity equation

$$\partial_t\rho_t + \nabla\cdot(\rho_t v_t) = 0 \tag{1.15}$$

pointwise on $(0,T)\times\mathbb{R}^d$.

Proof. Let $A \in \mathcal{B}(\mathbb{R}^d)$ be a Borel set representing a region of the state space at time t. Its preimage under the flow map $\Phi_{0,t}$ is defined by

$$\Phi_{0,t}^{-1}(A) := \left\{\, x_0 \in \mathbb{R}^d : \Phi_{0,t}(x_0) \in A \,\right\},$$

i.e., the set of initial states at time $t = 0$ that are mapped into A at time t (Fig **??**).

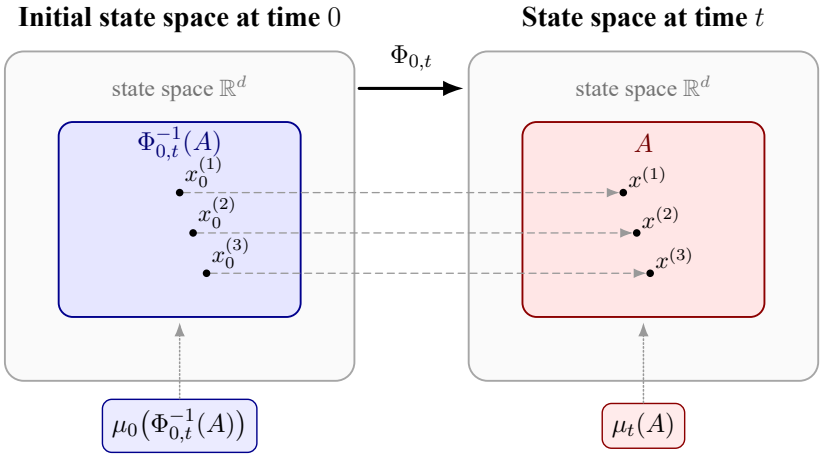


Figure 1.5: Measure transport under the flow $\Phi_{0,t}$: the mass of A at time t equals the mass of its preimage $\Phi_{0,t}^{-1}(A)$ at time 0.

Since $\Phi_{0,t}$ is continuousand A is Borel measurable, the preimage $\Phi_{0,t}^{-1}(A)$ is also a Borel set. The pushforward relation then reads

$$\mu_t(A) = \big((\Phi_{0,t})_{\#}\mu_0\big)(A) = \mu_0\big(\Phi_{0,t}^{-1}(A)\big), \quad A \in \mathcal{B}(\mathbb{R}^d).$$

Since μ_0 admits a Lebesgue density and $\Phi_{0,t}$ is a C^1 diffeomorphism, the change-of-variables formula implies that μ_t also admits a Lebesgue density.

$$\mu_t(\mathrm{d}x) = \rho_t(x)\,\mathrm{d}x, \quad \mu_0(\mathrm{d}x_0) = \rho_0(x_0)\,\mathrm{d}x_0,$$

we obtain

$$\mu_t(A) = \int_A \rho_t(x)\,\mathrm{d}x = \int_{\Phi_{0,t}^{-1}(A)} \rho_0(x_0)\,\mathrm{d}x_0.$$

Since we know the initial probability density ρ_0 and aim to determine the probability density ρ_t at time t, we need to express the integral in terms of the evolved state variable. Becuase $\Phi_{0,t}$ is a C^1 diffeomorphism, its inverse is also continuously differentiable. We can therefore rewrite the integral over the initial region $\Phi_{0,t}^{-1}(A)$ as an integral over the evolved region A by applying the change-of-variables formula with $x_0 = \Phi_{0,t}^{-1}(x)$:

$$\mathrm{d}x_0 = \left|\det D\Phi_{0,t}^{-1}(x)\right|\,\mathrm{d}x.$$

Therefore, we obtain

$$\mu_t(A) = \int_{\Phi_{0,t}^{-1}(A)} \rho_0(x_0)\,\mathrm{d}x_0 = \int_A \rho_0(\Phi_{0,t}^{-1}(x))\left|\det D\Phi_{0,t}^{-1}(x)\right|\,\mathrm{d}x.$$

The density at time t is

$$\rho_t(x) = \rho_0(\Phi_{0,t}^{-1}(x))\left|\det D\Phi_{0,t}^{-1}(x)\right|$$

for almost every $x \in \mathbb{R}^d$. Equivalently, using $x = \Phi_{0,t}(x_0)$ and the inverse Jacobian relation, we obtain

$$\rho_t(\Phi_{0,t}(x_0))\,|\det D\Phi_{0,t}(x_0)| = \rho_0(x_0)$$

for almost every $x_0 \in \mathbb{R}^d$.

By the Liouville–Jacobi formula (1.9), we have

$$J_{0,t}(x_0) = \exp\left(\int_0^t \nabla \cdot v_s(x_s)\,\mathrm{d}s\right).$$

Consequently, the density along trajectories satisfies

$$\rho_t(x_t) = \rho_0(x_0)\exp\left(-\int_0^t \nabla \cdot v_s(x_s)\,\mathrm{d}s\right).$$

This formula shows that spatial differentiability of ρ_0 is not required to establish differentiability along trajectories. Indeed, for almost every initial state x_0, the value $\rho_0(x_0)$ is finite and independent of time, while the time dependence of $\rho_t(x_t)$ is entirely determined by the exponential factor. Since the velocity field has jointly continuous spatial derivatives, this factor is continuously differentiable in time. Thus, for the representative of ρ_t defined by the

transport formula, $t \mapsto \rho_t(\Phi_{0,t}(x_0))$ is continuously differentiable for almost every x_0, even when ρ_0 is merely integrable.

If $\rho_0 \in L^1(\mathbb{R}^d)$ and the velocity field has jointly continuous spatial derivatives, then, provided the flow is a C^1 diffeomorphism, the transport formula defines a representative of ρ_t whose restriction to almost every trajectory is continuously differentiable in time. Indeed, along the trajectory $x_t = \Phi_{0,t}(x_0)$, the density transport formula gives

$$\widetilde{\rho}_t(x_t) J_{0,t}(x_0) = \rho_0(x_0).$$

For almost every initial state x_0, the right-hand side is finite and independent of time. Differentiating this conserved product and using the Liouville–Jacobi formula

$$\frac{\mathrm{d}}{\mathrm{d}t} J_{0,t}(x_0) = J_{0,t}(x_0) \nabla \cdot v_t(x_t),$$

we obtain

$$\begin{aligned} 0 &= \frac{\mathrm{d}}{\mathrm{d}t} \left[\widetilde{\rho}_t(x_t) J_{0,t}(x_0) \right] \\ &= J_{0,t}(x_0) \frac{\mathrm{d}}{\mathrm{d}t} \widetilde{\rho}_t(x_t) + \widetilde{\rho}_t(x_t) \frac{\mathrm{d}}{\mathrm{d}t} J_{0,t}(x_0) \\ &= J_{0,t}(x_0) \left[\frac{\mathrm{d}}{\mathrm{d}t} \widetilde{\rho}_t(x_t) + \widetilde{\rho}_t(x_t) \nabla \cdot v_t(x_t) \right]. \end{aligned}$$

Since $J_{0,t}(x_0) > 0$, division by the Jacobian yields

$$\frac{\mathrm{d}}{\mathrm{d}t} \widetilde{\rho}_t(x_t) = -\widetilde{\rho}_t(x_t) \nabla \cdot v_t(x_t).$$

Thus, no spatial differentiability of the initial density is required to establish differentiability of the transported density along almost every trajectory.

If the transport representative $\widetilde{\rho}(t, x)$ is C^1 in (t, x), the chain rule and the trajectory identity give

$$\partial_t \widetilde{\rho}(t, x_t) + v_t(x_t) \cdot \nabla \widetilde{\rho}(t, x_t) = -\widetilde{\rho}(t, x_t) \nabla \cdot v_t(x_t).$$

Equivalently,

$$\left[\partial_t \widetilde{\rho} + \nabla \cdot (\widetilde{\rho} v_t) \right](t, x_t) = 0.$$

For each fixed t, the expression

$$x_0 \mapsto \left[\partial_t \widetilde{\rho} + \nabla \cdot (\widetilde{\rho} v_t) \right](t, \Phi_{0,t}(x_0))$$

is continuous and vanishes for almost every x_0. A continuous function that vanishes almost everywhere with respect to Lebesgue measure must vanish everywhere. Since $\Phi_{0,t}$ is onto, the identity therefore holds at every state $x \in \mathbb{R}^d$. Hence,

$$\partial_t \widetilde{\rho} + \nabla \cdot (\widetilde{\rho} v_t) = 0.$$

□

Notice

The conserved quantity is probability mass: for every Borel region A,

$$\mu_t(\Phi_{0,t}(A)) = \mu_0(A).$$

The density itself generally changes along trajectories. Local volume expansion, measured by an increasing $J_{0,t}$, decreases the density carried by the same material element; contraction increases it. It is the product $\rho_t(\Phi_{0,t}(x_0))J_{0,t}(x_0)$ that remains constant. Thus, conservation of probability mass must be distinguished from constancy of density.

Statistical observables and weak probability transport. The statistical consequences of microscopic motion are measured through ensemble averages. A measurable observable $f : E \to \mathbb{R}$, with $E = \mathbb{R}^d$ here, represents a physical property such as a position component, energy, or an order parameter. Whenever f is integrable under μ_t, its expectation is

$$\langle f \rangle_t := \langle f, \mu_t \rangle = \int_E f(x)\, \mu_t(\,\mathrm{d}x) = \mathbb{E}[f(X_t)].$$

The trajectory specifies the microscopic state, whereas these expectations describe statistical properties of the ensemble. The pushforward identity connects the two descriptions:

$$\langle f, \mu_t \rangle = \int_{\mathbb{R}^d} f(\Phi_{0,t}(x_0))\, \mu_0(\,\mathrm{d}x_0).$$

For a C^1 observable, differentiation along each realization gives

$$\frac{\mathrm{d}}{\mathrm{d}t} f(X_t) = v_t(X_t) \cdot \nabla f(X_t) = (\mathcal{L}_t f)(X_t), \quad \mathcal{L}_t f = v_t \cdot \nabla f.$$

This is the deterministic generator introduced above, now acting on a measured physical property. On every compact time interval, assume that $f(\Phi_{0,t}(x_0))$ and its time derivative are dominated by μ_0-integrable functions of x_0. We may then differentiate the preceding integral to obtain

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle f, \mu_t \rangle = \langle \mathcal{L}_t f, \mu_t \rangle. \tag{1.16}$$

Thus, the rate of change of an ensemble average is the ensemble average of the microscopic rate of change of its observable. No differentiability of a density is needed for this identity.

Smooth observables as test functions. A physical observable need not be smooth or compactly supported. For a distributional formulation, however, it is useful to select $f \in C_c^\infty(\mathbb{R}^d)$. Such functions can represent smooth, spatially localized measurements and also serve as mathematical test functions. These roles are related, but this choice of test space

does not restrict all physical observables to that class.

For the following measure formulation, retain the given deterministic flow and assume the local flux integrability condition

$$\int_0^T \int_K |v_t(x)|\, \mu_t(\,\mathrm{d}x)\, \mathrm{d}t < \infty \quad \forall K \subset \mathbb{R}^d.$$

This holds, in particular, for a jointly continuous velocity field on $[0,T] \times \mathbb{R}^d$. The spatial differentiability and invertibility assumptions used for the density change of variables are not needed here.

Proposition 1.18 (Measure-valued continuity equation)**.** *The transported probability law $\mu_t = (\Phi_{0,t})_{\#}\mu_0$ satisfies*

$$\int_0^T \int_{\mathbb{R}^d} [\partial_t \psi(t,x) + v_t(x) \cdot \nabla_x \psi(t,x)]\, \mu_t(\,\mathrm{d}x)\, \mathrm{d}t = 0 \tag{1.17}$$

for every $\psi \in C_c^\infty((0,T) \times \mathbb{R}^d)$. Equivalently,

$$\partial_t \mu_t + \nabla \cdot (v_t \mu_t) = 0$$

in the distributional sense.

Proof. Fix $\psi \in C_c^\infty((0,T) \times \mathbb{R}^d)$ and define

$$F(t) := \int_{\mathbb{R}^d} \psi(t,x)\, \mu_t(\mathrm{d}x) = \int_{\mathbb{R}^d} \psi\big(t, \Phi_{0,t}(x)\big)\, \mu_0(\mathrm{d}x),$$

the second identity being the definition of the pushforward $\mu_t = (\Phi_{0,t})_{\#}\mu_0$.

Assume that, for μ_0-a.e. x, the trajectory $t \mapsto \Phi_{0,t}(x)$ is absolutely continuous with

$$\frac{\mathrm{d}}{\mathrm{d}t}\Phi_{0,t}(x) = v_t\big(\Phi_{0,t}(x)\big) \quad \text{for a.e. } t \in (0,T).$$

Then, for μ_0-a.e. x and a.e. t, the chain rule yields

$$\frac{\mathrm{d}}{\mathrm{d}t}\psi\big(t, \Phi_{0,t}(x)\big) = \partial_t \psi\big(t, \Phi_{0,t}(x)\big) + \nabla_x \psi\big(t, \Phi_{0,t}(x)\big) \cdot v_t\big(\Phi_{0,t}(x)\big).$$

Integrating in x and applying the pushforward formula once more,

$$\begin{aligned} F'(t) &= \int_{\mathbb{R}^d} \Big[\partial_t \psi\big(t, \Phi_{0,t}(x)\big) + \nabla_x \psi\big(t, \Phi_{0,t}(x)\big) \cdot v_t\big(\Phi_{0,t}(x)\big)\Big]\, \mu_0(\mathrm{d}x) \\ &= \int_{\mathbb{R}^d} \Big[\partial_t \psi(t,x) + v_t(x) \cdot \nabla_x \psi(t,x)\Big]\, \mu_t(\mathrm{d}x). \end{aligned} \tag{1.18}$$

Next, integrate (1.18) over $t \in [0,T]$. Since $\psi \in C_c^\infty((0,T)\times\mathbb{R}^d)$, it vanishes in a neighbourhood of the hyperplanes $t=0$ and $t=T$; hence

$$F(0) = \int_{\mathbb{R}^d} \psi(0,x)\,\mu_0(\mathrm{d}x) = 0, \quad F(T) = \int_{\mathbb{R}^d} \psi(T,x)\,\mu_T(\mathrm{d}x) = 0.$$

Consequently, by the fundamental theorem of calculus,

$$\begin{aligned} 0 &= F(T) - F(0) \\ &= \int_0^T F'(t)\,\mathrm{d}t = \int_0^T \int_{\mathbb{R}^d} \Big[\partial_t\psi(t,x) + v_t(x)\cdot\nabla_x\psi(t,x)\Big]\mu_t(\mathrm{d}x)\,\mathrm{d}t, \end{aligned}$$

which is (1.17). To deduce the distributional form, note that for every $\psi \in C_c^\infty((0,T)\times\mathbb{R}^d)$,

$$\Big\langle \partial_t\mu_t + \nabla\cdot(v_t\mu_t),\, \psi \Big\rangle = -\int_0^T \int_{\mathbb{R}^d} \Big[\partial_t\psi(t,x) + v_t(x)\cdot\nabla_x\psi(t,x)\Big]\mu_t(\mathrm{d}x)\,\mathrm{d}t = 0.$$

Hence $\partial_t\mu_t + \nabla\cdot(v_t\mu_t) = 0$ in $\mathcal{D}'((0,T)\times\mathbb{R}^d)$; conversely, this distributional identity clearly implies (1.17). This proves the assertion. □

Time-dependent observables and dynamical pictures. An observable may also depend explicitly on time:

$$f_t : E \to \mathbb{R}, \quad \langle f_t\rangle_t := \langle f_t, \mu_t\rangle = \int_E f_t(x)\,\mu_t(\,\mathrm{d}x), \quad E = \mathbb{R}^d.$$

At a fixed state x, the measured property can change because an external parameter or a measurement protocol changes. This explicit variation is distinct from the change in $f_t(X_t)$ caused by the motion of the state. It is also distinct from assigning dynamical time dependence to observables through a change of representation, as in Koopman evolution. We first keep the physical state coordinates fixed and account for both contributions.

Theorem 1.19 (Transport formula)**.**

Source exercise: Project 9, item 9. *Student Assessment Exercise* [Assessment] (adapted).

Let $F(t,x) = f_t(x)$ be C^1 in (t,x). On each compact time interval assume that $F(t,\Phi_{0,t}(x_0))$ and its time derivative are dominated by integrable functions of x_0 under μ_0. Then

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[F(t,X_t)] = \mathbb{E}[(\partial_t + v_t\cdot\nabla)F(t,X_t)],$$

or, equivalently,

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle f_t,\mu_t\rangle = \langle \partial_t f_t + \mathcal{L}_t f_t, \mu_t\rangle. \tag{1.19}$$

Proof. Since $X_t = \Phi_{0,t}(X_0)$ and $X_0 \sim \mu_0$, we have

$$\mathbb{E}[F(t, X_t)] = \int_{\mathbb{R}^d} F(t, \Phi_{0,t}(x_0))\, \mu_0(\mathrm{d}x_0).$$

For a fixed initial state x_0, the chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}t} F(t, \Phi_{0,t}(x_0)) = \partial_t F(t, \Phi_{0,t}(x_0)) + \nabla F(t, \Phi_{0,t}(x_0)) \cdot \frac{\mathrm{d}}{\mathrm{d}t} \Phi_{0,t}(x_0).$$

By the defining ODE of the flow,

$$\frac{\mathrm{d}}{\mathrm{d}t} \Phi_{0,t}(x_0) = v_t(\Phi_{0,t}(x_0)).$$

Consequently,

$$\frac{\mathrm{d}}{\mathrm{d}t} F(t, \Phi_{0,t}(x_0)) = \left[\partial_t F(t, x) + v_t(x) \cdot \nabla F(t, x)\right]_{x=\Phi_{0,t}(x_0)}.$$

By the assumed integrable domination, differentiation under the integral sign is justified. Hence,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[F(t, X_t)] &= \int_{\mathbb{R}^d} \frac{\mathrm{d}}{\mathrm{d}t} F(t, \Phi_{0,t}(x_0))\, \mu_0(\mathrm{d}x_0) \\ &= \int_{\mathbb{R}^d} \left[\partial_t F(t, x) + v_t(x) \cdot \nabla F(t, x)\right]_{x=\Phi_{0,t}(x_0)} \mu_0(\mathrm{d}x_0).\end{aligned}$$

Finally, using the pushforward relation

$$\mu_t = (\Phi_{0,t})_{\#}\mu_0,$$

we obtain, for every integrable function g,

$$\int_{\mathbb{R}^d} g(\Phi_{0,t}(x_0))\, \mu_0(\mathrm{d}x_0) = \int_{\mathbb{R}^d} g(x)\, \mu_t(\mathrm{d}x).$$

Applying this identity to $g(x) = \partial_t F(t, x) + v_t(x) \cdot \nabla F(t, x)$ yields

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[F(t, X_t)] &= \int_{\mathbb{R}^d} \left[\partial_t F(t, x) + v_t(x) \cdot \nabla F(t, x)\right] \mu_t(\mathrm{d}x) \\ &= \mathbb{E}\left[(\partial_t + v_t \cdot \nabla) F(t, X_t)\right].\end{aligned}$$

Since $F(t, x) = f_t(x)$ and $\mathcal{L}_t f_t = v_t \cdot \nabla f_t$, this is equivalent to

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle f_t, \mu_t\rangle = \langle \partial_t f_t + \mathcal{L}_t f_t, \mu_t\rangle.$$

□

The term $\partial_t f_t$ measures explicit variation at a fixed state, while $\mathcal{L}_t f_t$ measures variation

caused by the state's motion. Their sum is the material derivative,

$$\frac{Df_t}{Dt} := \partial_t f_t + v_t \cdot \nabla f_t.$$

When a smooth density exists and boundary terms vanish, this distinction can also be read from

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}^d} f_t \rho_t \, \mathrm{d}x = \int_{\mathbb{R}^d} \rho_t \partial_t f_t \, \mathrm{d}x + \int_{\mathbb{R}^d} f_t \partial_t \rho_t \, \mathrm{d}x.$$

Using (1.15), the second term equals $\int_{\mathbb{R}^d} \rho_t v_t \cdot \nabla f_t \, \mathrm{d}x$. It accounts for the changing distribution; the first term accounts for the changing observable. Compact spatial support, or sufficient decay and integrability, justifies this integration by parts.

1.2.3 Hamiltonian dynamics: a geometric application

Consider a conservative mechanical system with unit mass. Let $q, p \in \mathbb{R}^n$ denote the generalized position and momentum, respectively, and define the phase-space state $z = (q, p) \in \mathbb{R}^{2n}$. The Hamiltonian is

$$H(q,p) = \frac{1}{2}\|p\|^2 + \mathcal{V}(q), \quad \mathcal{V} \in C^2(\mathbb{R}^n).$$

Hamilton's equations yield

$$\dot{q}_t = \nabla_p H(q_t, p_t) = p_t, \quad \dot{p}_t = -\nabla_q H(q_t, p_t) = -\nabla \mathcal{V}(q_t).$$

Consequently, the dynamics in phase space are governed by the first-order ODE

$$\dot{z}_t = \begin{pmatrix} \dot{q}_t \\ \dot{p}_t \end{pmatrix} = \begin{pmatrix} p_t \\ -\nabla \mathcal{V}(q_t) \end{pmatrix}.$$

The coupling between position and momentum can be expressed more compactly by introducing the canonical matrix

$$\mathbb{J}_n = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

Writing

$$\nabla_z H(z) = \begin{pmatrix} \nabla_q H(q,p) \\ \nabla_p H(q,p) \end{pmatrix},$$

Hamilton's equations take the compact form

$$\dot{z}_t = \mathbb{J}_n \nabla_z H(z_t). \tag{1.20}$$

The matrix $\mathbb{J}_n$ is skew-symmetric and invertible, and defines the canonical symplectic form on phase space:

$$\omega(u, w) = u^\top \mathbb{J}_n w, \quad u, w \in \mathbb{R}^{2n}.$$

In particular, for $u = (u_q, u_p)$ and $w = (w_q, w_p)$,

$$\omega(u, w) = u_q^\top w_p - u_p^\top w_q.$$

This structure characterizes the geometric coupling between position and momentum in Hamiltonian mechanics. We next examine the conservation properties of the resulting flow, including energy conservation and preservation of the symplectic form.

Source exercise: Project 10, item 2–4. *Student Assessment Exercise* [Assessment] (adapted).

Theorem 1.20 (Energy conservation, symplecticity, and Liouville's theorem)**.** *Suppose $\mathcal{V} \in C^2(\mathbb{R}^n)$ and the Hamiltonian flow ϕ_t exists on an interval containing $t = 0$. Let $M_t(z) = D\phi_t(z)$ denote its variational matrix. Then the following properties hold.*

1. ***Energy conservation.*** *Along every trajectory,*

$$H(\phi_t(z)) = H(z).$$

2. ***Symplecticity.*** *The variational matrix preserves the canonical symplectic form:*

$$M_t(z)^\top \mathbb{J}_n M_t(z) = \mathbb{J}_n. \tag{1.21}$$

3. ***Liouville's theorem.*** *The Hamiltonian flow preserves phase-space volume:*

$$\det M_t(z) = 1.$$

Equivalently, the Hamiltonian vector field is divergence-free:

$$\nabla_z \cdot v(z) = 0.$$

Proof. Since $\mathbb{J}_n^\top = -\mathbb{J}_n$, the chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}t} H(z_t) = \nabla H(z_t)^\top \dot{z}_t = \nabla H(z_t)^\top \mathbb{J}_n \nabla H(z_t) = 0,$$

because $a^\top \mathbb{J}_n a = 0$ for every real vector a. The Hessian $\nabla^2 H(z_t)$ is symmetric. Differentiating $v(z) = \mathbb{J}_n \nabla H(z)$ yields $Dv(z_t) = \mathbb{J}_n \nabla^2 H(z_t)$. Thus the variational equation reads

$\dot{M}_t = \mathbb{J}_n \nabla^2 H(z_t) M_t$ with $M_0 = I_{2n}$. Differentiating $M_t^\top \mathbb{J}_n M_t$ and using $\mathbb{J}_n^\top = -\mathbb{J}_n$ and $\mathbb{J}_n^2 = -I_{2n}$ gives

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}(M_t^\top \mathbb{J}_n M_t) &= M_t^\top \left[(\mathbb{J}_n \nabla^2 H(z_t))^\top \mathbb{J}_n + \mathbb{J}_n(\mathbb{J}_n \nabla^2 H(z_t))\right] M_t \\ &= M_t^\top \left[\nabla^2 H(z_t)(-\mathbb{J}_n)\mathbb{J}_n + \mathbb{J}_n^2 \nabla^2 H(z_t)\right] M_t \\ &= M_t^\top (\nabla^2 H(z_t) - \nabla^2 H(z_t)) M_t = 0.\end{aligned}$$

The initial matrix is $M_0^\top \mathbb{J}_n M_0 = \mathbb{J}_n$, which proves (1.21). Taking determinants gives $(\det M_t)^2 = 1$. Continuity in time and $\det M_0 = 1$ fix $\det M_t = 1$. Independently, $\nabla \cdot v = \operatorname{tr}(\mathbb{J}_n \nabla^2 H) = 0$ since the trace of a product of an antisymmetric and a symmetric matrix vanishes. Liouville's formula therefore also yields $\det M_t = 1$. □

Proposition 1.21 (Two meanings of reversing a Hamiltonian trajectory)**.** *For any solution $z_t = (q_t, p_t)$ on $[0, T]$, the simple reversed curve $y_t = z_{T-t}$ satisfies $\dot{y}_t = -\mathbb{J}_n \nabla H(y_t)$, which is Hamilton's equation with Hamiltonian $-H$, not generally with H. For the mechanical Hamiltonian above define the involution $R(q, p) = (q, -p)$. Then*

$$\widetilde{z}_t = R z_{T-t} = (q_{T-t}, -p_{T-t})$$

solves the original *Hamilton equations for H.*

Proof. *Proof to be completed.* □

Example 1.22 (Harmonic oscillator: exact solution and a symplectic scheme)**.**

Problem

Take $\mathcal{V}(q) = \frac{1}{2}\omega^2 q^2$, $\omega > 0$, in one dimension. The system $\dot{q} = p$, $\dot{p} = -\omega^2 q$ is linear with $A = \left(\begin{smallmatrix} 0 & 1 \\ -\omega^2 & 0 \end{smallmatrix}\right)$.

Solution.

Exponentiating the oscillator matrix gives the exact trajectory below. Therefore $q_t = q_0 \cos(\omega t) + (p_0/\omega)\sin(\omega t)$, and $p_t = p_0 \cos(\omega t) - \omega q_0 \sin(\omega t)$. For comparison, velocity Verlet uses numerical states $(\widehat{q}_k, \widehat{p}_k)$ in a *kick–drift–kick* step:

$$\begin{aligned}\widehat{p}_{k+1/2} &= \widehat{p}_k - \tfrac{h}{2}\omega^2 \widehat{q}_k, \\ \widehat{q}_{k+1} &= \widehat{q}_k + h\widehat{p}_{k+1/2}, \\ \widehat{p}_{k+1} &= \widehat{p}_{k+1/2} - \tfrac{h}{2}\omega^2 \widehat{q}_{k+1}.\end{aligned} \tag{1.22}$$

Each kick or drift is the exact Hamiltonian flow of a purely potential or kinetic subsystem, respectively; their composition is symplectic. For the harmonic oscillator the resulting matrix is

$$V_h = \begin{pmatrix} 1 - \omega^2 h^2/2 & h \\ -h\omega^2(1 - \omega^2 h^2/4) & 1 - \omega^2 h^2/2 \end{pmatrix}, \quad \det V_h = 1.$$

In the stable regime $0 < h\omega < 2$, it preserves exactly the modified quadratic energy

$$\widetilde{H}_h(q,p) = \frac{1}{2}p^2 + \frac{1}{2}\omega^2(1-\omega^2h^2/4)q^2. \tag{1.23}$$

This explicitly explains the bounded $O(h^2)$ true-energy error on bounded oscillatory orbits. Explicit Euler, whose step matrix $I+hA$ has determinant $1+\omega^2h^2 > 1$, is not symplectic and generically produces secular energy growth; the numerical comparison follows in Table 1.2. ♦

Executed Hamiltonian experiment (Project 10): symplectic versus Euler evolution.

In the harmonic example take $\omega = 1$, $q_0 = 1$, $p_0 = 0$, step size $h = 0.1$ and $K = 1000$ steps, giving $T = 100$. The exact energy is $H_0 = 0.5$; write $\widehat{H}_k := H(\widehat{q}_k, \widehat{p}_k)$ for the computed true energy. Executing the updates of (1.22) and explicit Euler produces the results in Table 1.2. The bounds derived from (1.23) predict a maximum Verlet energy excursion of $\omega^4h^2q_0^2/8 = 0.00125$ for this particular initial state.

Table 1.2: Harmonic-oscillator energy comparison at $T = 100$ with $h = 0.1$.

Method	Energy at $T = 100$	Maximum $\lvert\widehat{H}_k - H_0\rvert$
Exact evolution	0.500000	0
Explicit Euler	10479.577819	10479.077819
Velocity Verlet	0.499724	0.001250

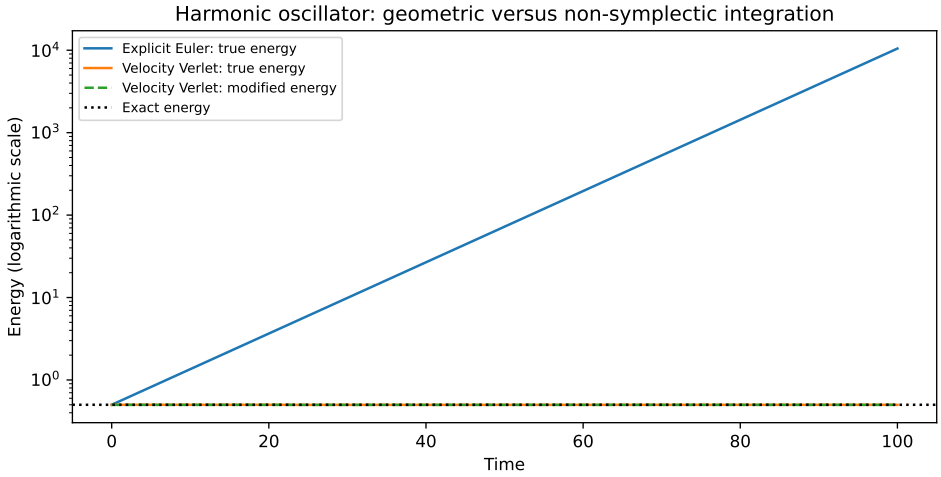


Figure 1.6: True and modified energies in the numerical Hamiltonian benchmark. The y-axis is logarithmic; the modified Verlet energy is constant up to roundoff, while Euler introduces an expanding phase-space map and secular energy growth.

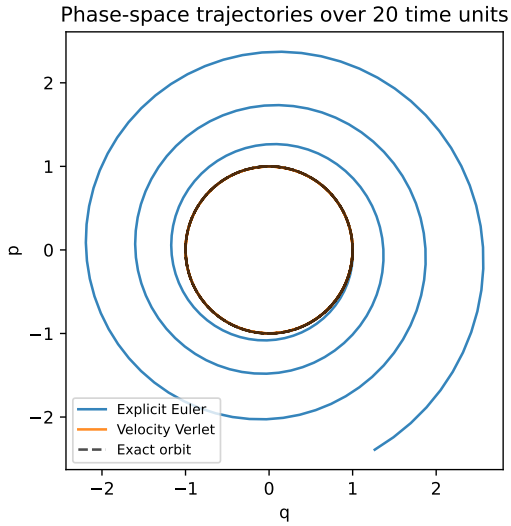


Figure 1.7: Exact, Euler, and Verlet trajectories in (q, p) over $0 \leq t \leq 20$. The differences arise from their discrete geometric structures, not from the continuous Hamiltonian system.

1.2.4 Entropy, variational principles, and free energy

1.2.4.1 Entropy and score under transport

For a smooth, strictly positive probability density ρ on $V = \mathbb{R}^d$, define its differential entropy by

$$S(\rho) = -\int_{\mathbb{R}^d} \rho(x) \log \rho(x) \, \mathrm{d}x,$$

provided that $\int_{\mathbb{R}^d} |\rho(x) \log \rho(x)| \, \mathrm{d}x < \infty$.

Theorem 1.23 (Evolution of differential entropy)**.**

Source exercise: Project 9, item 10. *Student Assessment Exercise* [Assessment] (adapted).

Let ρ_t be a strictly positive probability density that solves the continuity equation

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0$$

classically. Assume that differentiation under the integral is justified, all integrals below are finite, and the boundary terms arising from integration by parts vanish.
Define the differential entropy and the Stein score function by

$$S(\rho_t) = -\int_{\mathbb{R}^d} \rho_t(x) \log \rho_t(x) \, \mathrm{d}x, \quad s_t(x) := \nabla \log \rho_t(x).$$

Then

$$\frac{\mathrm{d}}{\mathrm{d}t}S(\rho_t) = \int_{\mathbb{R}^d} \rho_t(x)\nabla\cdot v_t(x)\,\mathrm{d}x = -\int_{\mathbb{R}^d} \rho_t(x)v_t(x)\cdot\nabla\log\rho_t(x)\,\mathrm{d}x$$
$$= \mathbb{E}[\nabla\cdot v_t(X_t)] = -\mathbb{E}[v_t(X_t)\cdot s_t(X_t)],$$

where X_t is a random variable with probability density ρ_t.

Proof. By the definition of differential entropy,

$$S(\rho_t) = -\int_{\mathbb{R}^d} \rho_t(x)\log\rho_t(x)\,\mathrm{d}x.$$

Differentiating with respect to time and applying the product rule, we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}S(\rho_t) = -\int_{\mathbb{R}^d} \big(1+\log\rho_t(x)\big)\partial_t\rho_t(x)\,\mathrm{d}x.$$

Using the continuity equation $\partial_t\rho_t + \nabla\cdot(\rho_t v_t) = 0$, we can rewrite this as

$$\frac{\mathrm{d}}{\mathrm{d}t}S(\rho_t) = \int_{\mathbb{R}^d} \big(1+\log\rho_t(x)\big)\nabla\cdot(\rho_t(x)v_t(x))\,\mathrm{d}x.$$

Integration by parts, together with the assumed vanishing of boundary terms, gives

$$\frac{\mathrm{d}}{\mathrm{d}t}S(\rho_t) = -\int_{\mathbb{R}^d} \rho_t(x)v_t(x)\cdot\nabla\big(1+\log\rho_t(x)\big)\,\mathrm{d}x$$
$$= -\int_{\mathbb{R}^d} \rho_t(x)v_t(x)\cdot\nabla\log\rho_t(x)\,\mathrm{d}x.$$

Since ρ_t is strictly positive, we have

$$\rho_t(x)\nabla\log\rho_t(x) = \nabla\rho_t(x).$$

Consequently, another integration by parts yields

$$\frac{\mathrm{d}}{\mathrm{d}t}S(\rho_t) = -\int_{\mathbb{R}^d} v_t(x)\cdot\nabla\rho_t(x)\,\mathrm{d}x = \int_{\mathbb{R}^d} \rho_t(x)\nabla\cdot v_t(x)\,\mathrm{d}x.$$

Finally, since X_t has probability density ρ_t, the expectation identity

$$\mathbb{E}[g(X_t)] = \int_{\mathbb{R}^d} g(x)\rho_t(x)\,\mathrm{d}x$$

implies that

$$\frac{\mathrm{d}}{\mathrm{d}t}S(\rho_t) = \mathbb{E}[\nabla\cdot v_t(X_t)] = -\mathbb{E}[v_t(X_t)\cdot s_t(X_t)],$$

where $s_t(x) = \nabla\log\rho_t(x)$ is the Stein score function. □

The entropy rate is therefore the ensemble average of local volume expansion: positive mean divergence increases differential entropy, negative mean divergence decreases it, and

volume-preserving flows leave it constant. Deterministic transport has no universal entropy increase law.

Proposition 1.24 (Lagrangian score dynamics)**.**

Source exercise: Project 9, item 11. *Student Assessment Exercise* [Assessment] (adapted).

Assume v_t is twice continuously differentiable in space, the positive density has $\ell_t = \log \rho_t$ of class C^1 in time and C^2 in space, and the mixed derivatives used below are continuous. Let $X_t = \Phi_{0,t}(x)$ satisfy

$$\frac{\mathrm{d}}{\mathrm{d}t} X_t = v_t(X_t), \quad X_0 = x.$$

Set

$$s_t^\rho = \nabla \log \rho_t, \quad G_t(x) = s_t^\rho(\Phi_{0,t}(x)).$$

Then, along each trajectory,

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} s_t^\rho(X_t) &= \frac{\mathrm{d}}{\mathrm{d}t} G_t(x) \\ &= -(Dv_t(X_t))^\top G_t(x) - \nabla(\nabla \cdot v_t)(X_t), \end{aligned} \tag{1.24}$$

with the initial condition

$$G_0(x) = \nabla \log \rho_0(x).$$

Here Dv_t denotes the Jacobian matrix of v_t, and $\nabla \cdot v_t = \mathrm{tr}(Dv_t)$.

Proof. *Proof to be completed.* □

Proposition 1.25 (Stein's integration-by-parts identity)**.**

Source exercise: Project 11, item 8. *Student Assessment Exercise* [Assessment] (adapted).

Let ρ be positive and C^1 and let $s^\rho = \nabla \log \rho$. If $f \in C^1$, $\mathbb{E}_\rho[\|\nabla f\| + \|f s^\rho\|] < \infty$, and the boundary terms in the coordinate integrations by parts vanish, then

$$\mathbb{E}_\rho[\nabla f(X) + f(X) s^\rho(X)] = 0.$$

For $X \sim \mathcal{N}(0, I_d)$ this reads $\mathbb{E}[X f(X)] = \mathbb{E}[\nabla f(X)]$.

Proof. *Proof to be completed.* □

1.2.4.2 KL divergence and its basic identities

In the information-theoretic results below, $(E, \mathcal{S})$ denotes an explicitly chosen measurable space, which may be V with its Borel structure but need not be a vector space. Write $B_b(E)$ for bounded $\mathcal{S}$-measurable real functions and $\mathcal{P}(E)$ for probability measures on $(E, \mathcal{S})$; when a topology is specified, its Borel sigma-algebra is understood. For proba-

bilities $\mu \ll \nu$ define

$$D_{\mathrm{KL}}(\mu\|\nu) = \int \log\left(\frac{\mathrm{d}\mu}{\mathrm{d}\nu}\right) \mathrm{d}\mu,$$

and set it to $+\infty$ when $\mu \not\ll \nu$. The convention $0 \log 0 = 0$ is used. Its negative part is integrable, since $-r \log r \le 1/e$ for $0 \le r \le 1$ and ν is a probability law; thus the extended value is well defined. For densities we use $D_{\mathrm{KL}}(\rho\|\eta)$ only as shorthand for $D_{\mathrm{KL}}(\rho(x)\,\mathrm{d}x\|\eta(x)\,\mathrm{d}x)$.

Theorem 1.26 (Gibbs' inequality)**.**

Source exercise: Project 11, item 1. *Student Assessment Exercise* [Assessment] (adapted).

For probability measures μ, ν, $D_{\mathrm{KL}}(\mu\|\nu) \ge 0$, with equality if and only if $\mu = \nu$.

Proof.

- If $\mu \not\ll \nu$, then by definition $D_{\mathrm{KL}}(\mu\|\nu) = +\infty$, and the inequality is immediate.
- Otherwise, since $\mu \ll \nu$, the Radon–Nikodym theorem gives the density

$$r = \frac{\mathrm{d}\mu}{\mathrm{d}\nu}, \quad \int_E r \,\mathrm{d}\nu = 1.$$

By the change-of-measure formula,

$$D_{\mathrm{KL}}(\mu\|\nu) = \int_E r \log r \,\mathrm{d}\nu,$$

with the convention $0 \log 0 = 0$. Consider the function $f(r) = r \log r,\ r \ge 0$. Since

$$f''(r) = \frac{1}{r} > 0, \quad r > 0,$$

f is strictly convex on $[0, \infty)$, where $f(0) = 0$ is defined by continuity. Applying Jensen's inequality with respect to the probability measure ν, we obtain

$$D_{\mathrm{KL}}(\mu\|\nu) = \int_E f(r) \,\mathrm{d}\nu \ge f\left(\int_E r \,\mathrm{d}\nu\right) = f(1) = 0.$$

Therefore, $D_{\mathrm{KL}}(\mu\|\nu) \ge 0$.

Moreover, since f is strictly convex, equality in Jensen's inequality holds if and only if r is constant ν-almost everywhere. Together with $\int_E r \,\mathrm{d}\nu = 1$, this gives

$$r = 1, \quad \nu\text{-a.e.}$$

Consequently, for every $A \in \mathcal{S}$,

$$\mu(A) = \int_A r \,\mathrm{d}\nu = \int_A 1 \,\mathrm{d}\nu = \nu(A),$$

and hence $\mu = \nu$. Conversely, if $\mu = \nu$, then $\mathrm{d}\mu/\,\mathrm{d}\nu = 1$ ν-almost everywhere, so

$$D_{\mathrm{KL}}(\mu\|\nu) = \int_E \log 1 \; \mathrm{d}\mu = 0.$$

Thus,

$$D_{\mathrm{KL}}(\mu\|\nu) \geq 0, \quad D_{\mathrm{KL}}(\mu\|\nu) = 0 \iff \mu = \nu.$$

□

Proposition 1.27 (Relative entropy of two transported densities)**.**

Source exercise: Project 9, item 12. *Student Assessment Exercise* [Assessment] (adapted).

Let positive smooth ρ_t, η_t solve continuity equations with velocities v_t, w_t, respectively, and suppose all boundary terms vanish. Set $s_t^\rho = \nabla \log \rho_t$ and $s_t^\eta = \nabla \log \eta_t$. Then

$$\frac{\mathrm{d}}{\mathrm{d}t} D_{\mathrm{KL}}(\rho_t\|\eta_t) = \int \rho_t (v_t - w_t) \cdot (s_t^\rho - s_t^\eta)\, \mathrm{d}x. \tag{1.25}$$

Proof. By the definition of the KL divergence,

$$D_{\mathrm{KL}}(\rho_t\|\eta_t) = \int_{\mathbb{R}^d} \rho_t \log \frac{\rho_t}{\eta_t}\, \mathrm{d}x.$$

Differentiating under the integral and applying the product and chain rules give

$$\frac{\mathrm{d}}{\mathrm{d}t} D_{\mathrm{KL}}(\rho_t\|\eta_t) = \int_{\mathbb{R}^d} \left[\partial_t \rho_t \left(\log \frac{\rho_t}{\eta_t} + 1 \right) - \frac{\rho_t}{\eta_t} \partial_t \eta_t \right] \mathrm{d}x.$$

Since ρ_t and η_t satisfy their respective continuity equations,

$$\partial_t \rho_t = -\nabla \cdot (\rho_t v_t), \qquad \partial_t \eta_t = -\nabla \cdot (\eta_t w_t),$$

we obtain

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} D_{\mathrm{KL}}(\rho_t\|\eta_t) = & - \int_{\mathbb{R}^d} \nabla \cdot (\rho_t v_t) \left(\log \frac{\rho_t}{\eta_t} + 1 \right) \mathrm{d}x \\ & + \int_{\mathbb{R}^d} \frac{\rho_t}{\eta_t} \nabla \cdot (\eta_t w_t)\, \mathrm{d}x. \end{aligned}$$

We evaluate the two terms separately. By integration by parts and the vanishing of boundary terms, the first term becomes

$$\begin{aligned} & - \int_{\mathbb{R}^d} \nabla \cdot (\rho_t v_t) \left(\log \frac{\rho_t}{\eta_t} + 1 \right) \mathrm{d}x \\ & = \int_{\mathbb{R}^d} \rho_t v_t \cdot \nabla \log \frac{\rho_t}{\eta_t}\, \mathrm{d}x \\ & = \int_{\mathbb{R}^d} \rho_t v_t \cdot (s_t^\rho - s_t^\eta)\, \mathrm{d}x. \end{aligned}$$

Similarly, integration by parts gives

$$\begin{aligned}
&\int_{\mathbb{R}^d} \frac{\rho_t}{\eta_t} \nabla \cdot (\eta_t w_t)\, \mathrm{d}x \\
&\quad = -\int_{\mathbb{R}^d} \eta_t w_t \cdot \nabla \left(\frac{\rho_t}{\eta_t}\right) \mathrm{d}x \\
&\quad = -\int_{\mathbb{R}^d} \rho_t w_t \cdot \nabla \log \frac{\rho_t}{\eta_t}\, \mathrm{d}x \\
&\quad = -\int_{\mathbb{R}^d} \rho_t w_t \cdot (s_t^\rho - s_t^\eta)\, \mathrm{d}x,
\end{aligned}$$

where we have used

$$\nabla \left(\frac{\rho_t}{\eta_t}\right) = \frac{\rho_t}{\eta_t} \nabla \log \frac{\rho_t}{\eta_t}.$$

Combining the two terms yields

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} D_{\mathrm{KL}}(\rho_t \| \eta_t) &= \int_{\mathbb{R}^d} \rho_t v_t \cdot (s_t^\rho - s_t^\eta)\, \mathrm{d}x \\
&\quad - \int_{\mathbb{R}^d} \rho_t w_t \cdot (s_t^\rho - s_t^\eta)\, \mathrm{d}x \\
&= \int_{\mathbb{R}^d} \rho_t (v_t - w_t) \cdot (s_t^\rho - s_t^\eta)\, \mathrm{d}x.
\end{aligned}$$

This completes the proof. □

Theorem 1.28 (KL disintegration and data processing)**.**

Source exercise: Project 11, item 7. *Student Assessment Exercise* [Assessment] (adapted).

Let E, F be standard Borel spaces, let μ, ν be probability measures on $E \times F$, and consider disintegrations

$$\begin{aligned}
\mu(\mathrm{d}x, \mathrm{d}y) &= \mu_X(\mathrm{d}x)\mu(\mathrm{d}y \mid x), \\
\nu(\mathrm{d}x, \mathrm{d}y) &= \nu_X(\mathrm{d}x)\nu(\mathrm{d}y \mid x).
\end{aligned}$$

Under $\mu \ll \nu$,

$$D_{\mathrm{KL}}(\mu\|\nu) = D_{\mathrm{KL}}(\mu_X\|\nu_X) + \int_E D_{\mathrm{KL}}\big(\mu(\cdot \mid x)\|\nu(\cdot \mid x)\big)\mu_X(\mathrm{d}x).$$

In particular, projection cannot increase KL divergence. For product laws,

$$D_{\mathrm{KL}}(\mu_X \otimes \mu_Y\|\nu_X \otimes \nu_Y) = D_{\mathrm{KL}}(\mu_X\|\nu_X) + D_{\mathrm{KL}}(\mu_Y\|\nu_Y),$$

with the same extended-real convention.

Proof. *Proof to be completed.* □

1.2.4.3 Convex geometry of relative entropy

The positivity of KL divergence does not make it a metric. Its direction specifies which law supplies the integration measure, and changing that law usually changes the penalty. This is important when an approximating model is trained by minimizing either $D_{\mathrm{KL}}(P\|Q)$ or $D_{\mathrm{KL}}(Q\|P)$.

Proposition 1.29 (Relative entropy is generally asymmetric)**.**

Source exercise: Project 11, item 2. *Student Assessment Exercise* [Assessment] (adapted).

Let $P = \mathcal{N}(0,1)$ and $Q = \mathcal{N}(0,4)$ on $\mathbb{R}$. Then

$$D_{\mathrm{KL}}(P\|Q) = \log 2 - \frac{3}{8}, \quad D_{\mathrm{KL}}(Q\|P) = \frac{3}{2} - \log 2.$$

In particular, the two relative entropies are different, although both are finite and both vanish only when their arguments agree.

Proof. *Proof to be completed.* □

Example 1.30 (Truncated Gaussian: an exactly solvable KL optimization)**.**

Source exercise: Project 11, item 3. *Student Assessment Exercise* [Assessment] (adapted).

Problem

Let $Z \sim \mathcal{N}(0,1)$ with density $\varphi(x) = (2\pi)^{-1/2} e^{-x^2/2}$, and put

$$\bar{\Phi}(b) := \int_b^\infty \varphi(x)\,\mathrm{d}x > 0, \quad \mu_b := \mathrm{Law}(Z \mid Z > b), \quad b \in \mathbb{R}.$$

Write $\nu = \mathcal{N}(0,1)$ and determine $D_{\mathrm{KL}}(\mu_b\|\nu)$.

To be completed.

Problem — Quadratic energy and constrained optimization

Let $\mathcal{V}(x) = \alpha x^2/2$ with $\alpha > 0$, and consider

$$\mathcal{J}_\alpha(b) := D_{\mathrm{KL}}(\mu_b\|\nu) + \int \mathcal{V}(x)\,\mu_b(\mathrm{d}x), \quad b \geq 0.$$

To be completed.

Theorem 1.31 (Joint and separate convexity of relative entropy)**.**

Source exercise: Project 11, item 4. *Student Assessment Exercise* [Assessment] (adapted).

Let $(E, \mathcal{S})$ be a measurable space, and let $\mathcal{P}(E)$ denote the space of probability measures on it. The relative entropy has the following convexity properties:

1. ***Joint convexity.*** *$D_{\mathrm{KL}}(P\|Q)$ is jointly convex in its two probability measure arguments $P, Q \in \mathcal{P}(E)$.*

2. ***Separate convexity.*** *For fixed $Q \in \mathcal{P}(E)$, $D_{\mathrm{KL}}(P\|Q)$ is convex in P. Similarly, for fixed $P \in \mathcal{P}(E)$, $D_{\mathrm{KL}}(P\|Q)$ is convex in Q.*

Proof. For $\vartheta = 0$ or $\vartheta = 1$, the inequality is an identity. We therefore consider $0 < \vartheta < 1$. If either $D_{\mathrm{KL}}(P_1\|Q_1)$ or $D_{\mathrm{KL}}(P_0\|Q_0)$ is infinite, the inequality is immediate. Hence, we may assume that both are finite, which implies $P_i \ll Q_i$ for $i = 0, 1$.

First, choose a common dominating measure

$$\lambda = P_0 + P_1 + Q_0 + Q_1.$$

By the Radon–Nikodym theorem, there exist nonnegative measurable densities

$$p_i = \frac{\mathrm{d}P_i}{\mathrm{d}\lambda}, \quad q_i = \frac{\mathrm{d}Q_i}{\mathrm{d}\lambda}, \quad i = 0, 1.$$

Consequently, the densities of the convex combinations P_ϑ and Q_ϑ are

$$p_\vartheta = \vartheta p_1 + (1-\vartheta)p_0,$$
$$q_\vartheta = \vartheta q_1 + (1-\vartheta)q_0.$$

Since $P_i \ll Q_i$, we have $p_i = 0$ λ-almost everywhere on the set where $q_i = 0$. Thus, whenever $q_\vartheta > 0$, the ratio of the mixture densities can be written as

$$\frac{p_\vartheta}{q_\vartheta} = \frac{\vartheta q_1}{q_\vartheta}\frac{p_1}{q_1} + \frac{(1-\vartheta)q_0}{q_\vartheta}\frac{p_0}{q_0}.$$

Here, terms with zero weights are omitted. The two weights are nonnegative and satisfy

$$\frac{\vartheta q_1}{q_\vartheta} + \frac{(1-\vartheta)q_0}{q_\vartheta} = 1.$$

Now consider the function

$$f(r) = r\log r, \quad r \geq 0,$$

with $f(0) = 0$ by continuity. Since

$$f''(r) = \frac{1}{r} > 0, \quad r > 0,$$

the function f is convex on $[0, \infty)$.

Applying Jensen's inequality with the weights above gives

$$q_\vartheta f\left(\frac{p_\vartheta}{q_\vartheta}\right) \leq \vartheta q_1 f\left(\frac{p_1}{q_1}\right) + (1-\vartheta)q_0 f\left(\frac{p_0}{q_0}\right).$$

Integrating both sides with respect to λ and using the density representation of relative entropy, we obtain

$$\begin{aligned} D_{\mathrm{KL}}(P_\vartheta \| Q_\vartheta) &= \int_E q_\vartheta f\left(\frac{p_\vartheta}{q_\vartheta}\right) \mathrm{d}\lambda \\ &\leq \vartheta \int_E q_1 f\left(\frac{p_1}{q_1}\right) \mathrm{d}\lambda + (1-\vartheta) \int_E q_0 f\left(\frac{p_0}{q_0}\right) \mathrm{d}\lambda \\ &= \vartheta D_{\mathrm{KL}}(P_1 \| Q_1) + (1-\vartheta) D_{\mathrm{KL}}(P_0 \| Q_0). \end{aligned}$$

Thus, relative entropy is jointly convex.

Finally, fixing $Q_0 = Q_1 = Q$ gives convexity in the first argument. Similarly, fixing $P_0 = P_1 = P$ gives convexity in the second argument. Hence, relative entropy is also separately convex. □

1.2.4.4 Gibbs tilts and convex duality

Theorem 1.32 (Gibbs variational identity)**.**

Source exercise: Project 11, item 5. *Student Assessment Exercise* [Assessment] (adapted).

Let $(E, \mathcal{S}, \nu)$ be a probability space, and let $U : E \to \mathbb{R}$ be a bounded measurable energy function. Define the normalization constant and the exponentially tilted probability measure by

$$Z_U = \int_E e^{-U} \, \mathrm{d}\nu, \quad \mathrm{d}\nu_U = Z_U^{-1} e^{-U} \, \mathrm{d}\nu.$$

Then, for every $\mu \in \mathcal{P}(E)$,

$$D_{\mathrm{KL}}(\mu \| \nu) + \int_E U \, \mathrm{d}\mu = D_{\mathrm{KL}}(\mu \| \nu_U) - \log Z_U,$$

with the extended-real convention for relative entropy. Consequently,

$$\inf_{\mu \in \mathcal{P}(E)} \left\{ D_{\mathrm{KL}}(\mu \| \nu) + \int_E U \, \mathrm{d}\mu \right\} = -\log Z_U,$$

and the infimum is attained uniquely at $\mu^* = \nu_U$.

Proof. Since U is bounded, the normalization constant satisfies $0 < Z_U < \infty$. By the definition of the exponentially tilted measure,

$$\frac{\mathrm{d}\nu_U}{\mathrm{d}\nu} = \frac{e^{-U}}{Z_U}.$$

Moreover, since e^{-U} is strictly positive, ν_U and ν are mutually absolutely continuous. Consequently, $\mu \ll \nu$ if and only if $\mu \ll \nu_U$.

If $\mu \not\ll \nu$, then both $D_{\mathrm{KL}}(\mu \| \nu)$ and $D_{\mathrm{KL}}(\mu \| \nu_U)$ are infinite. Since U is bounded, the identity holds in the extended-real sense. We therefore consider the case $\mu \ll \nu$.

By the chain rule for Radon–Nikodym derivatives,

$$\frac{\mathrm{d}\mu}{\mathrm{d}\nu_U} = \frac{\mathrm{d}\mu}{\mathrm{d}\nu}\frac{\mathrm{d}\nu}{\mathrm{d}\nu_U} = Z_U e^U \frac{\mathrm{d}\mu}{\mathrm{d}\nu}.$$

Taking logarithms gives

$$\log \frac{\mathrm{d}\mu}{\mathrm{d}\nu_U} = \log \frac{\mathrm{d}\mu}{\mathrm{d}\nu} + U + \log Z_U, \quad \mu\text{-a.e.}$$

Integrating both sides with respect to μ, we obtain

$$\begin{aligned} D_{\mathrm{KL}}(\mu\|\nu_U) &= D_{\mathrm{KL}}(\mu\|\nu) + \int_E U \ \mathrm{d}\mu + \log Z_U \int_E \mathrm{d}\mu \\ &= D_{\mathrm{KL}}(\mu\|\nu) + \int_E U \ \mathrm{d}\mu + \log Z_U, \end{aligned}$$

where we used $\mu(E) = 1$. Since U is bounded, this identity also holds when the relative entropies are infinite. Rearranging, we obtain the Gibbs variational identity

$$D_{\mathrm{KL}}(\mu\|\nu) + \int_E U \ \mathrm{d}\mu = D_{\mathrm{KL}}(\mu\|\nu_U) - \log Z_U.$$

By the nonnegativity of relative entropy, $D_{\mathrm{KL}}(\mu\|\nu_U) \geq 0$. Therefore,

$$D_{\mathrm{KL}}(\mu\|\nu) + \int_E U \ \mathrm{d}\mu \geq -\log Z_U.$$

Equality holds if and only if $D_{\mathrm{KL}}(\mu\|\nu_U) = 0$, which is equivalent to $\mu = \nu_U$. Since ν_U is itself a probability measure, the lower bound is attained uniquely at $\mu^* = \nu_U$. Hence,

$$\inf_{\mu\in\mathcal{P}(E)} \left\{ D_{\mathrm{KL}}(\mu\|\nu) + \int_E U \ \mathrm{d}\mu \right\} = -\log Z_U.$$

□

Theorem 1.33 (Donsker–Varadhan representation of relative entropy)**.**

Source exercise: Project 11, item 6. *Student Assessment Exercise* [Assessment] (adapted).

Let μ, ν be probability measures on a measurable space $(E, \mathcal{S})$, and let $B_b(E)$ denote the space of bounded measurable real-valued functions on E. Then the relative entropy admits the variational representation

$$D_{\mathrm{KL}}(\mu\|\nu) = \sup_{g\in B_b(E)} \left\{ \int_E g \ \mathrm{d}\mu - \log \int_E e^g \ \mathrm{d}\nu \right\}. \tag{1.26}$$

The supremum is attained if and only if μ and ν are equivalent and $\log(\mathrm{d}\mu/\mathrm{d}\nu) \in$

$L^\infty(\nu)$. In this case, all maximizers are given by

$$g_* = \log \frac{\mathrm{d}\mu}{\mathrm{d}\nu} + c, \quad c \in \mathbb{R},$$

up to ν-almost-everywhere equality.

Proof. *Proof to be completed.* □

Corollary 1.34 (Full Fenchel–Legendre duality for the log-partition functional)**.** *Fix a probability measure Q on $(E, \mathcal{S})$ and define, for $g \in B_b(E)$,*

$$\Lambda_Q(g) := \log \int_E e^g \, \mathrm{d}Q.$$

Then the following properties hold:

1. ***Convexity.*** *The functional Λ_Q is convex in g, and $P \mapsto D_{\mathrm{KL}}(P\|Q)$ is convex in P.*
2. ***First conjugacy relation.*** *For every $P \in \mathcal{P}(E)$,*

$$D_{\mathrm{KL}}(P\|Q) = \sup_{g \in B_b(E)} \left\{ \int_E g \, \mathrm{d}P - \Lambda_Q(g) \right\}.$$

3. ***Second conjugacy relation.*** *For every $g \in B_b(E)$,*

$$\Lambda_Q(g) = \sup_{P \in \mathcal{P}(E)} \left\{ \int_E g \, \mathrm{d}P - D_{\mathrm{KL}}(P\|Q) \right\}.$$

The supremum is attained uniquely at

$$P_g(\mathrm{d}x) = \frac{e^{g(x)}}{\int_E e^g \, \mathrm{d}Q} \, Q(\mathrm{d}x).$$

For the full convex-conjugate formulation, let $\mathcal{M}(E)$ denote the vector space of finite signed measures on $(E, \mathcal{S})$, and extend the relative entropy functional by

$$\mathcal{F}_Q(P) := \begin{cases} D_{\mathrm{KL}}(P\|Q), & P \in \mathcal{P}(E), \\ +\infty, & P \notin \mathcal{P}(E). \end{cases}$$

With respect to the function–measure pairing

$$\langle g, P \rangle := \int_E g \, \mathrm{d}P,$$

the two variational identities express the mutual Fenchel–Legendre conjugacy

$$\mathcal{F}_Q^* = \Lambda_Q, \quad \Lambda_Q^* = \mathcal{F}_Q.$$

Proof. *Proof to be completed.* □

f-divergences and density-ratio duality. The following definitions support the retained statements of the pending f-divergence and duality problems. Their extended derivations are deferred; the Gibbs and Donsker–Varadhan results above remain fully developed.

Definition 1.35 (f-divergence and convex conjugate)

Let $f : (0, \infty) \to \mathbb{R}$ be convex with $f(1) = 0$. For probability measures $P \ll Q$, with Radon–Nikodym derivative $r = \mathrm{d}P/\,\mathrm{d}Q > 0$ Q-almost everywhere, define

$$D_f(P\|Q) := \int f(r)\,\mathrm{d}Q. \tag{1.27}$$

The *convex conjugate* (Fenchel–Legendre transform) of f is

$$f^*(u) := \sup_{t>0}\{ut - f(t)\}.$$

The Fenchel–Young inequality reads

$$f(t) + f^*(u) \geq ut, \tag{1.28}$$

with equality when $u \in \partial f(t)$, where $\partial f(t)$ denotes the set of subgradients at t. For differentiable f, this condition becomes $u = f'(t)$.

The pending variational proof below starts from (1.28). The assumption $r > 0$ permits the elementary formulas below without introducing boundary conventions. For general probability measures, the extended f-divergence additionally requires the lower-semicontinuous extension at $r = 0$ and an appropriate singular-mass term when $P \not\ll Q$; it is *not* mathematically correct to apply (1.27) unchanged in such cases.

Theorem 1.36 (Joint convexity and the variational form of f-divergence)**.**

Source exercise: Project 11, item 9. *Student Assessment Exercise* [Assessment] (adapted).

Let P and Q have strictly positive densities p, q with respect to a common σ-finite measure m. Suppose f is proper, convex, continuously differentiable on $(0, \infty)$ with $f(1) = 0$, and that the displayed integrals exist. Then D_f is jointly convex as a function of (P, Q). Moreover, for every admissible measurable discriminator T whose objective is well defined (no $\infty - \infty$ subtraction),

$$\int T\,\mathrm{d}P - \int f^*(T)\,\mathrm{d}Q \leq D_f(P\|Q).$$

If $T_*(x) = f'(p(x)/q(x))$ is P-integrable and $f^*(T_*)$ is Q-integrable, equality is attained, and therefore

$$\boxed{D_f(P\|Q) = \sup_T \left\{\mathbb{E}_P[T(X)] - \mathbb{E}_Q[f^*(T(X))]\right\}.}$$

The supremum is over the measurable functions for which the objective is well defined, including T_*. If the candidate fails the stated integrability conditions, a truncation or approximation argument is needed; the attained formula does not cover that case.

Proof. *Proof to be completed.* □

Example 1.37 (KL as an f-divergence)**.**

Problem

Take $f(r) = r \log r$.

To be completed.

Theorem 1.38 (An exponential Fenchel dual for KL)**.**

Source exercise: Project 11, item 10. *Student Assessment Exercise* [Assessment] (adapted).

For probability measures P, Q,

$$\boxed{D_{\mathrm{KL}}(P\|Q) = 1 + \sup_{u \in B_b(E)} \{\mathbb{E}_P[u] - \mathbb{E}_Q[e^u]\}.} \tag{1.29}$$

The identity includes infinite entropy. A bounded maximizer exists precisely when $P \sim Q$ and $\log(\mathrm{d}P/\mathrm{d}Q)$ is bounded; it is then $u_* = \log(\mathrm{d}P/\mathrm{d}Q)$, uniquely Q-almost everywhere.

Proof. *Proof to be completed.* □

Problem — Deferred finite-state duality extension

For strictly positive probability vectors, relate the exponential Fenchel dual to the scalar conjugate of the exponential, and justify the associated minimax equality. Distinguish the domains of the log-partition and exponential-integral functionals.

To be completed.

Proposition 1.39 (Optimizing an additive constant recovers Donsker–Varadhan)**.**

Source exercise: Project 11, item 11. *Student Assessment Exercise* [Assessment] (adapted).

For every bounded measurable g and probability law Q, let $Z_g = \int e^g \,\mathrm{d}Q > 0$. Then

$$\sup_{c \in \mathbb{R}} \left\{1 + \mathbb{E}_P[g + c] - \mathbb{E}_Q[e^{g+c}]\right\} = \mathbb{E}_P[g] - \log Z_g. \tag{1.30}$$

Consequently, (1.30) implies that taking the supremum over g in the exponential representation (1.29) yields precisely (1.26).

Proof. *Proof to be completed.* □

1.2.4.5 Density-ratio estimation

Theorem 1.40 (Logistic classification recovers a density ratio)**.**

Source exercise: Project 11, item 12. *Student Assessment Exercise* [Assessment] (adapted).

Let P, Q have strictly positive Lebesgue densities p, q on $V = \mathbb{R}^d$. For a measurable score $g : V \to \mathbb{R}$ define the equally weighted logistic population risk

$$L_{\log}(g) := \mathbb{E}_P[\log(1 + e^{-g(X)})] + \mathbb{E}_Q[\log(1 + e^{g(X)})].$$

If $g_*(x) = \log(p(x)/q(x))$ is in the allowed function class and the risk is finite, it minimizes $L_{\log}$, uniquely up to $(P + Q)$-null sets. For this unrestricted population problem,

$$g_*(x) = \log \frac{p(x)}{q(x)}, \qquad \frac{1}{1 + e^{-g_*(x)}} = \frac{p(x)}{p(x) + q(x)}.$$

The minimum risk equals $2 \log 2 - 2\,\mathrm{JS}(P, Q)$, where

$$\mathrm{JS}(P, Q) = \tfrac{1}{2} D_{\mathrm{KL}}(P \| M) + \tfrac{1}{2} D_{\mathrm{KL}}(Q \| M), \quad M = \tfrac{1}{2}(P + Q).$$

Proof. *Proof to be completed.* □

What density-ratio estimation does and does not prove

The logistic score $g_* = \log(p/q)$ is a *population* solution when an unrestricted measurable discriminator is available. A neural network of restricted capacity or a finite training sample need not recover this function exactly. On regions where only one density is positive, the optimal log ratio is extended-valued, so a finite everywhere optimizer need not exist. The identity is therefore a mathematical target for learning, not a finite-data guarantee.

1.2.4.6 Entropy-regularized measure optimization

The final item of Project 11 is retained as a pending extension from Gibbs tilting to optimization over probability laws.

Theorem 1.41 (Reward-guided mean-field exponential tilt)**.**

Source exercise: Project 11, item 13. *Student Assessment Exercise* [Assessment] (adapted).

Let E be a Polish space and ν a fixed Borel probability law. Suppose $\mathcal{R} : \mathcal{P}(E) \to \mathbb{R}$ is concave, bounded above, and upper semicontinuous for weak convergence. Then

$$\sup_{\mu \in \mathcal{P}(E)} \{\mathcal{R}(\mu) - D_{\mathrm{KL}}(\mu \| \nu)\}$$

admits a unique maximizer $\mu_* \ll \nu$. Suppose additionally μ_* has a strictly positive density $r_* = \mathrm{d}\mu_*/\mathrm{d}\nu$, the differentiability conditions below hold, and $\mathcal{R}$ has a functional

derivative $\delta\mathcal{R}/\delta\mu(x;\mu_*)$ *characterized by*

$$\left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\right|_{\varepsilon=0} \mathcal{R}((1+\varepsilon\eta)\mu_*) = \int \frac{\delta\mathcal{R}}{\delta\mu}(x;\mu_*)\eta(x)\,\mu_*(\,\mathrm{d}x)$$

for every bounded η *with* $\int \eta\,\mathrm{d}\mu_* = 0$ *and sufficiently small* $|\varepsilon|$. *Assume the derivative and* $\log r_*$ *are integrable against* μ_*. *Then the maximizer satisfies*

$$\begin{aligned} \mu_*(\,\mathrm{d}x) &= \frac{\exp\left(\frac{\delta\mathcal{R}}{\delta\mu}(x;\mu_*)\right)}{Z(\mu_*)}\,\nu(\,\mathrm{d}x), \\ Z(\mu_*) &:= \int \exp\left(\frac{\delta\mathcal{R}}{\delta\mu}(x;\mu_*)\right)\nu(\,\mathrm{d}x). \end{aligned} \tag{1.31}$$

In (1.31)*, the functional derivative is only defined up to an additive constant, which is absorbed by* $Z(\mu_*)$.

Proof. *Proof to be completed.* □

Appendix A

ODE Foundations and Linear Evolution

This appendix collects the supporting ODE arguments used throughout Project 9. The source problems and their links remain in the main text; in particular, Example 1.10 contains the variation-of-constants calculation itself.

A.1 Existence, Uniqueness, and Continuation

We prove Theorem 1.9.

Notice

The proof is divided into three parts.

- First, we establish the local existence of a solution near the initial time.
- Second, we establish uniqueness, showing that the same initial condition determines at most one solution.
- Finally, we extend the local solution to the entire prescribed time interval, establishing global existence.

Together, these results establish the existence and uniqueness of a global solution.

Proof.

1. Local existence. Fix an initial condition $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^d$. We seek a local solution by reformulating the initial-value problem as a fixed-point equation for an integral operator. Since f is continuous, choose $a, \rho > 0$ and consider the compact cylinder

$$Q = [t_0 - a, t_0 + a] \times \overline{B}_\rho(x_0).$$

By continuity and compactness, f is bounded on Q. Define

$$M := \sup_{(t,y)\in Q} \|f(t, y)\| < \infty.$$

By the local Lipschitz assumption, there exists $L < \infty$ such that

$$\|f(t, y) - f(t, z)\| \leq L\|y - z\|$$

for all $t \in [t_0 - a, t_0 + a]$ and $y, z \in \overline{B}_\rho(x_0)$. Choose $0 < \delta \leq a$ sufficiently small such that

$$M\delta \leq \rho, \quad L\delta < 1.$$

Let $J_\delta = [t_0 - \delta, t_0 + \delta]$. We now construct the space of candidate trajectories:

$$\mathcal{C}_\delta := \left\{\gamma \in C(J_\delta, \mathbb{R}^d) : \|\gamma_t - x_0\| \leq \rho, \quad t \in J_\delta\right\},$$

equipped with the supremum norm

$$\|\gamma\|_\infty := \sup_{t \in J_\delta} \|\gamma_t\|.$$

Since $C(J_\delta, \mathbb{R}^d)$ is a Banach space and $\mathcal{C}_\delta$ is closed, $\mathcal{C}_\delta$ is complete under the supremum norm.

For the fixed initial data, define the Picard operator $\mathsf{Q} : \mathcal{C}_\delta \longrightarrow C(J_\delta, \mathbb{R}^d)$ by

$$(\mathsf{Q}\gamma)_t := x_0 + \int_{t_0}^{t} f(r, \gamma_r)\, \mathrm{d}r.$$

- We first show that Q maps $\mathcal{C}_\delta$ into itself. For any $\gamma \in \mathcal{C}_\delta$,

$$\|(\mathsf{Q}\gamma)_t - x_0\| = \left\| \int_{t_0}^{t} f(r, \gamma_r)\, \mathrm{d}r \right\| \leq M|t - t_0| \leq M\delta \leq \rho.$$

  Therefore, $\mathsf{Q}(\mathcal{C}_\delta) \subseteq \mathcal{C}_\delta$.

- Next, we establish the contraction property. For any $\gamma, \eta \in \mathcal{C}_\delta$, the Lipschitz condition gives

$$\begin{aligned} \|(\mathsf{Q}\gamma)_t - (\mathsf{Q}\eta)_t\| &= \left\| \int_{t_0}^{t} \big(f(r, \gamma_r) - f(r, \eta_r)\big)\, \mathrm{d}r \right\| \\ &\leq L|t - t_0| \|\gamma - \eta\|_\infty \\ &\leq L\delta \|\gamma - \eta\|_\infty. \end{aligned}$$

  Taking the supremum over $t \in J_\delta$, we obtain

$$\|\mathsf{Q}\gamma - \mathsf{Q}\eta\|_\infty \leq L\delta \|\gamma - \eta\|_\infty.$$

  Since $L\delta < 1$, the Picard operator is a contraction on the complete metric space $\mathcal{C}_\delta$.

By the Banach fixed-point theorem, there exists a unique fixed point $\gamma^* \in \mathcal{C}_\delta$ satisfying

$$\mathsf{Q}\gamma^* = \gamma^*.$$

Equivalently,

$$\gamma_t^* = x_0 + \int_{t_0}^{t} f(r, \gamma_r^*)\, \mathrm{d}r.$$

Since f and γ^* are continuous, the fundamental theorem of calculus implies that γ^* is continuously differentiable and satisfies

$$\dot{\gamma}_t^* = f(t, \gamma_t^*), \quad \gamma_{t_0}^* = x_0.$$

Thus, a local C^1 solution exists.

2. Uniqueness. Let $x(t)$ and $y(t)$ be any two solutions with the same initial condition, defined on a common interval containing t_0. We show that they agree throughout their common interval.

Fix an arbitrary time t in this interval. On the compact time interval between t_0 and t, both trajectories are continuous and therefore bounded. Hence, there exists $R > 0$ such that

$$x(\tau), y(\tau) \in \overline{B}_R(0)$$

throughout this interval. By the local Lipschitz assumption, there exists a constant $L < \infty$ such that

$$\|f(\tau, x) - f(\tau, y)\| \leq L\|x - y\|.$$

Choose $\delta > 0$ sufficiently small so that $L\delta < 1$.

Consider any subinterval $[a, b]$ of length at most δ on which the two solutions agree at one endpoint. Their integral representations then give

$$\|x(t) - y(t)\| \leq L \left| \int_a^t \|x(r) - y(r)\|\, \mathrm{d}r \right| \leq L\delta \|x - y\|_{C([a,b])}.$$

The same estimate holds when the common endpoint is b rather than a. Taking the supremum over $t \in [a, b]$ yields

$$\|x - y\|_{C([a,b])} \leq L\delta \|x - y\|_{C([a,b])}.$$

Since $L\delta < 1$, we conclude that

$$x(t) = y(t), \quad t \in [a, b].$$

We can divide the compact interval between t_0 and the arbitrary time t into finitely many subintervals of length at most δ. Starting from the common initial condition, the preceding argument can be applied successively on each subinterval, in either time direction. Consequently,

$$x(t) = y(t)$$

for every t in their common interval of definition.This establishes uniqueness and completes the proof. □

A.2 Composition, Reversibility, and Differentiable Dependence

Proof. It suffices to prove the claims on an arbitrary compact time window, which we write as $[0, T]$ after translating the time coordinate. Uniqueness makes the constructions agree on overlapping windows, giving the flow throughout I. By Theorem 1.9, each initial condition determines a unique solution on $[0, T]$, including evolution to times earlier than the initial time. The existence, uniqueness, and continuation arguments are given in Appendix A.1. We use this result to construct the flow and establish its properties.

1. Construction of the flow and its algebraic properties. We first formulate the flow in terms of the solution set. Let

$$\mathcal{S}_v := \left\{\eta \in C^1([0,T], V) : \dot{\eta}_t = v_t(\eta_t)\right\}.$$

For each $r \in [0, T]$, define the evaluation map

$$E_r : \mathcal{S}_v \to V, \quad E_r(\eta) := \eta_r.$$

By global existence, every initial state $x \in V$ determines a solution on $[0, T]$, so E_r is surjective. By uniqueness, two solutions agreeing at time r must coincide throughout the interval, so E_r is injective. Hence E_r is a bijection. We can therefore define the two-time flow by

$$\Phi_{s,t} := E_t \circ E_s^{-1}.$$

In other words, E_s^{-1} assigns to an initial state its unique solution trajectory, while E_t evaluates that trajectory at time t. Thus, writing $x_\cdot = E_s^{-1}(x)$, we recover $\Phi_{s,t}(x) = x_t$, whose terminal-time evolution and initial condition are given by (1.10).

The identity, composition, and reversibility properties now follow directly from the algebra of maps. First,

$$\Phi_{s,s} = E_s \circ E_s^{-1} = \mathrm{id}_V .$$

For any $r, s, t \in [0, T]$,

$$\Phi_{r,t} \circ \Phi_{s,r} = (E_t \circ E_r^{-1}) \circ (E_r \circ E_s^{-1}) = E_t \circ E_s^{-1} = \Phi_{s,t}.$$

Similarly, taking the inverse of the defining composition,

$$\Phi_{s,t}^{-1} = (E_t \circ E_s^{-1})^{-1} = E_s \circ E_t^{-1} = \Phi_{t,s}.$$

Thus every $\Phi_{s,t}$ is a bijection, with

$$\Phi_{s,s} = \mathrm{id}_V, \quad \Phi_{r,t} \circ \Phi_{s,r} = \Phi_{s,t}, \quad \Phi_{s,t}^{-1} = \Phi_{t,s}.$$

2. Differentiability with respect to the initial state. The preceding construction establishes the algebraic properties of the flow. To show that it is a C^1 diffeomorphism, we must additionally establish its differentiability with respect to the initial state.

Fix $s \in [0,T]$ and $x \in V$, and write

$$x_t := \Phi_{s,t}(x).$$

The candidate derivative is determined by the linear variational equation

$$\begin{cases} \partial_t M_{s,t}(x) = D_x v_t(x_t) M_{s,t}(x), \\ M_{s,s}(x) = I_V. \end{cases}$$

Since $D_x v_t(x_t)$ is continuous in time, this linear matrix equation has a unique solution on $[0,T]$, including both forward and backward times.

We now verify that $M_{s,t}(x)$ is indeed the derivative of the flow. For $\delta x \in V$, define

$$\Delta_t(\delta x) := \Phi_{s,t}(x+\delta x) - \Phi_{s,t}(x).$$

The integral equations for the two trajectories and the global Lipschitz condition, with uniform constant L, imply, by Gronwall's inequality,

$$\sup_{t\in[0,T]} \|\Delta_t(\delta x)\| \le e^{LT}\|\delta x\|.$$

Set

$$A_{x,t} := D_x v_t(x_t).$$

The continuity of $D_x v$ implies uniform continuity on every compact neighborhood of the reference trajectory $\{(t,x_t) : t \in [0,T]\}$. Consequently, the first-order Taylor remainder

$$R_t^{\text{lin}}(\delta x) := v_t(x_t + \Delta_t(\delta x)) - v_t(x_t) - A_{x,t}\Delta_t(\delta x)$$

satisfies

$$\sup_{t\in[0,T]} \|R_t^{\text{lin}}(\delta x)\| = o(\|\delta x\|).$$

Subtracting the integral equation for the reference trajectory from that of the perturbed trajectory gives

$$\Delta_t(\delta x) = \delta x + \int_s^t \left[A_{x,r}\Delta_r(\delta x) + R_r^{\text{lin}}(\delta x)\right] \, \mathrm{d}r.$$

On the other hand, the variational equation gives

$$M_{s,t}(x)\delta x = \delta x + \int_s^t A_{x,r} M_{s,r}(x)\delta x \, \mathrm{d}r.$$

Subtracting these two relations yields

$$\Delta_t(\delta x) - M_{s,t}(x)\delta x = \int_s^t A_{x,r}\big[\Delta_r(\delta x) - M_{s,r}(x)\delta x\big]\ \mathrm{d}r + \int_s^t R_r^{\mathrm{lin}}(\delta x)\ \mathrm{d}r.$$

Since A_x is bounded on $[0,T]$, another application of Gronwall's inequality, in both time directions, gives

$$\sup_{t\in[0,T]} \|\Delta_t(\delta x) - M_{s,t}(x)\delta x\| = o(\|\delta x\|).$$

Therefore,

$$D_x\Phi_{s,t}(x) = M_{s,t}(x).$$

Moreover, continuous dependence of solutions on their coefficients, together with the continuous dependence of $\Phi_{s,t}(x)$ on x and the joint continuity of $D_x v$, implies that $M_{s,t}(x)$ varies continuously with x. Hence $\Phi_{s,t}$ is continuously differentiable with respect to the initial state.

Since the same argument applies with s and t interchanged, both $\Phi_{s,t}$ and its inverse $\Phi_{t,s}$ are C^1. Thus

$$\Phi_{s,t} \in \mathrm{Diff}^1(V).$$

3. Autonomous flows and the one-parameter group. Suppose now that $v_t = v$ is autonomous and that solutions exist for all real times. Define

$$\phi_t := \Phi_{0,t}.$$

For any initial time s, the trajectory $\tau \mapsto \Phi_{s,s+\tau}(x)$ satisfies

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\Phi_{s,s+\tau}(x) = v\big(\Phi_{s,s+\tau}(x)\big), \quad \Phi_{s,s}(x) = x.$$

By autonomy and uniqueness, it coincides with $\phi_\tau(x)$. Therefore,

$$\Phi_{s,t} = \phi_{t-s}.$$

The composition law then becomes

$$\phi_{t+r} = \Phi_{0,t+r} = \Phi_{r,t+r} \circ \Phi_{0,r} = \phi_t \circ \phi_r.$$

In particular, setting $r = -t$ gives $\phi_{-t} = \phi_t^{-1}$. Thus the autonomous flow defines a one-parameter group of C^1 diffeomorphisms of V.

4. Time reversal. Finally, let x_t be any solution on $[0,T]$ and define the reversed trajectory

$$y_t := x_{T-t}.$$

By the chain rule,

$$\dot{y}_t = -\dot{x}_{T-t} = -v_{T-t}(x_{T-t}) = -v_{T-t}(y_t).$$

Moreover, $y_0 = x_T$. Hence the reversed trajectory satisfies

$$\begin{cases} \dot{y}_t = -v_{T-t}(y_t), \\ y_0 = x_T. \end{cases}$$

Equivalently, if $\widetilde{\Phi}_{s,t}$ denotes the flow of the reversed vector field $\widetilde{v}_t = -v_{T-t}$, then

$$\widetilde{\Phi}_{s,t} = \Phi_{T-s,T-t}.$$

This completes the proof. □

A.3 Construction and Completeness of the Fundamental Matrix

Let $A : I \to \mathbb{R}^{d\times d}$ be continuous on a time interval I. Consider the homogeneous system $\dot{y}_t = A_t y_t$. Use the Euclidean vector norm and its induced matrix norm. Let $e_1, \ldots, e_d$ be the standard basis of $\mathbb{R}^d$. Theorem 1.9, applied on compact time intervals, provides the column solutions u_j on I such that

$$\dot{u}_j(t) = A_t u_j(t), \quad u_j(s) = e_j.$$

Define the normalized fundamental matrix by its columns:

$$U(t,s) = \begin{pmatrix} u_1(t) & \cdots & u_d(t) \end{pmatrix}.$$

Its initial columns and their differential equations give

$$\partial_t U(t,s) = A_t U(t,s), \quad U(s,s) = I_d. \tag{A.1}$$

Columnwise uniqueness also proves uniqueness of this matrix initial-value problem.

Theorem A.1 (Fundamental matrix)**.** *The normalized fundamental matrix $U(t,s)$ is invertible for all $s,t \in I$. Every homogeneous solution is $y_t = U(t,s)y_s$, and $U(t,r)U(r,s) = U(t,s)$ with $U(t,s)^{-1} = U(s,t)$.*

Proof. We prove completeness and invertibility here; the algebraic identities are established in the next section. For constant scalars $c_1, \ldots, c_d$, linearity implies

$$\frac{\mathrm{d}}{\mathrm{d}t} \sum_{j=1}^{d} c_j u_j(t) = A_t \sum_{j=1}^{d} c_j u_j(t).$$

This is the superposition principle for homogeneous solutions. Given any homogeneous

solution y_t, linear algebra writes its initial vector uniquely as $y_s = \sum_j c_j e_j$. The curve $\sum_j c_j u_j(t) = U(t,s)y_s$ solves the same equation and has the same initial value. ODE uniqueness, rather than linear algebra alone, now forces it to equal y_t at every time. In particular, evaluation at s is a linear bijection between the vector space of homogeneous solutions and $\mathbb{R}^d$. Its inverse sends c to the solution $U(\cdot, s)c$; thus the solution space has dimension d and the d column solutions form a basis.

To prove independence at an arbitrary time $t_0 \in I$, suppose $U(t_0, s)c = 0$. The solution $z_t = U(t,s)c$ then has zero value at t_0. Uniqueness with initial time t_0, in both time directions, makes it the zero solution throughout I. At s this gives $c = 0$. Hence $U(t_0, s)$ has trivial kernel, and a square matrix with trivial kernel is invertible. This proves that the columns remain independent at every time, without assuming an inverse or a composition identity in advance. The remaining assertions follow from Appendix A.4. □

A.4 Algebraic Identities for the Evolution Matrix

Fix any $r, s \in I$. As functions of t, the matrices $F(t) = U(t,r)U(r,s)$ and $G(t) = U(t,s)$ satisfy $F' = A_t F$ and $G' = A_t G$. At $t = r$ both equal $U(r,s)$. Columnwise uniqueness with initial time r therefore yields

$$U(t,r)U(r,s) = U(t,s).$$

No ordering of the three times is needed, since the solutions exist on all of I. Taking the terminal time equal to s, and then interchanging s and t, gives

$$U(s,t)U(t,s) = I_d, \quad U(t,s)U(s,t) = I_d.$$

Consequently $U(t,s)^{-1} = U(s,t)$, consistently with the independent kernel argument above. In particular,

$$U(t,s)U(\tau,s)^{-1} = U(t,s)U(s,\tau) = U(t,\tau).$$

These are identities for the linear homogeneous evolution matrix. They do not rely on properties of a general state flow.

For later differentiation under the integral, joint continuity can be checked directly: for a fixed $a \in I$, $U(t,\tau) = U(t,a)U(\tau,a)^{-1}$. Both factors are continuous in their respective variables, since inversion is continuous on invertible matrices. Thus $U(t,\tau)$ and $\partial_t U(t,\tau) = A_t U(t,\tau)$ are jointly continuous.

A.5 Matrix Exponentials and Time Ordering

The fundamental matrix packages all homogeneous initial vectors into one matrix: it avoids solving a new system for each y_s. Its definition as an initial-value problem always applies; an ordinary matrix exponential is an additional representation requiring suitable

coefficients.

Constant matrix. For $A_t = A$, the power series

$$E(t) = e^{A(t-s)} = \sum_{n=0}^{\infty} \frac{(t-s)^n A^n}{n!}$$

and its derivative series converge uniformly on compact time intervals by comparison with scalar exponential series in $\|A\|$. Termwise differentiation gives $E' = AE$ and $E(s) = I_d$. Uniqueness of (A.1) proves $U(t,s) = e^{A(t-s)}$.

Pairwise commuting coefficients. Assume $A_t A_r = A_r A_t$ for all $r, t \in I$ and put $B_A(t,s) = \int_s^t A_\tau \, \mathrm{d}\tau$. Then $\partial_t B_A(t,s) = A_t$ and $B_A(t,s)A_t = A_t B_A(t,s)$, by integrating the commutation relation. The product rule, before using commutativity, reads

$$\frac{\mathrm{d}}{\mathrm{d}t} B_A(t,s)^n = \sum_{k=0}^{n-1} B_A(t,s)^k A_t B_A(t,s)^{n-1-k}.$$

Commutativity makes this $nA_t B_A(t,s)^{n-1}$. On a compact interval, $\|B_A\|$ and $\|A\|$ are bounded, so both the exponential series and its derivative series converge uniformly. It follows that

$$\frac{\mathrm{d}}{\mathrm{d}t} e^{B_A(t,s)} = A_t e^{B_A(t,s)}, \quad e^{B_A(s,s)} = I_d, \quad U(t,s) = \exp\left(\int_s^t A_\tau \, \mathrm{d}\tau\right).$$

For noncommuting coefficients the terms in the product-rule sum cannot in general be combined. Thus differentiating $e^{B_A(t,s)}$ does not in general give $A_t e^{B_A(t,s)}$; time dependence alone does not justify the ordinary exponential formula.

Peano–Baker series from the integral equation. Integrating (A.1) gives

$$U(t,s) = I_d + \int_s^t A_r U(r,s) \, \mathrm{d}r. \tag{A.2}$$

Set $P_0(t,s) = I_d$ and recursively define

$$P_n(t,s) = \int_s^t A_{t_1} P_{n-1}(t_1,s) \, \mathrm{d}t_1, \quad n \geq 1.$$

For $t \geq s$, repeated substitution yields

$$\begin{aligned} U(t,s) = I_d &+ \int_s^t A_{t_1}\,\mathrm{d}t_1 \\ &+ \int_s^t \int_s^{t_1} A_{t_1} A_{t_2}\,\mathrm{d}t_2\,\mathrm{d}t_1 \\ &+ \sum_{n=3}^{\infty} \int_{s \leq t_n \leq \cdots \leq t_1 \leq t} A_{t_1} \cdots A_{t_n}\,\mathrm{d}t_n \cdots \mathrm{d}t_1. \end{aligned}$$

To justify the series, take a compact $K \subset I$ containing s and put $M = \max_K \|A_t\|$. Induction in the recursive integral gives, in either time direction,

$$\|P_n(t,s)\| \leq \frac{M^n |t-s|^n}{n!}, \quad t \in K.$$

In forward time, the factor $|t-s|^n/n!$ is the volume of the ordered integration simplex. In backward time the same bound follows by taking absolute values in each oriented integral. The scalar exponential series therefore proves absolute uniform convergence of $S(t,s) = \sum_{n=0}^{\infty} P_n(t,s)$ on K. Passing its uniformly convergent sum through the integral gives $S = I_d + \int_s^t A_r S(r,s)\,\mathrm{d}r$. The fundamental theorem of calculus gives $\partial_t S = A_t S$ and $S(s,s) = I_d$, so uniqueness identifies $S = U$. Equivalently, $\partial_t P_n = A_t P_{n-1}$ and

$$\|\partial_t P_n(t,s)\| \leq \frac{M^n |t-s|^{n-1}}{(n-1)!}, \quad n \geq 1,$$

which also justifies termwise differentiation on K.

This construction is exactly Picard iteration of (A.2): starting with $U^{[0]} = I_d$, $U^{[m+1]} = I_d + \int_s^t A_r U^{[m]}(r,s)\,\mathrm{d}r$ gives $U^{[m]} = \sum_{n=0}^{m} P_n$. Thus the ordered series records the successive matrix Picard iterates, rather than a new existence principle.

Meaning of the time-ordered exponential. For forward time, the notation

$$U(t,s) = \mathcal{T} \exp\left(\int_s^t A_\tau\,\mathrm{d}\tau \right), \quad t \geq s,$$

denotes precisely the Peano–Baker series. The symbol $\mathcal{T}$ places later-time matrices on the left: at second order the product is $A_{t_1} A_{t_2}$ for $t_1 \geq t_2$, so the earlier contribution acts first on a column vector. Equivalently, the nth term is

$$P_n(t,s) = \frac{1}{n!} \int_{[s,t]^n} \mathcal{T}\big[A_{t_1} \cdots A_{t_n}\big]\,\mathrm{d}t_1 \cdots \mathrm{d}t_n.$$

The cube splits into $n!$ ordered simplices, apart from boundaries of zero volume. Sorting the factors makes all these integrals equal, which explains the factor $1/n!$. With commuting coefficients the ordering can be removed, and the cube integral is $B_A(t,s)^n$; without commutativity this simplification is unavailable.

For $t < s$, use the same recursive oriented integrals for P_n or the already proved identity $U(t,s) = U(s,t)^{-1}$. In that direction the nested products place earlier physical times on the left (reverse chronological ordering), with a sign $(-1)^n$ in the nth term after reorienting the integrals. The forward $\mathcal{T}$ convention above should therefore not be read as unchanged chronological ordering for a backward interval.

The solution required by the source question remains (1.7); the ordered exponential specifies its homogeneous matrix when the coefficients do not commute. Return to Example 1.10 for the concise worked solution.

A.6 Uniform Single-Step Jacobian Estimates

This section supplies the technical estimates used in parts (a) and (c) of the numerical flow-Jacobian analysis linked at the discrete Jacobian problem. Retain its assumptions: v is C^2 on a neighborhood of $[0,1] \times B$, where $B \subset V = \mathbb{R}^d$ is compact and convex and contains the exact and sufficiently fine Euler trajectories. For $h = 1/K$, $t_k = kh$, and $V_k(x) = x + hv_{t_k}(x)$, the single-step derivative is $DV_k(x) = I_d + hDv_{t_k}(x)$.

To expand this determinant, consider a matrix $A \in \mathrm{End}(V)$ with columns $a_1, \dots, a_d$. By multilinearity in the columns $e_j + ha_j$, $\det(I_d + hA)$ is a polynomial in h. Its constant term is $\det I_d = 1$. Choosing ha_j from exactly one column contributes $h \det(e_1, \dots, a_j, \dots, e_d) = hA_{jj}$, so the linear coefficient is $\mathrm{tr}(A)$. Define the determinant remainder by

$$R_A^{\det}(h) := \det(I_d + hA) - 1 - h\,\mathrm{tr}(A) = \sum_{j=2}^{d} c_j(A) h^j.$$

Each remaining term selects at least two columns from hA, and hence contains at least h^2. The coefficients $c_j(A)$ are polynomials in the entries of A. We use the Euclidean vector norm and its induced matrix norm, as in the main analysis. For a fixed bound $\|A\| \le L_v$, continuity on the compact matrix ball gives the finite constant

$$C_{d,L_v} := \sum_{j=2}^{d} \sup_{\|A\| \le L_v} |c_j(A)|, \quad |R_A^{\det}(h)| \le C_{d,L_v} h^2 \quad (|h| \le 1).$$

The sum is zero when $d = 1$. This constant depends only on d and L_v, and is independent of A and h.

By the stated regularity on the compact set $[0,1] \times B$, fix L_v bounding $\|Dv_t(x)\|$ there. Taking $A = Dv_{t_k}(x)$ and using $\mathrm{tr}(Dv_{t_k}(x)) = \nabla \cdot v_{t_k}(x)$ gives

$$\begin{aligned} R_k^{\det}(h,x) &:= R^{\det}_{Dv_{t_k}(x)}(h), \\ \det DV_k(x) &= 1 + h\nabla \cdot v_{t_k}(x) + R_k^{\det}(h,x), \quad |R_k^{\det}(h,x)| \le C_{d,L_v} h^2. \end{aligned}$$

Thus the determinant expansion has an $O(h^2)$ remainder uniform in k and $x \in B$. The same derivative bound gives $|\nabla \cdot v_t(x)| \le dL_v$; write $B_{\mathrm{div}} := dL_v$. Set

$$u_k(h,x) := h\nabla \cdot v_{t_k}(x) + R_k^{\det}(h,x), \quad h_0 := \min\left\{1, \frac{1}{2(1 + B_{\mathrm{div}} + C_{d,L_v})}\right\}.$$

For $0 < h < h_0$, uniformly in k and $x \in B$,

$$|u_k(h,x)| \le (B_{\mathrm{div}} + C_{d,L_v})h < \frac{1}{2}, \quad \det DV_k(x) = 1 + u_k(h,x) \ge \frac{1}{2} > 0.$$

Consequently all these determinants have well-defined real logarithms for the same range of step sizes.

For the logarithmic expansion, define the scalar remainder

$$r_{\log}(u) := \log(1+u) - u.$$

On $|u| \le 1/2$, the second derivative of $\log(1+u)$ has absolute value at most 4. Taylor's theorem therefore gives $|r_{\log}(u)| \le C_{\log} u^2$ with the universal constant $C_{\log} = 2$. Define the combined logarithmic remainder separately by

$$r_k^{\log}(h,x) := R_k^{\det}(h,x) + r_{\log}(u_k(h,x)).$$

Then

$$\log\det DV_k(x) = h\nabla \cdot v_{t_k}(x) + r_k^{\log}(h,x),$$
$$|r_k^{\log}(h,x)| \le \big[C_{d,L_v} + C_{\log}(B_{\mathrm{div}} + C_{d,L_v})^2\big]h^2 =: Ch^2.$$

Here C depends only on d and the fixed derivative bound L_v, and is independent of k, $x \in B$, and $0 < h < h_0$. This proves the uniform $O(h^2)$ logarithmic expansion without identifying the determinant and logarithmic remainders.

Notation and Conventions

The tables distinguish mathematical objects by type and scope. Global entries apply throughout unless a local definition explicitly qualifies them; topic-specific entries belong to the indicated construction. They are an index to the definitions and hypotheses in the text, not additional existence, regularity, or convergence assumptions.

In these notes, hats usually indicate numerical approximations or statistical estimates, bars usually identify defined reduced models, and tildes usually indicate transformations or modified objects. These are common conventions within this manuscript, not universal rules. Every exception is defined locally. A change of model, a sample from an exact law, and a time-discretization error are distinct operations.

Operators use upright symbols such as D_A, M_A, and Q. Scalar functionals use ordinary letters: S for differential entropy, F for free energy, $H_{m_{\mathrm{ref}}}$ for negative entropy, and $L_{\log}$ for logistic risk. Calligraphic letters continue to denote the explicitly defined spaces and collections.

Global Notation

States, time, and approximation

Symbol	Mathematical object	Definition and convention
$V = \mathbb{R}^d$	State space	The full Euclidean state space; d is its dimension. A general measurable state space is written $(E, \mathcal{S})$.
$\mathcal{Z} = \mathbb{R}^m$	Reduced space	The range of a collective variable $\xi : V \to \mathcal{Z}$, with $m \leq d$ when reduction is intended.
$x,\ x_0$	States	A generic deterministic state and a prescribed initial state. The fixed initial time is stated locally.
$x_t,\ x_{t_k}$	Exact states	The exact trajectory and its value at a grid point: $x_t = \Phi_{0,t}(x_0)$ when the initial time is zero.
$X_t,\ Z_t$	Random states	A full random state and the projected process $Z_t = \xi(X_t)$. Random initial conditions do not turn an ODE into an SDE.

States, time, and approximation (continued)

Symbol	Mathematical object	Definition and convention
$x^i,\ X_t^i$	Components	Coordinate indices are superscripts; vectors are columns. Parenthesized superscripts $x^{(\ell)}$ instead label sample realizations.
s, t, r, τ	Continuous times	Usually s is the initial time and t the terminal time; integration variables are local dummy variables.
$T,\ K,\ h,\ t_k$	Time mesh	On $[0,T]$, $h = T/K$ and $t_k = kh$. From an initial time s to T, use $h = (T-s)/K$ and $t_k = s + kh$.
$N_x,\ \Delta x,\ x_i$	Spatial mesh	The cell count, cell width, and spatial cell centers. Spatial and temporal refinement parameters are distinct.
N_{samp}	Sample count	The number of observations or samples; it is neither a time-step count nor a spatial cell count.
$\widehat{x}_k$	ODE approximation	A numerical state at time t_k, with $\widehat{x}_0 = x_0$ when the initial datum is represented exactly.
$\widehat{X}_k$	SDE approximation	A time-discrete approximation to a random state. A pathwise error comparison with X_{t_k} requires a specified common coupling.
e_k	State error	$e_k := \widehat{x}_k - x_{t_k}$. Its norm is a scalar error; the vector itself is not an error magnitude.
$\widehat{\cdot}$, $\bar{\cdot}$, $\widetilde{\cdot}$	Common decorations	In these notes, usually an approximation/estimate, a defined reduced model, and a transformation/modification, respectively. These are local conventions, not universal rules; exceptions are defined where used.

Flows, derivatives, and evolution

Symbol	Mathematical object	Definition and convention
$v_t(x)$	Velocity field	$v_t(x) := v(t,x)$, with $\dot{x}_t = v_t(x_t)$. It is an instantaneous field, not a finite-time state map.
$\Phi_{s,t},\ \phi_t$	Exact flow maps	$\Phi_{s,t}$ maps a state at s to its value at t. In a specified autonomous complete system, $\phi_t := \Phi_{0,t}$.
$A_t,\ U(t,s)$	Linear evolution	$\partial_t U(t,s) = A_t U(t,s)$ and $U(s,s) = I_d$. The traditional matrix notation puts terminal time first: $\Phi_{s,t}(x) = U(t,s)x$ in the homogeneous linear case.
$D_x F(x)$	State derivative	The derivative matrix, with entries $\partial F^i/\partial x^j$. The subscript x may be omitted when the differentiated variable is unambiguous.
$M_{s,t}(x)$	Flow derivative	$M_{s,t}(x) := D_x \Phi_{s,t}(x)$ is a square matrix. Write $M_t := M_{0,t}$ only after fixing the initial time and state.

Flows, derivatives, and evolution (continued)

Symbol	Mathematical object	Definition and convention
$J_{s,t}(x)$	Flow determinant	$J_{s,t}(x) := \det M_{s,t}(x)$ is a scalar. It is positive under the stated smooth-flow assumptions; $J_t := J_{0,t}$ is an explicit abbreviation.
$V_k(x)$	Euler map	$V_k(x) = x + hv_{t_k}(x)$ and $\widehat{x}_{k+1} = V_k(\widehat{x}_k)$. Capital V_k and lowercase v_t have different meanings.
$\widehat{\Phi}_{h,k}$	Numerical flow	For Euler, $V_{k-1} \circ \cdots \circ V_0$, from the initial state to t_k. Other methods carry a method label when compared.
$\widehat{M}_{h,k},\ \widehat{J}_{h,k}$	Numerical derivatives	The derivative of the numerical flow and its determinant, with the initial-state argument retained where needed. Integrating a variational system gives this derivative only when the updates are consistent.
$\mathrm{id}_V,\ I_d$	Identities	The identity state map and the $d \times d$ identity matrix. An omitted dimension must be unambiguous.
$\mathcal{A} = B_b(V)$	Observable algebra	Bounded Borel measurable real functions, with pointwise algebra operations and supremum norm; unbounded observables require their own integrability conditions.
$\mathcal{K}_F,\ \mathcal{K}_{s,t}$	Koopman operators	$\mathcal{K}_F f = f \circ F$ and $\mathcal{K}_{s,t} f = f \circ \Phi_{s,t}$. These act on observables and reverse the order of state-map composition.
$\mathsf{T}_F,\ \mathsf{T}_{s,t}$	Law transport	$\mathsf{T}_F \mu = F_{\#}\mu$ and $\mathsf{T}_{s,t}\mu = (\Phi_{s,t})_{\#}\mu$. These act on measures, not on state vectors.
$\mathsf{K}_{s,t}$	Transition kernel	$\mathsf{K}_{s,t}(x, \mathrm{d}y)$ is a Markov probability kernel. The typeface distinguishes it from the time-step count K.
$\mathsf{P}_{s,t}$	Markov operator	$\mathsf{P}_{s,t} f(x) = \int f(y) \mathsf{K}_{s,t}(x, \mathrm{d}y)$. For a deterministic flow it equals $\mathcal{K}_{s,t}$.
$\mathcal{L}_t$	Generator	An infinitesimal operator on observables. Its domain and limit topology are specified locally; its differential expression depends on the underlying dynamics.
X_{obs}	Observable space	A selected invariant Banach space for a norm-generator; it is not the trajectory space $\mathcal{X}$ used in the linear ODE example.

Probability, analysis, and errors

Symbol	Mathematical object	Definition and convention
$(\Omega, \mathcal{F}, \mathbb{P})$	Probability space	The sample space, its sigma-algebra, and its probability measure. A moving physical region is instead written $\mathcal{D}_t$.
$\mu_t,\ \nu_t$	Probability laws	When full and reduced states are compared, $\mu_t = \mathrm{Law}(X_t)$ and $\nu_t = \mathrm{Law}(Z_t)$.
$\rho_t,\ p_{\xi,t}$	Densities	Only when the laws are absolutely continuous: $\mu_t(\,\mathrm{d}x) = \rho_t(x)\,\mathrm{d}x$ and $\nu_t(\,\mathrm{d}z) = p_{\xi,t}(z)\,\mathrm{d}z$. Static marginals may be written p_ξ.
$\pi,\ \rho_{\mathrm{eq}}$	Equilibrium law	π denotes an invariant probability law where established; if it has a density, $\pi(\,\mathrm{d}x) = \rho_{\mathrm{eq}}(x)\,\mathrm{d}x$. Invariance is a model-dependent conclusion.
$\mathbb{E}_\mu,\ \mathrm{Law}$	Expectations and laws	Expectation under a specified law and the law of a random variable. The expectation subscript may be omitted on a fixed probability space.
$\langle f, \mu\rangle$	Integration pairing	$\int f\ \mathrm{d}\mu$, distinct from the Euclidean inner product $x^\top y = x \cdot y$.
$\|x\|,\ \|A\|$	Default norms	Euclidean vector norm and its induced matrix norm. Function-space norms carry an explicit subscript, such as $\|f\|_\infty$.
$\|A\|_{\mathrm{F}},\ \|a\|$	Other magnitudes	Frobenius matrix norm and scalar absolute value. For a measurable region, $\|\mathcal{D}\|$ denotes its Lebesgue volume.
$\nabla,\ \nabla^2,\ \Delta,\ \nabla\cdot$	Spatial derivatives	The column gradient, Hessian matrix, Laplacian, and divergence. For a vector field v, its divergence is $\nabla \cdot v$.
$\mathrm{d},\ \partial_t,\ \delta$	Differential symbols	Integration/SDE differential, time partial derivative, and a defined variation. A dot denotes an ordinary time derivative.
$\mu_n \xrightarrow{\mathrm{w}} \mu$	Weak convergence of laws	On the declared topological space: convergence of integrals of all bounded continuous test functions. It does not mean function-space weak convergence.
$f_n \rightharpoonup f$	Function-space weak limit	Weak convergence in an explicitly named Banach or Hilbert space, tested against its continuous dual. No such convergence is inferred from law convergence alone.
$\Longrightarrow$	Logical implication	Reserved here for logical implication; it is not the notation for either of the preceding limits.
$\ll,\ \sim$	Measure relations	$\mu \ll \nu$ means absolute continuity; $P \sim Q$ means mutual absolute continuity, whereas $X \sim \mu$ specifies a distribution. Scalar asymptotic equivalence is defined in its own context.
$\delta_x,\ \mathbf{1}_A,\ \#$	Measure operations	A Dirac probability law, an indicator, and pushforward. A formal Dirac-delta density on a level set is interpreted under integration.

Probability, analysis, and errors (continued)

Symbol	Mathematical object	Definition and convention
$O(h^p),\ o(h^p)$	Asymptotic orders	The limit parameter and fixed data are stated. A uniform bound identifies the variables excluded from its constant; order notation is not the name of a particular remainder.
$C,\ L_x,\ L_t$	Estimate constants	Constants are local unless identified explicitly. Their sources, parameter dependence, and uniformity are stated at first use; a reused letter does not assert an unchanged numerical value.

Topic-specific Notation

Linear ODEs, geometry, and mechanics

Symbol	Mathematical object	Definition and convention
B_t	Affine forcing	The continuous vector-valued external term in $\dot{x}_t = A_t x_t + B_t$, with $B : I \to V$. Brownian motions use W with a stated superscript when needed.
$\mathcal{X},\ \mathcal{Y}$	Trajectory spaces	$C^1(I;V)$ and $C(I;V)$ in the affine ODE example. Generic path evaluation $\gamma(t)$ is the same value as γ_t.
$\mathsf{M}_A,\ \mathsf{D}_A$	Trajectory operators	$(\mathsf{M}_A\gamma)(t) = A_t\gamma(t)$ and $\mathsf{D}_A = \partial_t - \mathsf{M}_A$. Neither is the observable algebra or generator.
$\mathcal{H},\ E_t,\ S_t$	Homogeneous solutions	$\mathcal{H} = \ker \mathsf{D}_A$; E_t evaluates a homogeneous trajectory, and $S_t = E_t^{-1}$ supplies its complete solution. S_t is not a Hessian.
$\gamma_{\mathrm{p}},\ \mathcal{K}_s$	Affine geometry	The chosen zero-initial-value particular solution and the space of paths vanishing at s. The affine solution set and initial-value slice are defined in the example.
T_t	Temperature	Temperature in the cooling example, with environment value T_{env} and relaxation rate λ. It is not the terminal time T.
$\mathcal{D}_t,\ \Theta_w(t)$	Volume and weighted action	$\mathcal{D}_t = \Phi_{0,t}(\mathcal{D}_0)$ is a moving region. For the weighted-path problem only, $\Theta_w(t) = \int_0^t w_r^{-1}\,\mathrm{d}r$.
$\mathcal{V},\ H(q,p)$	Potential and Hamiltonian	The potential is model-specific. The Hamiltonian and physical hypotheses are stated in each model; $\frac{1}{2}\|p\|^2 + \mathcal{V}(q)$ is used only for the unit-mass mechanical model in these notes.
$q, p,\ z = (q,p)$	Canonical variables	In mechanics $q, p \in \mathbb{R}^n$ and $d = 2n$. In the particle introduction, subscripts on $q_i, p_i \in \mathbb{R}^3$ label particles.

Linear ODEs, geometry, and mechanics (continued)

Symbol	Mathematical object	Definition and convention
$\mathbb{J}_n,\ \nabla^2 H(z_t)$	Symplectic geometry	$\mathbb{J}_n$ is the standard $2n \times 2n$ block matrix with upper-right block I_n and lower-left block $-I_n$. The Hamiltonian Hessian is written directly.
$\widehat{q}_k,\ \widehat{p}_k$	Numerical mechanics	Computed position and momentum, including a hatted half-step momentum in Verlet. $\widehat{H}_k = H(\widehat{q}_k, \widehat{p}_k)$ is the computed true energy.
$V_h,\ \widetilde{H}_h$	Verlet quantities	The locally defined Verlet update matrix and its modified energy; the latter is not a numerical approximation to the state.
$\delta x,\ \Delta_t(\delta x)$	Initial-data sensitivity	The vector perturbation and difference of two exact flow trajectories in the differentiability proof; distinct from numerical error e_k.
$\mathsf{Q}_{s,x},\ \mathsf{Q}_\delta$	Picard operators	The integral map for initial data (s, x) and its restriction to the local trajectory set on J_δ. With the interval fixed, the latter is abbreviated Q; it is distinct from the Markov operator $\mathsf{P}_{s,t}$.
$P_n(t,s),\ U^{[n]}(t,s)$	Ordered series	A Peano–Baker term and a Picard partial sum. $B_A(t,s) = \int_s^t A_r \,\mathrm{d}r$ is an integrated coefficient matrix, not a Hamiltonian.
$\mathcal{T}$	Time ordering	The forward ordering places later-time matrices on the left; backward intervals use the explicitly stated reversed ordering.

Entropy, convexity, and information

Symbol	Mathematical object	Definition and convention
$S(\rho),\ \ell_t$	Entropy and log density	$S(\rho) = -\int \rho \log \rho \,\mathrm{d}x$ when finite; $\ell_t = \log \rho_t$ requires positive density.
$s_t^\rho,\ H_t^\rho$	Score and log Hessian	$s_t^\rho = \nabla \log \rho_t$ and $H_t^\rho = \nabla^2 \ell_t$. Neither is a solution map or the Hamiltonian.
$F(\rho),\ \beta$	Free energy	The diffusion free-energy functional and inverse temperature, under the local potential, integrability, and boundary assumptions. This functional is not the sigma-algebra $\mathcal{F}$.
$P, Q,\ p, q$	Information-theory laws	Locally, P, Q are probability measures and p, q their densities under a declared reference measure. These density symbols are not mechanical position and momentum.

Entropy, convexity, and information (continued)

Symbol	Mathematical object	Definition and convention
$D_{\mathrm{KL}},\ D_f$	Divergences	The directed relative entropy and the specified f-divergence. Density arguments are explicitly declared shorthand for measure arguments.
$r,\ f^*,\ \partial f$	Convex duality	$r = \mathrm{d}P/\,\mathrm{d}Q$ is a density ratio; f^* is the scalar convex conjugate and ∂f its subdifferential. A star on an optimizer has a separate, locally stated role.
$\vartheta,\ m_{\mathrm{ref}}$	Convex mixtures and base measure	The scalar mixing weight and the fixed reference measure. θ is instead used for model parameters.
$H_{m_{\mathrm{ref}}}$	Negative entropy	$\int p \log p \ \mathrm{d}m_{\mathrm{ref}}$ on its declared density space; its Bregman representation is used only under the stated directional differentiability conditions.
$c,\ Z_c,\ \nu_c$	Gibbs tilt	$Z_c = \int e^{-c}\,\mathrm{d}\nu$ and $\nu_c = Z_c^{-1} e^{-c}\nu$. An abstract tilt cost equals $\beta\mathcal{V}$ only when that identification is specified.
$\Lambda_Q,\ \Psi_Q$	Partition functionals	The log-exponential integral and exponential integral, respectively; their natural conjugate domains differ.
$\mathcal{J}_\alpha,\ m(b)$	Truncated Gaussian example	The constrained objective and inverse Mills ratio; b is a threshold, φ the standard Gaussian density and $\bar{\Phi}$ its upper tail. The bar here does not denote a reduced dynamical model.
$L_{\log},\ \mathcal{R}$	Risk and reward	The logistic population risk and the concave measure-valued reward functional. Neither is the generator $\mathcal{L}_t$.

Conditioning, learned transport, and numerics

Symbol	Mathematical object	Definition and convention
$\mu_z,\ \mu_{t,z}$	Conditional kernels	A static regular conditional law and a selected version of $\mathrm{Law}(X_t \mid \xi(X_t) = z)$. Null-set and geometric regularity qualifications are retained.
$\mathcal{M}_z,\ G_\xi$	Level geometry	$\mathcal{M}_z = \{x : \xi(x) = z\}$ and $G_\xi = D\xi D\xi^\top$. Surface measure is $\mathcal{H}^{d-m}$ on this level set.
J_ξ	Coarea factor	$\sqrt{\det G_\xi}$, positive under the stated full-rank assumption. It is not the square-flow derivative determinant $J_{s,t}$.
$F_\xi,\ B_i^\xi$	Mean-force objects	$F_\xi = -\beta^{-1} \log p_\xi$ for the specified static Gibbs marginal. The lifting field satisfies $D\xi B_i^\xi = e_i$.
$Y_\sigma,\ \varphi_{\sigma,m}$	Noisy observations	A noisy collective-variable observation and its Gaussian smoothing kernel. The observation noise parameter σ is not a time index.

Conditioning, learned transport, and numerics (continued)

Symbol	Mathematical object	Definition and convention
$W_t,\ W_t^{\mathrm{red}},\ d_W$	Brownian motions	The full noise, a scalar reduced noise, and full noise dimension. The scalar noise W_t^{ξ} is constructed in the projection argument. Coupling or independence is specified locally.
$b_t,\ \sigma_t,\ a_t$	SDE coefficients	Full drift and noise matrix, or in the scalar projected model a conditional drift and conditional squared-gradient factor. Averaging a noise amplitude is not averaging its square.
$A_\xi,\ Z_t^0,\ \bar{X}_t$	Projected models	The scalar projected drift before conditioning, the exact projected process, and a specified closure model. Matching marginals does not assert matching path laws.
$\boldsymbol{j}_t,\ v_t^{\mathrm{cur}}$	Probability current	$v_t^{\mathrm{cur}} = \boldsymbol{j}_t/\rho_t$ only where $\rho_t > 0$; no velocity is defined by this quotient at a zero of the density.
$\gamma,\ I_t,\ v_t^{\mathrm{int}}$	Stochastic interpolation	In this subsection only, γ is an endpoint coupling and $I_t = \alpha_t X_0 + \lambda_t X_1$. The conditional interpolation velocity is defined almost everywhere under its time marginals.
$g_{\theta,t},\ \widehat{\theta}$	Learning notation	The time-dependent parameterized family (equivalently the network interface $g_\theta(t,z)$) and fitted parameters. Write $\widehat{g}_t = g_{\widehat{\theta},t}$ and $\widehat{b}_t$ for learned functions.
$F_{\mathrm{tr}},\ F_t^{\mathrm{tr}}$	Transport maps	A quantile/triangular transport and its time interpolation. Coordinate components are F_{tr}^i; CDFs F_0, F_w and the weight function w are defined in that example.
$\widehat{\mu}_{w,N_{\mathrm{samp}}}$	Weighted empirical law	The normalized weighted sample measure; finite-sample approximation is not a smooth diffeomorphic image of a Gaussian.
$\psi_t,\ \mathcal{E},\ \mathfrak{L}$	Kinetic control	The scalar multiplier, terminal functional and Lagrangian of the formal stationarity system. ψ_t is not the autonomous flow ϕ_t.
$R_A^{\mathrm{det}},\ r_{\log}$	Explicit remainders	The determinant-polynomial and scalar logarithmic remainders; their definitions and uniform estimates are in Appendix A.6.
$r_k^{\log},\ r_k^{\mathrm{quad}}$	Stepwise remainders	The combined logarithmic remainder and interval quadrature remainder. State dependence is retained until evaluation at $\widehat{x}_k$.
$R_t^{\mathrm{lin}},\ B_{\mathrm{div}}$	Differentiability and bounds	The initial-data Taylor remainder, distinct from determinant remainders; B_{div} bounds the divergence in the Euler argument.
$D_{\mathrm{diff}},\ q_{i\to j}^{\Delta x}$	Spatial discretization	The scalar diffusion constant and outgoing transition rate from cell i to cell j. Fixed-mesh formulas abbreviate the rate as $q_{i\to j}$.
$Q^{\Delta x},\ \pi_i^{\Delta x}$	Finite-state evolution	The mass-column generator (columns sum to zero) and stationary cell masses. The backward matrix on observables is its transpose; fixed-mesh formulas may write π_i.

Conditioning, learned transport, and numerics (continued)

Symbol	Mathematical object	Definition and convention
$m_i^{\Delta x}(t)$	Semidiscrete mass	The exact continuous-time solution of the spatially discrete system, approximating a PDE cell mass. The local abbreviation is $m_i(t)$; a further time approximation would be $\widehat{m}_{k,i}^{\Delta x}$.
$A_{\mathrm{OU}},\ R_t$	OU benchmark	The positive-definite matrix and full two-dimensional state in $\mathrm{d}R_t = -A_{\mathrm{OU}} R_t\,\mathrm{d}t + \sqrt{2\beta^{-1}}\,\mathrm{d}W_t$. The drift sign is part of the model.
$m_t^{\mathrm{OU}},\ \Sigma_t$	OU Gaussian statistics	The mean and covariance; Σ_∞ is the established equilibrium covariance. The mean is distinct from the finite-volume mass vector.
$C_{\mathrm{full}},\ C_{\mathrm{closure}}$	Time correlations	The locally defined stationary autocovariances; equality of one-time laws does not imply equality of these functions.

The principal definitions are developed in Sections 1.1, 1.2, **??**, **??**, and **??**; the ODE constructions are detailed in Appendix A. A new symbol should be defined at first use, assigned an object type and scope, and checked against these tables. Standard local notations such as mechanical p, q and information-theory densities p, q are retained only with their scope made explicit.

References

[Assessment] *Student Assessment Exercise*. Unpublished doctoral screening problem portfolio, 33-page PDF. Original problem source supplied with this project as StudentAssessmentExercise_unlocked.pdf; cited by Project and item number. This manuscript selects questions from Projects 8–14. No verifiable author or publication year specified in the supplied copy.